[T]-Theory: A Universal Field Theory of Mind, Body, and Cosmos

An Introduction to the Fractal Programme

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2026

— ◊ ◊ ◊ —

"This is your last chance. After this, there is no turning back.

If this book is blue – the story ends, you wake up in your bed and believe whatever you want to believe.

If the book is red – you stay in Wonderland, and I show you how deep the rabbit hole goes.

Remember, all I'm offering is the truth."

— ADAPTED FROM THE MATRIX —

WHAT COLOR IS THE BOOK?

— ◊ ◊ ◊ —

[T]-Theory: The Blue Book

P1 · Formal Gateway · For: The Generalist

One equation. Twenty scales. Mind, body, and cosmos.

What this book claims:

- A single wave equation governs everything from synapses to the Hubble horizon
- Consciousness, trauma, dark matter, and musical emotion are the same field at different scales
- These are not metaphors — they are verified proofs in Lean 4

Three predictions (Planck 2018):

- $\Omega_\Lambda = 7/11$ — dark energy: 93% match
- $\Omega_c = 3/11$ — dark matter: 97% match
- $w = -1$ — equation of state: exact

Zero free parameters. Zero fine-tuning.

I · The One Equation

$$(\nabla^2 + k^2)\, G(x, x') = -\delta^3(x - x')$$

This is the **Green's function equation**. G is the propagator — the answer to: *"if something happens here, what is felt there?"*

At every scale from quantum foam (10^{-35}~m) to the observable universe (10^{26}~m), the same structural equation applies. Only the substrate and wavenumber k change.

The somatic memory kernel — how long a feeling persists:

$$K(\tau) = K_0\, e^{-\tau/\tau_m}, \quad \tau_m = \frac{1}{k v_s}$$

As $k \to 0$, the correlation length $\ell = 1/k \to \infty$: the field becomes globally integrated. This is the mathematical signature of consciousness onset.

II · The 20-Scale Dial

Scale	Substrate	G-Identification
0–1	Quantum / String	Graviton & worldsheet propagator
2–3	Nuclear / Atomic	Gluon & Coulomb kernel
4–5	Molecular / Cellular	Folding landscape · Synaptic response
6–7	Brain / Mind	Cortical EMF · Conscious percept pole
8–9	Swarm / Society	Active-matter field · Rapport propagator
10–13	Geo / Stellar	Seismic · Helioseismic Green's function
14–17	Galactic / Cosmic	Lensing kernel · BAO propagator
18–20	Observable Universe	Gravitational wave · Universal propagator

III · Why It Matters

The programme has three layers you can enter from:

As a reader: Chapters are self-contained. Read in any order. Start wherever your life has questions.

As a scientist: The Lean 4 proofs are at `github.com/ITI-Theory/U`. All 24 papers are on Zenodo (CC BY 4.0). Build the code. Break the claims.

As a clinician: The BRECVEMA field (Books 9–10) maps directly to intervention protocols. The trauma-well geometry has clinical predictions.

IV · Where to Go Next

- **The physics**: Book 2 (Physics) — the master equation in full
- **The mind**: Book 6 (Consciousness) — the percept pole derivation
- **The clinic**: Book 9 (Psychology/Trauma) — somatic field therapy
- **The proofs**: Book 7 (Computing) — Lean 4 verification architecture

G-ID: *The Universal Propagator* — $G(x, x')$ at the Hubble scale.

ORCID: 0009-0007-2194-0850 · · Zurich 2026

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The Green Propagator

G-ID: *The Universal Propagator — $G(x,x')$ at the Hubble scale*

The Green Propagator answers the most fundamental question in physics: if something happens here, what is felt there? In this gateway book, it is the invisible thread connecting every chapter — from the vibration of a guitar string to the emotional state of a person in crisis to the expansion of the observable universe. You do not need the mathematics to feel it. Look for the moments when the book shows two very different things behaving in exactly the same way: that sameness is the propagator at work. The whole programme has one equation; this book shows you what it means.

Introduction: What If Everything Feels?

There is a question that serious scientists rarely ask aloud: why does anything feel like anything? Not just humans — why does a bruised knee hurt, why does music make you cry, why does standing at the edge of a cliff produce something that isn't quite fear and isn't quite exhilaration but is unmistakably *something*? The standard picture of science tells us to ask how neurons fire, how hormones travel, how evolutionary pressures shaped behaviour. But it has very little to say about the experience itself — the raw, undeniable fact that there is something it is like to be you, reading these words, right now.

This book is an introduction to a framework that takes that question seriously — mathematically seriously.

The framework is called the **Universal Somatic Field**, and it belongs to a broader scientific programme called **[T]-Theory**. The word *somatic* comes from the Greek for body. The *Universal Somatic Field* is, at its core, a physical field — like the electromagnetic field, or the gravitational field — that pervades space, that has energy, that obeys equations, and that is experienced from the inside as sensation, emotion, and awareness. It is not a metaphor. It is a mathematical object, derived from the same foundational equations that underlie string theory and quantum field theory, and its predictions can be tested.

What Kind of Thing Is a Field?

Most of us learned about fields in secondary school without quite realising it. The Earth has a gravitational field: at every point in space around the planet, you can assign a number (and a direction) representing the gravitational pull. If you place a ball at any of those points, the field tells you exactly how it will move. The electromagnetic field works the same way — it fills space, it has energy, and when you flip a light switch, it is the electromagnetic field that carries the signal from the switch to the bulb.

The Universal Somatic Field is a field of this kind. It fills space. It has energy. It evolves according to equations. But it is a *tensor* field — which means that instead of just a number at each point in space, it carries a rich internal structure, something more like a small grid of numbers encoding both magnitude and direction in several different senses at once. That internal structure is what encodes the difference between, say, the feeling of warm sunlight on your face and the feeling of stepping on a piece of Lego at three in the morning.

Why Would Physics Produce Something Like This?

Modern physics has a peculiar secret. The most successful framework we have for describing everything from sub-atomic particles to the large-scale structure of the universe — a framework called *M-theory* or string theory — requires the universe to have more dimensions than the three of space and one of time that we experience. The extra dimensions are compactified: rolled up so tightly that we cannot see them directly. But they leave traces. The geometry of those hidden dimensions shapes the physical constants of our universe in ways that physicists are still working out.

The Universal Somatic Field is one of those traces. When you carry out the mathematics of compactifying M-theory down to four dimensions along a specific geometric pathway, you find a field equation that looks remarkably like the equation for a nervous system. The field has attractor states — stable configurations that it tends to settle into, just as a ball rolling on a hilly landscape tends to settle into valleys. Those attractor states, in the language of the framework, are emotions. Happiness, grief, awe, contentment — each is a valley in the energy landscape of the somatic field.

This is not poetry. It is algebra. The equations are published, the parameters can be estimated from experimental data, and the attractor structure has been computed for a realistic model of human experience.

The Shape of an Emotion

Imagine the surface of a lake on a completely windless day. Now imagine you drop a stone into it. Ripples spread outward. If you drop several stones in different places at the same time, the ripples interact: they add together in some places and cancel in others, producing a complex interference pattern.

The somatic field works similarly, but in a much richer space. Your emotional state at any moment is a configuration of the field — a particular pattern of energy across the landscape of possible experience. That configuration evolves over time according to the field equations, pushed by incoming sensory data, by memories, by the state of your nervous system. When the configuration settles into a stable valley, you feel something distinct and nameable. When it is between valleys — turbulent, uncertain — you feel what most of us call anxiety, or restlessness, or that unnamed sense that something is not quite right.

The framework gives precise mathematical descriptions of: how attractor states form and dissolve; what happens when traumatic experience carves a very deep valley (a *trauma well*); how two people sharing an interaction can synchronise their fields (the formal basis of empathy and rapport); and what distinguishes healthy emotional variability from stuck, repetitive patterns.

What You Will Find in This Book

The chapters that follow present the scientific case in three layers, each building on the last.

The first layer is the theory itself. You will meet the core equations in the most accessible form possible — not as a wall of symbols, but as a set of pictures and physical intuitions. You will see why the field has to be a tensor rather than a scalar, what the attractor structure looks like geometrically, and what the formal definition of a *felt emotion* is within the framework.

The second layer is evidence. The framework makes predictions: about which brain states should produce which phenomenal experiences, about the neural correlates of emotional transitions, about how trauma alters the field structure, about what musical chord sequences do to the energy landscape. Several of these predictions have been tested, and this book reports what was found.

The third layer is implication. If the Universal Somatic Field is a real physical field — if emotions have the same ontological status as gravitational fields — then a great deal changes. In medicine, psychological suffering becomes a field pathology with a geometry, not a malfunction to be suppressed. In physics, consciousness becomes continuous with the rest of nature, not an embarrassing anomaly to be explained away. In daily life, the way you understand your own experience — and the experience of the people around you — acquires a rigour and a compassion that casual introspection alone cannot provide.

This Is Part of a Larger Programme

This book is the gateway volume of the **[T]-Theory Fractal Programme** — a set of fifteen books, each applying the Universal Somatic Field to a different academic domain. Physicists will find a volume grounding the field in quantum field theory and M-theory. Neuroscientists will find a volume connecting it to the electromagnetic theory of consciousness. Clinicians will find a volume on trauma and therapy. Lawyers, economists, geophysicists, musicologists — each field has its own book, written in its own language, showing how the same master equation governs phenomena that discipline had studied in isolation.

Recent work extended the framework to cosmology with striking results. The same eleven-dimensional structure that fixes the geometry of emotional attractor states also predicts, without any free parameters, two of the largest numbers in physics: the cosmological constant — the energy driving the accelerating expansion of the universe — falls within 7% of the measured value; and the fraction of the universe made of dark matter falls within 3% of the Planck satellite's measurement. These predictions come from counting dimensions (seven compact, three spatial, one temporal), not from fitting the data. The equation that describes how a memory gets stuck is the same equation that describes why 27% of the universe is invisible.

This is the gateway. It asks only that you follow the logic, and stay curious.

The stone is in the water. Watch the ripples.

The Programme

This is a document about structure.

Not about feelings — though feelings are what the programme is ultimately for. Not about therapy — though therapy is one of the principal applications. Not about physics — though physics is where the mathematics comes from. It is about a single recurring observation: that the equations governing emotional dynamics are the same equations that govern quantum fields, and that this is not a metaphor.

When an identification like that is made precisely — when you can say not "this is *like* a wave" but "this *is* a wave in the technical sense, with the same propagator, the same energy function, the same topology, and therefore the same theorems" — a compressed body of work becomes possible. You are not building from scratch. You are navigating.

This document describes what was built by navigating, and why the pieces form a whole.

The Gap the Programme Addresses

Every large language model deployed today is a classical system. Its training is gradient descent. Its inference is deterministic or thermally noisy sampling. The architecture was designed to model the neocortex — pattern recognition, sequence prediction, error minimisation.

The complementary system — the limbic system, responsible for valuation, threat detection, arousal modulation, and the somatic state reinstatement that underlies trauma — had no formal mathematical treatment before this work. The clinical literature described it richly (Porges, van der Kolk, Levine). The neuroscience described its anatomy. Neither provided a model from which predictions could be derived and tested.

Simultaneously, the psychology of music had reached a similar ceiling. A 991-page handbook (Juslin and Sloboda, 2010) treated music-induced emotion almost entirely through Russell's valence–arousal circumplex: a static two-dimensional map. The circumplex describes *where* a listener is, not *how* they move, what traps them, or what allows escape. No dynamical model of music-induced affect existed.

The programme fills both gaps with the same model, via the same method.

The Structure of the Argument

The argument has three movements and several extensions:

Paper	Movement	Contribution
Mathematical Co-identification (2026)	Method	Names and formalises the procedure
The Soma-Field (2026)	Model	Applies it to emotional dynamics
Quantum Soma and the Penrose Gap (2026)	Empirical test	Confirms the central claim
Field Notes from the Inside (2026)	Lived case	Primary-source clinical grounding
A Dynamical Field Model of Music-Induced Affect (2026)	Extension	Demonstrates domain generality
The Tensor (2026)	Extension	Applies the framework to abstract film

The popular account (*A Voyage into Trauma*, 2026) provides the same argument in accessible form, for readers without a physics background.

The Method: Mathematical Co-identification

What It Is

The history of mathematical science contains a recurring event. At a certain moment, a scientist recognises that the quantity they are studying is not *like* a quantity already understood in another domain — it *is* the same mathematical object, under a change of label. When this identification is made precisely, every theorem proved about the source object becomes available in the target domain immediately, without re-derivation.

This event has happened many times:

- Hopfield (1982) recognised that a neural network's energy-minimisation dynamics are the same as a spin-glass Hamiltonian. Every result from statistical mechanics of spin glasses — ground states, phase transitions, capacity bounds — imported.
- Veneziano (1968) recognised that the Euler Beta function, a result in pure mathematics, described the scattering amplitudes of hadrons. String theory began.
- Black and Scholes (1973) recognised that an option pricing equation was the heat diffusion equation. Every analytical tool from thermodynamics imported.

The paper *Mathematical Co-identification: A Method for Structural Import Across Scientific Domains* (Johnson, 2026a) names this procedure, formalises it as a distinct scientific method with its own validity criteria and failure modes, and distinguishes it from analogy, metaphor, and modelling. The key distinction:

> **Analogy**: A is *like* B in certain respects. Illuminating, not transferable.
>
> **Co-identification**: A *is* B under relabelling. Every theorem about B is a theorem about A.

Why It Matters as Method

A co-identification can be wrong. The identification is only valid if the mathematical type matches: the same dimensionality, the same algebraic structure, the same boundary conditions, the same symmetry group. The paper provides a falsifiability protocol — a formal procedure for pre-registering an import claim and specifying what observation would disconfirm it.

This matters because the failure mode of co-identification is not sloppy reasoning — it is overly precise reasoning applied to the wrong type. The paper catalogues seven historical examples to distinguish the valid from the invalid pattern.

The Soma-Field Model is the worked example throughout. The identification was not discovered by reading physics textbooks and looking for something that felt similar. It was discovered by writing down the equations the emotional system was observed to satisfy and recognising the form.

When MCI Is Over: The Verification Threshold

Here is something no paper on scientific method says clearly enough.

Every scientist who produces a major structural identification — Hopfield recognising the Ising Hamiltonian, Veneziano recognising the Euler Beta function — does so by being, for a period, a *type astronaut*. They are scanning the existing mathematical universe for a structure whose type signature matches the phenomenon they are studying. This is abductive reasoning: not deduction from first principles, not induction from data alone, but the systematic search of a known solution space for a structure that fits. It is, to be direct, a form of academic hacking. And it is how most of the connective work of mathematical science actually gets done.

What no one says — because scientists do not typically publish the search, only the result — is that the moment of structural identity is also the moment the method becomes irrelevant.

When Veneziano published his scattering amplitude, he did not need to cite the library where he found the Beta function. The formula was true. Every theorem about the Beta function was now a theorem about hadronic scattering. The library visit was the ladder; the amplitude was the building. The ladder came down.

The present work makes this transition explicit, because being explicit about it is itself a contribution. Every reader of the earlier papers — particularly *Mathematical Co-identification* — has correctly understood that MCI was the search engine. What should now be equally clear is that the search is over.

The exact moment the search ended was when the Lean 4 kernel accepted the proof. Not when a reviewer accepted a paper. Not when a human mathematician checked the algebra. When a formal proof-checking engine with no stake in the outcome closed the goal. That is the verification threshold. Before it: exploration. After it: structural fact.

What the work therefore rests on is not MCI. It rests on four things:

1. **Inductive structural necessity.** The 11-dimensional decomposition of a body-field-mind system is not an analogy with M-theory. It is the minimum geometry required to account for the functional degrees of freedom of a conscious organism. The isomorphism to M-theory is a *theorem*, not a design choice.

2. **Structural identity, not analogy.** A co-identification is not “A is like B.” It is “A *is* B under relabelling.” Every theorem about B becomes a theorem about A — immediately, without re-derivation. This is categorically different from saying that emotional dynamics *resemble* a Hopfield network. They *are* one.

3. **Scale invariance.** The same Helmholtz Green’s function equation governs 20 scales of physical reality, from quantum foam to the cosmic web. This is a discovered law. MCI was used to find it; it stands whether or not MCI is remembered.

4. **Kernel verification.** The Lean 4 proofs are the epistemological gold standard. A human reviewer can miss a subtle error. The kernel cannot. Kernel-verified theorems are exactly as true as the axioms they depend on — and the axioms are explicit and listed.

The *mathematical-co-identification* paper remains an accurate account of how the work was done — and naming the search process honestly is itself a contribution, in a discipline where the search is usually erased. But readers coming to the thesis or omnibus now should understand that they are reading the *results*, not the method. The method is in the history. The results are in the formal proofs.

The Model: The Soma-Field

Five Co-identifications

The Soma-Field Model (Johnson, 2026b) is built from five sequential co-identifications, each importing a body of mathematics from physics into emotional dynamics:

Co-identification 1: The Hopfield identification. The brain's emotional attractor dynamics satisfy the same energy function as a Hopfield neural network. The energy function is:

$$H(\mathbf{e}) = -\tfrac{1}{2}\mathbf{e}^\top W\mathbf{e} - \mathbf{b}^\top\mathbf{e}$$

where $\mathbf{e} \in \mathbb{R}^N$ is the emotional state vector, W is the coupling matrix, and $\mathbf{b}$ is a bias vector encoding baseline arousal. The local minima of H are the named attractor states: regulated calm, fight, flight, freeze, flow, dissociation.

Co-identification 2: The QFT identification. The emotional field propagates as a quantum field. The conscious emotional percept is the one-dimensional impulse response — the Green's function — of an eleven-dimensional coupling manifold. The same object that describes a massive particle in quantum field theory describes a conscious emotion: a pole in the propagator of the field.

$$G(\omega) = \frac{1}{\omega^2 - m^2 + i\epsilon}$$

This is not a metaphor. The threshold T at which a sub-perceptual field fluctuation becomes a conscious emotional percept is the mass parameter m in the propagator. Below threshold: virtual. Above threshold: real.

Co-identification 3: The brane identification. The body and the nervous system are not the same manifold. The body is a 3-brane embedded in the 11-dimensional coupling manifold. Somatic pain states and the body schema are field modes on this brane, not on the bulk manifold. This is the formal statement of the somatic grounding of emotion.

Co-identification 4: The G_2 holonomy identification. The seven compactified extra dimensions of the coupling manifold are a G_2 manifold. The G_2 holonomy group is the one that gives rise to topological obstructions — loops through the moduli space that cannot be continuously contracted to a point. In emotional terms: trauma configurations from which smooth continuous change cannot escape. The topological barrier is not a metaphor for being stuck. It is a mathematical object with a winding number.

Co-identification 5: The renormalisation group identification. Developmental trajectory maps onto the renormalisation group flow. The age at which a traumatic modification was introduced corresponds to the energy scale at which the coupling constant was set. High-energy (early developmental) modifications are renormalisation-group relevant — they affect all subsequent scales. Low-energy (later life) modifications are irrelevant in the technical sense. This gives the formal account of why early trauma is not simply a more intense version of later trauma: it is a different class of object.

What the Model Predicts

From these five identifications, several predictions follow that are not derivable from any existing clinical model:

1. **Threshold crossings are phase transitions.** The transition from sub-perceptual to conscious emotion is a second-order phase transition in the field. This predicts hysteresis — it is easier to stay in a state than to enter it, and easier to stay out than to leave.

2. **Complex PTSD is a topological configuration.** The coupling matrix W for a CPTSD nervous system has a specific structure: a winding-number-protected attractor landscape in which the Fear basin is separated from the Awe basin by a barrier that low-noise classical gradient descent cannot cross. This is a prediction about matrix structure, not a description of symptoms.

3. **Autism Spectrum Condition modifies the threshold operator.** The threshold parameter T in the ASC nervous system has a different coupling to the field modes than in the neurotypical case — specifically, the threshold is non-uniform across sensory modalities, producing the characteristic pattern of simultaneous hypo- and hyper-sensitivity.

4. **ADHD modifies the effective temperature.** The stochastic term in the Langevin dynamics governing the ADHD nervous system has higher effective temperature T_{eff}. This is not a deficit of attention; it is a higher rate of escape from local minima — an advantage in landscapes where rapid sampling is valuable and a liability where sustained convergence is required.

5. **Quantum mechanisms are required for certain transitions.** For trauma configurations with topological barriers (non-zero winding number), low-noise classical gradient descent cannot reach the global minimum. A quantum mechanism is required. This is the prediction that QUANT-EXP-1 was designed to test.

The Empirical Test: QUANT-EXP-1

The Prediction

The soma-field model makes a specific, falsifiable claim: for a Hopfield landscape with a topological trauma barrier, low-noise classical Langevin dynamics starting from the Fear attractor cannot reach the Awe attractor. Quantum annealing — a physically realisable mechanism — can.

This is not a claim about whether people should use quantum computers in therapy. It is a claim about reachability: that the mathematical structure of the barrier distinguishes the quantum and classical regimes in a measurable way.

The prediction was registered in the Zenodo v1 deposit of the Soma-Field paper (doi:10.5281/zenodo.20350515) before the experiment was run.

The Experiment

Quantum Soma and the Penrose Gap (Johnson, 2026c) reports QUANT-EXP-1: an exact 8-qubit statevector simulation on a 256-dimensional Hilbert space, implementing the Soma-Field Hopfield Hamiltonian with a transverse-field quantum annealing schedule.

The experimental design:

- **System**: 8-qubit Hopfield model encoding four emotional modes (Fear, Calm, Awe, Grief) plus sub-modes. Coupling matrix W set to produce a topological barrier between Fear and Awe.
- **Quantum dynamics**: Transverse-field annealing with schedule $H(s) = (1-s)H_X + sH_{\text{problem}}$, $s \in [0, 1]$.
- **Classical baseline**: Overdamped Langevin dynamics at low temperature ($T_{\text{eff}} = 0.01$), same starting state, same landscape.
- **Primary outcome**: Peak Awe-dominant occupancy (quantum) versus success rate of cold-classical crossings.

Results

Results are presented against the pre-registered barrier ladder:

Barrier strength	Classical cold rate	Classical cold CI [95%]	Quantum peak
$W = -6$	0.000	[0.000, 0.019]	0.389
$W = -8$	0.000	[0.000, 0.019]	0.408
$W = -10$	0.000	[0.000, 0.019]	0.408
$W = -12$	0.000	[0.000, 0.019]	0.409
$W = -14$	0.000	[0.000, 0.019]	0.416

Bootstrap confidence intervals (n = 200 seeds) confirm that the classical cold success rate is bounded above by 1.9% at all tested barrier strengths. Quantum peak occupancy is stable at 0.389–0.416 across the full range.

Pre-registered hardening protocol — all checks passed:

- **Bootstrap** (n = 200): cold CI = [0.000, 0.019]; quantum peak 0.408–0.410. Intervals do not overlap at any barrier strength.
- **Control A** (start from Awe, barrier intact): classical 16/16 stay in Awe. PASS. Confirms that the barrier is directional: it blocks Fear → Awe, not the reverse.
- **Control B** (barrier removed, $W[\text{Fear,Awe}] = +0.4$): classical 16/16 reach Awe. PASS. Confirms that the barrier, not the landscape geometry, is what blocks classical dynamics.
- **Spectral gap**: gap narrows monotonically with barrier strength (B8: 0.0095, B10: 0.0089, B12: 0.0085) and reaches its minimum at $s \approx 0.999$, confirming the tunnelling bottleneck is late in the anneal.

Verdict: The strong reachability claim stands. QUANT-EXP-1 is a PASS.

The Penrose Connection

The paper situates this result in the context of Penrose's argument about non-computability and consciousness. The connection is not that consciousness requires quantum mechanics in general. The connection is more specific:

Penrose identified a *gap* between what classical computation can reach and what consciousness can do. The soma-field identifies a corresponding *topological gap* in the emotional landscape between what classical gradient descent can reach and what a genuinely new state of the nervous system requires. QUANT-EXP-1 provides the computational demonstration that the gap exists and is crossable by a quantum mechanism.

The contribution is not to resolve Penrose's claim about consciousness. It is to *instantiate* the gap in a concrete, testable, mathematical setting.

The Lived Case: Field Notes from the Inside

Field Notes from the Inside: A Patient-Constructed Model of Emotional Dynamics (Johnson, 2026d) performs a function that the formal papers cannot perform: it provides the primary-source clinical grounding.

The paper is written by the person who has Autism Spectrum Condition (Level 2), Attention Deficit Hyperactivity Disorder, and Complex Post-Traumatic Stress Disorder — and who also has a degree in physics. The model was not developed by observing patients. It was developed by having the conditions and finding the existing models inadequate.

The epistemological contribution of this paper is often undervalued. Every formal model of a human system is, in the end, derived from observation of that system. When the observer and the observed are the same entity, and that entity has the training to translate observation into formal mathematics, the resulting model has a different epistemic status from one derived by observation from the outside. The paper makes this explicit, situates it within the autoethnographic research tradition, and argues that the resulting model is *more* constrained, not less — because any prediction the model makes that does not match the primary observer's experience is immediately falsified.

The formal content is a set of operator modifications for the three conditions:

- **ASC**: The threshold operator T is replaced by a modality-dependent operator T_k for each sensory channel k, with different coupling strengths. The result is the characteristic simultaneous hypo- and hyper-sensitivity: some channels are below threshold where the neurotypical channel is above it, others are above where the neurotypical channel is below.
- **ADHD**: The Langevin noise term $\sqrt{2T_{\text{eff}}}\,\eta(t)$ has elevated T_{eff}. This is a quantitative modification, not a qualitative one. The system is not broken; it is sampling the energy landscape at higher temperature. The therapeutic implication is not to reduce the noise but to design the landscape so that high-temperature sampling is an advantage.
- **CPTSD**: The coupling matrix W has the topological structure described in §3.2: a winding-number-protected barrier between Fear and regulated states. The barrier was installed before language, before narrative memory, before the self that can explain the barrier was formed. The modification is not a layer added to a pre-existing structure. It is the structure.

Extensions: Music, Film, and the Domain Generality of the Model

Music-Induced Affect

A Dynamical Field Model of Music-Induced Affect: Beyond the Valence–Arousal Circumplex (Johnson, 2026e) applies the soma-field framework to a domain where the empirical literature is rich and the theoretical models are weak.

Juslin and Sloboda's *Handbook of Music and Emotion* (2010) — 991 pages — contains the circumplex as its dominant quantitative framework. The circumplex is a static map. It describes where a listener is; it does not model how they move. The soma-field is the first dynamical model of music-induced affect.

The key predictions that the circumplex cannot make but the field model does:

1. **Phase transitions, not continuous shifts.** State changes in music-induced affect are not smooth movements across the circumplex. They are threshold crossings — sudden re-configurations

of the attractor landscape. The field model predicts the conditions under which a transition occurs and the hysteresis that prevents immediate return.

2. **The adaptive function of high effective temperature.** In the ADHD nervous system (elevated T_{eff}), music that holds a neurotypical listener in a stable state may drive repeated transitions. This is not a bug; it is the same high sampling rate that characterises the ADHD cognitive profile. The model gives this a formal account.

3. **Basin depth asymmetry.** The freeze attractor basin is deeper than the regulated calm basin. This means it is harder to leave freeze than it is to leave calm — asymmetric with respect to the direction of transition. Music that successfully moves a listener from freeze to calm is doing qualitatively different work than music that moves a calm listener to a more activated state.

The paper also specifies a real-time instrument implementation: a MIDI controller array driving a Python field server at 50 Hz, with audio output via Ableton Live and 3D fractal visual output (Mandelbulb projection onto HoloGauze screen). The instrument is not described; it is specified formally, with pre-registered hypotheses and disconfirmation criteria.

The Tensor: An Abstract Film

The Tensor: An Abstract Film Definition (Johnson, 2026f) extends the framework to abstract film. A film is defined not by its pixels but by its **emotional score**: a vector-valued trajectory $\mathbf{e}^*(t)$ through the emotional field, parameterised by story-time $t \in [0, 1]$.

The rendering — the actual audiovisual output a viewer experiences — is generated at runtime from this trajectory, the viewer's own soma-field state, and a set of control parameters. In the limit where the viewer's biofeedback is available, the film adapts to where the viewer is: the trajectory is not what the viewer experiences, but what the film proposes. The work is not the pixels. It is the map.

This is a significant claim about what an artwork is. A conventional film is fixed: the same sequence of frames for every viewer at every screening. The tensor film is a field: a mathematical object that takes the viewer's state as input and produces an output adapted to it. The artistic statement is in the trajectory, not the realisation.

The paper does not describe how to make such a film. It defines the abstract structure that any realisation of such a film must instantiate — the way a musical score defines a symphony without being the performance.

The Argument as a Whole

The six papers form a single argument, and it can be stated in a paragraph:

> The limbic system and its coupling to the body are governed by the same mathematical equations as a quantum field on a manifold with G_2 holonomy. This identification is not a metaphor; it is a co-identification in the technical sense, with all the theorems of each source domain importing into the target. Among those theorems is one that has clinical consequences: topological barriers in the emotional attractor landscape cannot be crossed by low-noise classical gradient descent. A quantum mechanism can cross them. This has been computationally confirmed (QUANT-EXP-1) against a pre-registered hardening protocol. The model correctly describes the structure of Autism Spectrum Condition, ADHD, and Complex PTSD as operator modifications, and generalises to music-induced affect and abstract film with no change to the underlying mathematics.

What makes this a research programme rather than a single paper is the **generativity**: the method (co-identification) produces results in any domain where an attractor landscape with topological structure can be identified. The soma-field is one instantiation. Music-induced affect is a second. Abstract film is a third. A fourth — currently in design — is **H-AL**: a holographic avatar whose body is a live Mandelbulb rendering of the emotional field state, projected at human scale through a hologauze screen and accompanied by a synthesised voice narrating the field in real time. The geometry of the fractal changes as the field changes; regulated calm and trauma produce visually distinct and mathematically characterisable forms. The same functor architecture (§A.4 of the main paper) supports this output with no changes to the field computation. Each of these instantiations generates falsifiable predictions from the same mathematical core.

What makes this a *novel* research programme is the **gap it fills**: no formal dynamical model of the limbic system existed before this work. The Hopfield framework gave the neocortex its formal model in 1982. The soma-field gives the limbic system its formal model in 2026. Together they constitute the first complete formal description of the two principal computational substrates of the vertebrate brain.

What Remains

The body of work described here is computationally complete. All pre-registered hardening checks have been executed. The claims that can be confirmed by simulation have been confirmed.

Three categories of work remain outside the scope of these papers:

Physical hardware confirmation. QUANT-EXP-1 uses exact statevector simulation. Running the same 8-qubit experiment on IBM Quantum free-tier hardware would produce the sentence "confirmed on physical quantum hardware." This is feasible, is the logical next step for any journal submission targeting a hardware-inclusive venue, and is not required to support any claim in the current corpus.

Peer review. The three published papers are currently archived on Zenodo as open preprints. Peer review in ranked journals is a separate track, ongoing. The relevant venues are: *Frontiers in Computational Neuroscience* (Hypothesis and Theory article type) for the soma-field paper; *Synthese* or *Philosophy of Science* for the co-identification paper; *Music Perception* or *Frontiers in Psychology* for the music-affect paper.

Empirical clinical application. The model makes predictions about specific clinical populations (ASC, ADHD, CPTSD) that require empirical testing outside the computational domain. This constitutes a research programme for clinical collaborators. The predictions are pre-specified in §3.2 of this document and in the relevant papers; they are not vague.

Physical substrate. The model is formally complete but physically silent on the tissue substrate in which the soma-field is instantiated in living organisms. A companion paper, *The Physical Substrate of the Soma-Field* (Johnson, 2026g), develops this layer across three converging research traditions: biotensegrity (Ingber, Levin) as the mechanical architecture through which the somatic wave propagates globally; fascial-interstitial continuity (Langevin, Schleip, Oschman) as the active signalling tissue and physical locus of attractor-depth encoding; and biofield physiology (Popp, Ho, McCraty, Rubik) as the candidate physical correlate of the field itself. The most clinically significant result is the quantitative correspondence between fascial stiffness and attractor depth: chronic fascial armoring measurable by shear-wave elastography is the physical implementation of the energy barriers that QUANT-EXP-1 shows to be quantum-resistant. Myofascial release is thus barrier *lowering* — not barrier crossing — and therapist-client physiological entrainment is the physical mechanism of co-identification.

Data and Code Availability

All papers, simulation code, result tables, figures, and Lean 4 formal proofs are archived at the following Zenodo records (open access):

Paper	DOI
The Soma-Field	10.5281/zenodo.20350515
Mathematical Co-identification	10.5281/zenodo.20287981
Quantum Soma and the Penrose Gap	10.5281/zenodo.20351230

The unreviewed papers (*Field Notes from the Inside*, *Music-Induced Affect*, *The Tensor*, and this synthesis document) will be deposited on Zenodo as part of the next release of the research archive.

A Voyage into Trauma

The Soma-Field Theory of Emotional Life

Alistair Johnson

2026

> *For everyone who was told their body was overreacting. It wasn't. It was solving the right problem.*

Preface: The T's

This book began as a physics paper.

It ended as a map.

The paper was called the Soma-Field Model, and it was written in the language of physicists: Hamiltonians, propagators, coupling matrices, Wick rotations. It was precise. It was, I think, correct. And it was almost entirely unreadable to the people it was most about.

This book is the translation.

There are four T's running through what follows, and they are not accidental. The first is **Trauma** — the subject. The second is **Threshold** — a specific parameter in the model, written T, that marks the boundary between what becomes conscious and what stays body. The third is **Time** — in particular, developmental time, the age at which a modification occurred, which turns out to matter enormously. The fourth is **Transformation** — not recovery, not a return, but a going forward into a wider landscape.

There is a fifth T that I notice only in retrospect: **Trance** — in two senses simultaneously. *A Voyage into Trance* is a 1995 Goa trance compilation by Paul Oakenfold; the title of this book is borrowed from it. A trance state is, in the language of the Soma-Field Model, a phase transition of the emotional field — a threshold crossing guided by sound and rhythm rather than threat. The trance state produced by extended rhythmic music at 140 BPM and the freeze response of a traumatised nervous system are not the same experience. They are governed by the same mathematics: the same threshold crossings, the same phase transitions, the same field dynamics. The T's of that album and the T's of this book are the same T's.

I should tell you what happened in 1968 before we go any further, because it is the reason this model exists and the reason I am the one writing it. I was approximately eighteen months old. I developed septic arthritis in my left hip — a bacterial infection of the joint that, without treatment, destroys the socket entirely. It was treated, and I recovered. The treatment involved three months in hospital under what were then called "no-touch protocols": the infection risk was judged to require isolation from physical contact. Three months is a long time at eighteen months. The body learns quickly at that age, and what mine learned — in the absence of any other available hypothesis — was that the world was a place where pain arrived without warning and comfort did not follow.

That is not a complaint. The clinicians saved the joint. But the learning happened, and it happened before language, before narrative memory, before the self that can tell the story was formed. The modification was not added to an existing structure. It was the structure.

This book is, among other things, an attempt to say that formally — to give that experience a mathematical description precise enough to make predictions, to inform clinical practice, and to explain why the goal is not to return to a self that never existed, but to build forward into one that can.

I should also tell you that I promised, years ago, to write a book about a campsite in the Glarus Alps of Switzerland — a valley with parabolic limestone walls that make sound oscillate like a natural resonator, adjacent to one of the world's great tectonic structures. The book about the campsite and the book about the soma-field found each other. The Interlude between Part II and Part III is where they meet.

The physics is real. The equations correspond to something. And the voyage is the proof of it.

Alistair Johnson May 2026

How to Read This Book

This book is written for three kinds of reader, and you can navigate it differently depending on which you are.

If you are new to all of this — no physics background, no clinical background, just a body that has been confusing you — read Part I first. Chapters 1 through 3 are written for you. The mathematics in those chapters is kept to a minimum; the ideas are introduced through physical intuition and lived experience. When equations do appear, they are explained in words immediately. Nothing is assumed except curiosity.

If you are a mental health professional who wants to understand what this model adds to your existing framework — you can begin with the Part I overview and then move directly to Parts III and IV. The Going Deeper boxes throughout the book are written for you. The appendices contain the full mathematics as it appears in the academic paper.

If you are a physicist, mathematician, or computationalist who has arrived here by accident or curiosity — you will recognise the Hamiltonian formulation immediately. The novel content for you is in Chapters 6, 7, and Appendix A. The Lean 4 type sketches in Appendix B may be of particular interest; they are incomplete proofs, marked with `sorry` where the hard work remains, and they represent a research programme.

A note on boxes. Throughout the book you will find four types:

> **LEARNING OBJECTIVES** — what this chapter sets out to establish, listed at the start.

AUTHOR'S NOTE — personal first-person sections. The research and the life it emerged from are not separate, and I have not pretended otherwise.

GOING DEEPER — technical sections with more mathematics or formal detail. They can be skipped without losing the main argument. They can also be read first if that is your preference.

KEY TERMS — precise definitions for terms that carry specific meaning in this model. A full glossary is in Appendix D.

PART I: THE BODY KNOWS

What the Body Remembers

```
> "The body keeps the score."
> — Bessel van der Kolk, 2014
>
```

LEARNING OBJECTIVES

By the end of this chapter, you will understand:

- Why the body responds to safety and danger independently of what the conscious mind believes
- What the freeze response is and why it exists
- The difference between a body that is "overreacting" and a body that has learned accurately
- Why the first step in understanding trauma is understanding that survival responses are not mistakes

---

The Waiting Room

Picture a waiting room. You are sitting in a chair, waiting for a routine appointment. Nothing unusual is happening. The lighting is fluorescent. There is a plant in the corner that needs watering. Someone across the room is reading a magazine with the particular focused patience of someone who is not, in fact, reading a magazine.

For most people in this waiting room, the body is doing nothing remarkable. Heart rate is steady. Breathing is even. The background hum of vigilance that keeps us alive is running at its ordinary level.

For some people, this waiting room is already an event. Heart rate has been elevated since the appointment was booked. Breathing is shallower than usual. There is a low-frequency alertness — a readiness — that is using energy, keeping muscles slightly contracted, keeping the hearing slightly sharpened, keeping the eyes moving. The mind may know that this is a routine appointment. The body has reached a different conclusion and is acting on it.

This is not anxiety in the clinical sense, though it may be diagnosed as such. It is not a failure of rational thinking. It is a body doing exactly what it learned to do — and doing it accurately, given what it knows.

The question this book asks is: what does the body know, and how did it learn it, and what does it mean to change that learning?

What Trauma Is (and Is Not)

The word *trauma* is used in many ways. In this book, it has a specific meaning.

Trauma is a **permanent modification of the nervous system's prediction model** in response to an experience that exceeded the system's capacity to process and integrate at the time of the experience.

Notice what this definition does and does not say.

It does **not** say that trauma is a weakness. A bridge that bends under a load it was not designed for is not a weak bridge — it is a bridge responding appropriately to a force that exceeds its design specifications.

It does **not** say that trauma is a mental event. The modification happens at the level of the nervous system — in the way sensory signals are filtered, in the way the body prepares for action, in the way energy is distributed across physiological systems. These are physical processes.

It does **not** say that the traumatic event needs to be dramatic. A three-month absence of contact comfort at eighteen months of age is not a bomb going off. It is, by most conventional standards, a minor medical intervention. What matters is the match between the experience and the system's current capacity — and at eighteen months, the nervous system has no framework whatsoever for "temporary separation for medical reasons."

What trauma **is**: a successful adaptation. The nervous system encountered a situation it could not model, and it updated its model in the direction that maximised survival. If the world is a place where pain arrives without warning and comfort does not follow, then a nervous system set to high alert — always scanning, always slightly contracted, always ready — is a nervous system correctly calibrated to that world. The problem is not that the calibration is wrong. The problem is that the world has changed and the calibration has not.

> **AUTHOR'S NOTE: The Hospital, 1968**
>
> I do not remember it. I was too young for explicit memory to have formed.
>
> What I have are the downstream signals: a body that has always treated routine medical environments as emergencies. A nervous system that identifies "caring professional approaching to help" and responds with the physiology of threat. A skeleton of responses so deeply embedded that they long predate any narrative I have been able to construct about them.

> My developmental age at the time was approximately eighteen months. The modification that happened then was not added to an existing nervous system. It was the nervous system being formed. That is a distinction that will matter a great deal in Chapter 6.
>
> For now: the body knows things that the mind never learned. This book is an attempt to write those things down in a language precise enough to work with.

The Polyvagal Ladder

In the 1990s, neuroscientist Stephen Porges developed what he called Polyvagal Theory — an account of the autonomic nervous system that begins not with the familiar fight-or-flight response but with the evolutionary history of the structures involved.

The key observation is this: the autonomic nervous system is not a single dial that runs from "calm" to "alarmed." It is a hierarchy of three systems, each older than the one above it, each more primitive, each mobilised in sequence as the perceived threat increases.

```
┌──────────────────────────────────────────────────────────────────┐
│ VENTRAL VAGAL STATE           Social engagement branch           │
│ Safe, connected, curious      Myelinated vagus nerve             │
│ Window of Tolerance           Heart rate regulated               │
│ ───────────────────────── ← most recently evolved                │
├──────────────────────────────────────────────────────────────────┤
│ SYMPATHETIC STATE             Mobilisation branch                │
│ Alert, energised, defensive   Spinal cord pathway                │
│ Fight or flight               Heart rate elevated                │
│ ───────────────────────── ← older                               │
├──────────────────────────────────────────────────────────────────┤
│ DORSAL VAGAL STATE            Immobilisation branch              │
│ Shutdown, collapse, freeze    Unmyelinated vagus nerve           │
│ Dissociation, numbing         Heart rate dropped                 │
│ ───────────────────────── ← most ancient                        │
└──────────────────────────────────────────────────────────────────┘
```

Figure 1.1. The polyvagal hierarchy. Under conditions of safety, the most evolved system (ventral vagal) governs — enabling social connection, learning, and curiosity. As perceived threat increases, the sympathetic system activates, preparing the body for action. If the threat is overwhelming or escape is impossible, the oldest system (dorsal vagal) takes over: immobilisation, shutdown, disconnection. Trauma often

```
involves the system being stuck at a lower rung long after the original threat has
passed.
```

The critical word in that last sentence is *perceived*. The hierarchy responds to what the body detects as dangerous, not to what the thinking mind judges as dangerous. These are different processes. The thinking mind can be entirely convinced that there is no danger — and the body can, at exactly the same moment, be running the threat response at full intensity. Both are responding to real information. They are just reading different signals.

The Freeze Response

The freeze response is the least understood of the three states and, for many trauma survivors, the most characteristic.

It is not the absence of a response. It is a full physiological engagement — the body is doing something, and doing it with considerable energy. What it is doing is playing dead.

This is a deeply ancient response. In evolutionary terms, freezing when threatened by a predator is sometimes the optimal move: many predators respond to movement, and a motionless prey item may not register as prey at all. The freeze response in mammals also involves the release of endogenous opioids — nature's way of making the potential experience of being killed slightly less intolerable. This is why dissociation during overwhelming trauma is sometimes described as a kind of mercy.

For humans in modern environments, the freeze response is triggered not by literal predators but by anything the nervous system has learned to classify as equivalent. A raised voice. A medical environment. A particular combination of sensory signals that, at some point in the past, preceded something overwhelming. The body does not distinguish between the original context and the contemporary one. It responds to the signal, not the story.

The result is a person who, in the middle of a conversation or a clinic appointment or an otherwise ordinary moment, suddenly goes quiet and still and seems to be looking at something slightly to the left of wherever they are. They are not being difficult. They are not choosing not to engage. They are playing dead because the body has concluded that this is the appropriate moment to play dead.

Why This Matters for Treatment

If trauma is a modification of a prediction model — an accurate learning from an overwhelming experience — then the therapeutic question is not *how do we fix the broken response* but *how do we update the prediction model with new information*.

This is a different question, and it has a different answer.

Fixing a broken response implies that the body is malfunctioning. Updating a prediction model implies that the body has been doing its job correctly, and that the job now requires new data.

The Soma-Field Model, which is the subject of this book, provides a mathematical framework for what "updating the prediction model" means precisely: what structures change, what the before and after states look like, and — critically — what kind of change is possible given when the original learning occurred.

That last point is where this model adds something that existing frameworks do not provide, and it is the subject of Chapter 6.

KEY TERMS

Soma — the body as experienced from the inside; the totality of interoceptive (internally sensed) signals.

Polyvagal hierarchy — the three-level autonomic nervous system described by Porges, ordered from most evolutionarily ancient (dorsal vagal) to most recently evolved (ventral vagal).

Window of Tolerance — the range of arousal within which the nervous system can function flexibly, process information, and engage socially. Above the window: hyperarousal (sympathetic). Below the window: hypoarousal (dorsal vagal shutdown).

Freeze response — the immobilisation state of the dorsal vagal system; the body's oldest threat response, involving disconnection, stillness, and endogenous opioid release.

CHAPTER SUMMARY

Trauma is a modification of the nervous system's prediction model — a successful adaptation to overwhelming experience, not a failure or weakness. The polyvagal hierarchy describes how three evolutionary layers of the autonomic nervous system respond to perceived threat. The freeze response is the most ancient: immobilisation as survival strategy. For treatment to work, it must address the body's model, not argue with the thinking mind. The Soma-Field Model provides a mathematical language for this.

A Field of Feeling

```
> "The body is the unconscious mind."
> — Candace Pert, 1997
>
```

LEARNING OBJECTIVES

By the end of this chapter, you will understand:

- What a physical field is and why emotion behaves like one
- Why emotions are body phenomena, not brain phenomena
- What interoception is and why it is fundamental
- The meaning of "the soma-field" as a technical term

What a Field Is

Imagine the gravitational field of the Earth.

You cannot see it. You cannot touch it. You cannot pick up a handful of gravity and examine it under a microscope. But it is real — as real as anything in physics — and you have felt it every moment of your existence. It is everywhere in space around the Earth. It has a *strength* at every point (stronger close to the Earth, weaker further away). It has a *direction* at every point (towards the centre of the Earth). And it exerts a force on everything that is in it.

A field, in physics, is precisely this: a quantity that has a value at every point in space. Temperature is a field. The wind is a field (a vector field — direction and magnitude at every point). The electromagnetic field is a field. Quantum fields, which are the foundation of modern physics, are fields.

The key insight of the Soma-Field Model is this: **emotion is a field phenomenon**.

Not a metaphor. A precise claim about how emotional signals distribute themselves in the body, interact with each other, and evolve over time.

Emotions in the Body

Antonio Damasio, in his somatic marker hypothesis (1994), proposed that emotions are fundamentally body states: that what we call "emotion" is the brain's representation of a pattern of physiological activation — heartrate, muscle tension, gut movement, hormonal state, skin conductance, respiratory rhythm. We do not feel an emotion and then notice body signals. The body signal *is* the emotion; what the brain does is read it.

This is deeply counterintuitive if you have spent your life inside a culture that treats the mind as the real thing and the body as its vehicle. But the neuroscience supports it consistently. Patients with damage to the parts of the brain that receive and integrate body signals do not make better decisions because they are freed from emotional interference — they make worse decisions, because they have lost access to the somatic markers that tell them which options feel safe and which feel dangerous.

The body is not an obstacle to clear thinking. It is the substrate of it.

Interoception is the technical name for the body's ability to sense its own internal state: heartbeat, breath, gut sensation, muscle tone, the position of limbs, the temperature of organs. Interoceptive accuracy — how precisely a person can read their own body signals — varies widely between individuals and is significantly disrupted by trauma.

```
┌──────────────────────────────────────────────────────────────────────┐
│                    INTEROCEPTIVE BODY MAP                            │
│                                                                      │
│     ┌──────────┐          Fear:      rapid heartbeat, tight chest    │
│     │   HEAD   │          Shame:     face flush, stomach drop        │
│     └────┬─────┘          Calm:      slow breath, warm belly         │
│          │                Anger:     jaw clench, shoulder tension    │
│     ┌────┴─────┐          Grief:     throat constriction, chest heavy│
│     │  CHEST   │          Joy:       chest expansion, light limbs    │
│     │ ♥  lungs │          Freeze:    whole-body stillness, cold      │
│     └────┬─────┘          Disgust:   gut recoil, throat closing      │
│          │                                                           │
│     ┌────┴─────┐          Each emotion has a characteristic          │
│     │  BELLY   │          distribution across the body —             │
│     │  gut     │          a spatial pattern of activation.           │
│     └────┬─────┘          This pattern is what the Soma-Field        │
│          │                Model calls the emotional field state.     │
│     ┌────┴─────┐                                                     │
│     │  PELVIS  │                                                     │
│     │  limbs   │                                                     │
│     └──────────┘                                                     │
└──────────────────────────────────────────────────────────────────────┘
```

Figure 2.1. The body map of emotional activation. Emotions are not events in the head; they are distributed patterns of physiological arousal across the body. Research by Nummenmaa et al. (2014) mapped these patterns by asking participants to colour body silhouettes where they felt each emotion. The patterns are consistent across cultures.

Figure 2.1. The body–brain coupling stack. Interoceptive signals from the body feed into the brainstem and autonomic nervous system, which couples bidirectionally to the limbic soma-field (coupling matrix $\mathbf{W}$). The field gates input to the prefrontal cortex via the threshold θ; what crosses becomes conscious percept. *Author's original figure.*

Figure 1: Figure 2.1. The body–brain coupling stack. Interoceptive signals from the body feed into the brainstem and autonomic nervous system, which couples bidirectionally to the limbic soma-field (coupling matrix $\mathbf{W}$). The field gates input to the prefrontal cortex via the threshold θ; what crosses becomes conscious percept. *Author's original figure.*

GOING DEEPER: The Foot That Isn't There

Pain is not in the foot. It is in the brain's model of the foot.

The clearest proof is phantom limb pain. When a limb is amputated, many patients continue to feel it — and feel it *hurting*. The foot is gone. The pain is real. It wakes people at night, responds to analgesics, and can be agonising for years. What is in pain is the brain's neural map of the foot, which persists in the cortex long after the tissue is gone.

Ramachandran's solution was a mirror box. A mirror placed along the body's midline creates a reflection of the intact hand where the absent hand should be. The patient watches the reflection move. The brain's model updates: *the hand is there, the hand is moving, the hand is fine.* For many patients, the pain decreases or disappears. The model changed. The suffering reduced. Nothing in the body changed at all.

This is not a curiosity. It is the normal condition of all somatic experience. The brain does not receive raw body signals and display them. It maintains a continuous predictive model of the body and generates what you *feel* from that model. The felt body is the predicted body.

For the Soma-Field Model, this is load-bearing. The field $\mathbf{e}(t)$ is not a readout of the physical body. It is the nervous system's model of the body. When the model is updated — by new sensory experience, somatic therapy, or the slow accumulation of safety — what is felt changes. Not because the body changed. Because the prediction changed.

Therapy does not fix the body. It updates the model.

The Soma-Field: A Technical Definition

In the Soma-Field Model, we represent the body's emotional state as a vector of activation levels across a set of emotional dimensions. Call this vector **e**:

$$\mathbf{e} = (e_1, e_2, \dots, e_n)$$

Each e_i is a real number representing the activation level of a somatic emotional mode at a given moment: the level of fear-readiness in the body, the level of grief-contraction, the level of social-engagement openness, and so on. The exact labelling of the modes is secondary to the structure; what matters is that there is a space of such states and a dynamics on that space.

The soma-field is this vector **e**, evolving in time. It is not a single number (arousal level) or a pair of numbers (valence and arousal). It is a multi-dimensional state that captures the full texture of somatic experience at a given moment.

Three things are immediate from this definition:

1. **The field has a position**: the current emotional state is a point in an n-dimensional space.
2. **The field has dynamics**: it moves through this space over time.
3. **The field has a structure**: some positions are stable (attractors), and the dynamics drives the field toward them.

The next chapter is about that structure.

> **GOING DEEPER: Quantum Fields and Why They Are Relevant**
>
> In quantum field theory (QFT), the fundamental objects are not particles but fields — wave-like disturbances propagating through space. Particles are what you see when a field vibrates at a high enough amplitude to be detected: an electron is a ripple in the electron field, a photon is a ripple in the electromagnetic field.
>
> The soma-field is not a quantum field in the literal sense. Emotional dynamics are classical, not quantum. What the Soma-Field Model borrows from QFT is the *mathematical language*: the same equations that describe how quantum fields couple to each other turn out to describe how emotional modes couple to each other. This is not because emotion is quantum-mechanical. It is because coupling dynamics — the mathematics of how things that interact shape each other's behaviour — takes the same form wherever it appears.
>
> This correspondence is the subject of Chapter 7. For now: the QFT connection is a mathematical tool, not a metaphysical claim.

Figure 2.2. The soma-field oscillates continuously. Most of the time its modes remain below the perception threshold T (dashed line) — sub-threshold activity that drives physiology and behaviour invisibly. Only a sufficiently large excitation crosses T into felt experience. Interoceptive training and somatic therapy work, in part, by lowering T. *Author's original figure.*

Figure 2: Figure 2.2. The soma-field oscillates continuously. Most of the time its modes remain below the perception threshold T (dashed line) — sub-threshold activity that drives physiology and behaviour invisibly. Only a sufficiently large excitation crosses T into felt experience. Interoceptive training and somatic therapy work, in part, by lowering T. *Author's original figure.*

KEY TERMS

Field — a quantity with a value at every point in space (or, in the soma-field context, at every point in the body's state space).

Interoception — the nervous system's process of sensing the internal state of the body.

Interoceptive accuracy — the precision with which a person can consciously read their own interoceptive signals.

Soma-field — the vector **e** of somatic activation levels across emotional modes; the state of the body's emotional field at a given moment.

State space — the set of all possible values of **e**; the arena within which emotional dynamics occurs.

The Energy Landscape

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> "Nature does not create mountains and valleys at random.
> They are shaped by the forces beneath them."
>
```

LEARNING OBJECTIVES

By the end of this chapter, you will understand:

- Why some emotional states are stable and others are transient
- The meaning of “attractor” and “basin of attraction”
- What the Hamiltonian is and why it organises the model
- Why you keep returning to familiar emotional states even when you don’t want to

---

Hills and Valleys

Imagine placing a ball on a hilly landscape. If you place it at the bottom of a valley and give it a small push, it rolls away from where you pushed it — and then rolls back. The valley is stable. The bottom of the valley is an *attractor*: the ball is drawn toward it from nearby positions.

If you place the ball at the top of a hill and give it a small push, it rolls away from the hilltop — and keeps going. The hilltop is *unstable*. Small perturbations grow into large departures.

The same geometry applies to emotional states.

Some emotional states are at the bottom of valleys in the body’s landscape: they are stable, they are where the system tends to rest, and perturbations that push the body away from them are followed by a return. Other states are at hilltops: they are unstable configurations that the system passes through on its way between valleys.

The crucial question — the question that distinguishes a regulated nervous system from a dysregulated one, and distinguishes one person’s landscape from another’s — is: where are the valleys? How deep are they? How wide? How many are there?

Figure 3.1. The emotional energy landscape (2D contour). Four attractor basins are visible: Calm (wide, deepest — the global minimum of a regulated nervous system), Freeze (narrow and very deep — easy to fall into, hard to leave), Fight and Flight (intermediate depth). The system rolls downhill to the nearest basin; the depth controls escape difficulty and the width controls resilience to perturbation. *Author's original figure.*

Figure 3: Figure 3.1. The emotional energy landscape (2D contour). Four attractor basins are visible: Calm (wide, deepest — the global minimum of a regulated nervous system), Freeze (narrow and very deep — easy to fall into, hard to leave), Fight and Flight (intermediate depth). The system rolls downhill to the nearest basin; the depth controls escape difficulty and the width controls resilience to perturbation. *Author's original figure.*

Figure 3.2. Basin of attraction map. Each point in state space is coloured by the attractor it flows to under gradient descent: blue = Calm, purple = Freeze, orange = Fight, green = Flight. The calm basin dominates a regulated landscape. Freeze occupies a small area but is disproportionately deep — a narrow funnel. The boundaries between basins are the separatrices: invisible thresholds in state space that determine which valley a given perturbation resolves to. *Author's original figure.*

Figure 4: Figure 3.2. Basin of attraction map. Each point in state space is coloured by the attractor it flows to under gradient descent: blue = Calm, purple = Freeze, orange = Fight, green = Flight. The calm basin dominates a regulated landscape. Freeze occupies a small area but is disproportionately deep — a narrow funnel. The boundaries between basins are the separatrices: invisible thresholds in state space that determine which valley a given perturbation resolves to. *Author's original figure.*

Attractors and Basins

An **attractor** is a stable state — a bottom of a valley. A **basin of attraction** is the set of all points from which the system rolls toward a given attractor: the "catchment area" of the valley.

For a regulated nervous system, the primary attractor is some version of calm social engagement — the ventral vagal state of Polyvagal Theory. The basin is wide: a large range of perturbations (emotions, sensations, social situations) all resolve back to this resting state. The system is resilient.

For a trauma-modified nervous system, the landscape has changed. A second attractor — hypervigilance, alert-readiness, the sympathetic mobilisation state — may have become deep and wide. The calm attractor may still exist but its basin has narrowed: it takes very little to tip the system out of calm and into alertness. And a third attractor — the freeze state, the dorsal vagal shutdown — may be very deep indeed: once the system tips into it, escape requires a large input of energy.

This is not a metaphor for how trauma "feels." It is a description of the actual dynamics of the system.

The Hamiltonian

The landscape has a name in physics: the **Hamiltonian**. Denoted H, it is a function that assigns an energy value to every possible state of the system.

For the soma-field, the Hamiltonian takes the form:

$$H(\mathbf{e}) = -\frac{1}{2}\sum_{i,j} W_{ij}\, e_i\, e_j - \sum_i \theta_i\, e_i$$

Let us read this in plain English.

The first term, $-\frac{1}{2}\sum_{i,j} W_{ij}\, e_i\, e_j$, captures the *interactions between emotional modes*. W_{ij} is the coupling between mode i and mode j — how strongly they influence each other. When fear is high, does shame rise with it? When calm is present, does anger fall? The matrix W encodes all of these mutual influences. The minus sign means that aligned coupling (modes reinforcing each other) lowers the energy — makes the state more stable.

The second term, $-\sum_i \theta_i\, e_i$, captures the *individual thresholds* of each mode. θ_i is the bias of mode i — how much the system tends toward or away from it in the absence of coupling. A mode with a large positive θ_i has a natural tendency toward high activation.

The dynamics — the way the field moves through state space over time — follows from this energy function. The field always moves *downhill*: toward lower values of H.

$$\dot{\mathbf{e}} = -\nabla H(\mathbf{e}) + \eta(t)$$

This equation says: the rate of change of the emotional state ($\dot{\mathbf{e}}$) equals the negative gradient of the energy (the direction of steepest descent on the landscape) plus a noise term $\eta(t)$ representing the small random fluctuations of physiological and environmental variation. The system is always rolling toward the nearest valley, with a small amount of noise that occasionally kicks it over a hill into a different basin.

The noise term has a deeper structure. The *level* of noise — how wide the fluctuations are — is set by the autonomic nervous system, specifically by heart rate variability (HRV): high coherence in the cardiac rhythm narrows the noise, stabilising the field; low HRV widens it. But there is a second, more predictive cardiac quantity: the **cardiac acceleration** $\dot{H}$ — the rate at which heart rate is *changing*. A rising heart rate predicts approach to a threshold; a falling heart rate predicts retreat from one. The current BPM tells you where you are. The acceleration of BPM tells you where you are going next.

> **GOING DEEPER: Gravity and the Heartbeat**
>
> Gravity, in SI units, is measured in metres per second squared (m/s^2) — it is an *acceleration*, not a speed. It tells you not where a falling object is, but how fast its velocity is changing: where it will be next.
>
> Cardiac acceleration — the rate of change of heart rate — has units beats/s^2. Same type, different physical dimension. And the same logical character: it tells you not what the BPM is, but where it is heading. N+1, not N.
>
> In the soma-field, cardiac acceleration acts as a **landscape tilt**: it tips the energy

function toward activation or rest before any emotional threshold is crossed. When the heart accelerates, the field is being pulled toward higher-energy states by a force it cannot see and cannot always attribute correctly. Some anxiety that feels emotionally caused is cardiac in origin — the field cannot distinguish the two from the inside. This is the somatic equivalence principle: you cannot tell, from your own experience, whether your emotional landscape tilted because something happened, or because your heart accelerated first.

Clinically: monitoring the *direction* of heart rate change, not just its level, gives earlier warning of threshold approach than any other non-invasive signal.

GOING DEEPER: Why Physicists Love the Hamiltonian

The Hamiltonian was introduced by William Rowan Hamilton in the 1830s as a way of rewriting Newton's equations in a more elegant form. What Hamilton discovered is that the trajectory of any physical system — the path it takes through its state space over time — can be derived entirely from a single scalar function H. You do not need to describe all the forces. You just need the energy landscape, and the dynamics follows.

In quantum mechanics, the Hamiltonian operator $\hat{H}$ plays the same role: it determines how a quantum state evolves over time through Schrödinger's equation, $i\hbar\,\partial_t\psi = \hat{H}\psi$. The eigenvalues of $\hat{H}$ are the allowed energy levels.

In the Soma-Field Model, $H(\mathbf{e})$ is neither Newtonian nor quantum: it is the Hamiltonian of a classical stochastic system (a Langevin system), where the dynamics is gradient descent with noise. But the mathematical structure — a scalar energy function that determines everything else — is identical.

This is not a coincidence. It is because "a system has stable states to which it returns" is a very general physical principle, and the Hamiltonian is the most general way to formalise it.

The Coupling Matrix

The matrix W — the coupling matrix — is the central object of the model. It encodes the emotional architecture of a nervous system: which modes excite each other, which inhibit each other, how strongly, and in which direction.

For a neurotypical, regulated nervous system, W has a specific mathematical property: it is *symmetric*. $W_{ij} = W_{ji}$: the influence of mode i on mode j equals the influence of mode j on mode i. This symmetry is not incidental. It is what guarantees the existence of an energy function: if W is not symmetric, the dynamics cannot be written as gradient descent, and the system may not have stable fixed points at all. It may cycle indefinitely.

Trauma, in this model, is a modification of W that breaks this symmetry. A traumatised nervous system has couplings that do not balance: fear activates shame more strongly than shame activates fear; hypervigilance activates the freeze response more readily than the freeze response resolves back to hypervigilance. The asymmetric couplings create directional flows in the landscape — attractors that are easy to fall into and hard to climb out of.

This is the formal basis of the clinical observation that trauma often feels like a one-way ratchet.

KEY TERMS

Attractor — a stable state in the energy landscape; a valley that the field rolls toward from nearby positions.

Basin of attraction — the region of state space from which the system flows toward a given attractor.

Hamiltonian — the energy function $H(\mathbf{e})$ that organises the dynamics; the mathematical description of the landscape.

Coupling matrix W — the matrix encoding the interactions between emotional modes; shapes the landscape by determining which states lower the energy.

Threshold θ_i — the individual bias of emotional mode i; shifts its natural resting level.

CHAPTER SUMMARY

Emotional states are points in a landscape shaped by the Hamiltonian H. Attractors are stable states (valley bottoms); basins of attraction are the regions from which the system rolls toward each attractor. The dynamics — gradient descent with noise — always moves toward lower energy. The coupling matrix W encodes the interactions that shape the landscape. Symmetry of W guarantees stable attractors; asymmetry (introduced by trauma) creates directional flows that are hard to reverse.

Figure 3.3. 1D energy cross-section along a principal axis of the landscape. The height of each barrier between basins determines transition probability: a deep Freeze well with a high approach barrier (right) requires substantial energy input to escape — corresponding clinically to a freeze response that does not self-resolve without intervention. Barrier asymmetry (left-to-right ≠ right-to-left) is the signature of trauma modification. *Author's original figure.*

Figure 5: Figure 3.3. 1D energy cross-section along a principal axis of the landscape. The height of each barrier between basins determines transition probability: a deep Freeze well with a high approach barrier (right) requires substantial energy input to escape — corresponding clinically to a freeze response that does not self-resolve without intervention. Barrier asymmetry (left-to-right ≠ right-to-left) is the signature of trauma modification. *Author's original figure.*

PART II: HOW THE FIELD CHANGES

The Weight on the Field

> "The question is not why the behaviour persists,
> but what it was optimised for."
>

LEARNING OBJECTIVES

By the end of this chapter, you will understand:

- What the C-PTSD operator is and how it modifies the field
- Why hypervigilance is not an error but an optimisation
- What "the landscape has changed" means precisely
- The difference between a perturbation and a structural modification

---

The Modification

Complex PTSD (C-PTSD) is distinguished from single-incident PTSD by the presence of repeated, prolonged, or developmental trauma — particularly trauma that occurred in relationships on which the person depended for survival. The result is not a discrete memory that can be "processed" and resolved. It is a pervasive reorganisation of the emotional field architecture: a new landscape, not a scar on an old one.

In the Soma-Field Model, C-PTSD is represented as a modification of the coupling matrix:

$$W_{\text{C-PTSD}} = W_0 + \Delta W_{\text{trauma}}$$

where W_0 is the baseline coupling matrix and ΔW_{trauma} is the modification — an asymmetric additive term that reshapes the landscape. Crucially, ΔW_{trauma} is not symmetric: it introduces directional flows. Certain states become easy to fall into and hard to leave. Others become difficult to access from the modified landscape even though they exist.

This can be visualised as a landscape that has been tilted and deformed: new deep valleys in places that were not attractors before, old deep valleys raised, and the topology of connectivity between states changed.

Figure 4.1. Four neurotype landscapes (1D cross-section). *Typical* (upper left): a deep wide Calm basin with accessible secondary states. *C-PTSD* (lower left): Calm shallowed and narrowed, Freeze dominant — the resting state shifts toward high-vigilance. *ADHD* (upper right): all basins flattened, low barriers, rapid transitions — high-temperature dynamics. *ASD* (lower right): narrow steep wells with high barriers between states — strong attractor stability, low noise tolerance, high cost of transitions. *Author's original figure.*

Figure 6: Figure 4.1. Four neurotype landscapes (1D cross-section). *Typical* (upper left): a deep wide Calm basin with accessible secondary states. *C-PTSD* (lower left): Calm shallowed and narrowed, Freeze dominant — the resting state shifts toward high-vigilance. *ADHD* (upper right): all basins flattened, low barriers, rapid transitions — high-temperature dynamics. *ASD* (lower right): narrow steep wells with high barriers between states — strong attractor stability, low noise tolerance, high cost of transitions. *Author's original figure.*

Why Hypervigilance Is an Optimisation

A nervous system that has adapted to an environment of chronic threat has correctly learned that:

1. Danger is frequent and unpredictable.
2. The cost of missing a threat is very high.
3. The cost of false alarms is low (relative to the cost of missing a real threat).

Given these parameters, the optimal configuration is exactly what we see in C-PTSD: a bias toward high vigilance, a wide definition of "potential threat," a fast-responding sympathetic system, and a slow-to-settle calm state. The hypervigilance attractor is deep because a deep attractor is appropriate to the environment it was optimised for.

The modification is not an error. It is a correct solution to the wrong problem — where "the wrong problem" means the original environment, which no longer exists (or no longer exists in the same form).

This reframing is not merely philosophical. It changes the clinical question from "how do we extinguish the hypervigilance response" to "how do we update the landscape to incorporate evidence that the current environment is different." These are very different operations, with very different implications for what kind of therapeutic intervention is useful.

Thresholds and Consciousness

There is a parameter in the model that has not yet been introduced, and it does a great deal of work. This is the **threshold** T — denoted with the capital T that recurs throughout this book.

The threshold is a level of field activation above which an emotional state becomes conscious experience — enters awareness as a felt emotion — rather than remaining as sub-threshold somatic activation. Below T, the field is active but not felt; the activation is present in the body, influencing behaviour and physiology, but not represented in consciousness.

This has immediate clinical consequences. A person with a very high threshold T may have a strongly activated soma-field — may be physiologically in a fear state, with all the somatic correlates

— while experiencing nothing that they would call fear. The activation is real. The consciousness of it is absent. Somatic therapy, interoceptive training, and bodywork all operate, in part, by lowering T: bringing below-threshold somatic content into awareness.

A person with a very low threshold T experiences the opposite: everything is felt, amplified, present. This is associated with high interoceptive sensitivity, certain presentations of anxiety, and some forms of neurodivergence.

The threshold is where the physics and the clinical presentation most visibly connect.

AUTHOR'S NOTE: The Landscape I Inherited

There is a version of this chapter that is abstract: modifications to coupling matrices, reshaping of landscapes, asymmetric W. And then there is the version that is what it feels like to live in a modified landscape.

What it feels like is this: calm is always provisional. Not shallow, exactly — but unsecured. Like a surface that holds weight when you step carefully but gives way if you shift too quickly. Alert is never far away. And underneath alert, the freeze state is a gravity well that does not announce itself before you are already in it.

The modification in my case is not a perturbation on a pre-existing normal landscape. That would require a W_0 to perturb. The timeline does not allow for that. That is the subject of Chapter 6.

KEY TERMS

C-PTSD operator — the modification ΔW to the coupling matrix that reshapes the energy landscape; the mathematical representation of the effect of complex trauma.

Threshold T — the activation level above which a soma-field state becomes conscious experience. The central parameter distinguishing felt emotion from sub-threshold somatic activation.

Hypervigilance attractor — the deep stability basin in the modified landscape corresponding to high-arousal, high-alert states.

Figure 4.2. The perception threshold T. Mode i (grey) oscillates continuously but never crosses T — it is sub-perceptual, influencing behaviour and physiology without entering felt experience. Mode j (blue) rises through T and becomes a consciously felt emotion. The threshold is the key parameter that distinguishes somatic activation from emotional awareness; its value varies across individuals and can be modified by interoceptive practice, arousal level, and therapeutic work. *Author's original figure.*

Figure 7: Figure 4.2. The perception threshold T. Mode i (grey) oscillates continuously but never crosses T — it is sub-perceptual, influencing behaviour and physiology without entering felt experience. Mode j (blue) rises through T and becomes a consciously felt emotion. The threshold is the key parameter that distinguishes somatic activation from emotional awareness; its value varies across individuals and can be modified by interoceptive practice, arousal level, and therapeutic work. *Author's original figure.*

Memory Written in the Body

LEARNING OBJECTIVES

By the end of this chapter, you will understand:

- The difference between narrative memory and somatic memory
- What the memory kernel is and what it does to the field dynamics
- Why trauma memory persists — and why some trauma memories persist much longer
- What therapeutic processing means in terms of the memory kernel

Two Kinds of Memory

When you remember a conversation from last week, you are using **episodic memory** — the explicit, narrative record of events that occurred at specific times and places. Episodic memory is context-dependent, verbally expressible, and subject to conscious recall and revision. It is stored primarily in the hippocampus.

When you flinch at a sound that resembles the sound that preceded something terrible, you are using **procedural** or **somatic memory** — a form of memory that is not stored as narrative but as pattern: as a configured readiness in the body to respond in a particular way to particular signals. Somatic memory is not verbally expressible (you cannot "tell the story" of a procedural response; you can only notice it happening). It does not require conscious recall — it is not a replay of an event but an embodied preparation. It is stored across the body: in muscle tone, in the brainstem, in the autonomic nervous system, in the way sensory signals are gated before they reach cortical processing.

Trauma creates primarily somatic memory. This is why it is not resolved by talking about it. The body has stored information in a form that language does not reach.

The Memory Kernel

In the Soma-Field Model, the effect of past activation on present dynamics is captured by a **memory kernel** $K(\tau)$. This is a function that says: an activation of the field τ time units ago continues to influence the field now, with a weight proportional to $K(\tau)$.

For C-PTSD, the memory kernel takes the form:

$$K_{\text{trauma}}(\tau) = \sum_k A_k \, e^{-|\tau|/\tau_k}$$

This is a sum of decaying exponentials. Each term represents a distinct trauma trace: A_k is the amplitude (how strongly the trace affects the current field) and τ_k is the decay time (how long the trace persists before fading).

```
REGULATED: No significant memory kernel

┌──────────────────────────────────────────────────────────────┐
│  Field  ▲                                                    │
> │      │  │   (episode resolves; field returns
>
│         │                           baseline                 │
│         └─────────────────────────────────────────────→      │
│                            time                              │
└──────────────────────────────────────────────────────────────┘


C-PTSD: Significant memory kernel — traces persist

┌──────────────────────────────────────────────────────────────┐
│  Field  ▲                                                    │
>
│         │  ────┘    └────┘  └──────┘         └──────       │
│         │                                                    │
│         └─────────────────────────────────────────────→      │
│                            time                              │
│  Baseline elevated; episodes bleed into one another;         │
│  field rarely returns to original rest level                 │
└──────────────────────────────────────────────────────────────┘
```

Figure 5.1. The effect of the trauma memory kernel on field dynamics. In a regulated

```
system (top), a field activation episode resolves and the field returns to a low
resting level. In the C-PTSD-modified system (bottom), the memory kernel elevates
the baseline between episodes, so that subsequent episodes begin from a higher resting
activation. Over time, the field cycles at an elevated level without returning to rest.
```

Why Early Traces Persist

The decay time τ_k is central: it determines how long a trace remains active.

For trauma that occurs early in development — before language, before narrative memory capacity — the decay time tends to be much longer. There are two reasons.

First, **somatic memory has no verbal layer**. For trauma occurring after language develops, the episodic and somatic memories co-encode: the narrative version partially "covers" the somatic trace, providing a context that can be accessed verbally. Verbal processing in therapy can then shorten the effective lifetime of the trace. For pre-verbal trauma, the somatic trace has no narrative companion. It cannot be reached by talking. The decay time is governed by purely somatic processes, which are much slower.

Second, **the trace cannot be separated from the structure**. For pre-verbal trauma, the memory is not a modification of an already-formed architecture. The architecture itself was shaped by the conditions of the traumatic period. This is addressed more formally in Chapter 6.

What Therapy Does

In the language of the memory kernel, effective somatic therapy does two things:

1. It reduces the amplitudes A_k: the traces continue to influence the field, but with less force. Activation episodes are smaller and resolve more completely.
2. It increases the decay times τ_k: the traces fade more quickly after episodes. The field returns to rest more rapidly.

The goal is not to eliminate the traces — the nervous system cannot un-learn an experience, and attempting to make it do so is not the right model. The goal is to reduce their influence to a level that allows the field to return to rest between episodes: to restore the gap between activations in which recovery occurs.

> **GOING DEEPER: The Memory Kernel and the QFT Propagator**
>
> This may seem like a digression, but it is one of the most striking features of the model. The memory kernel for C-PTSD — $K(\tau) = \sum_k A_k e^{-|\tau|/\tau_k}$ — is mathematically identical to the **Euclidean propagator** in quantum field theory.

In QFT, the Euclidean propagator $G_E(\tau)$ describes how a disturbance in a quantum field at time 0 correlates with the field at time τ:

$$G_E(\tau) = \langle \phi(0)\, \phi(\tau) \rangle = \frac{1}{2m}\, e^{-m|\tau|}$$

The mass m of the QFT particle corresponds to $1/\tau_k$ in the memory kernel. A heavier particle creates a shorter-range propagator; a shorter-lived trauma trace has a larger $1/\tau_k$ (i.e., smaller τ_k, faster decay).

This identity is not an analogy. The two expressions are the same function with different names for the parameters. The Wick rotation — the substitution $t \to -i\tau$ that takes quantum mechanics into statistical mechanics — is the formal bridge between them, and it is the subject of Chapter 7.

KEY TERMS

Episodic memory — explicit, narrative memory of events at specific times and places; accessible to conscious recall and verbal expression.

Somatic (procedural) memory — embodied memory stored as configured physiological readiness; not verbally expressible; activated by sensory signals that match the original encoding context.

Memory kernel $K(\tau)$ — the function describing how field activations at time τ in the past continue to influence the current field state.

Amplitude A_k — the strength of a trauma trace's influence on the current field.

Decay time τ_k — the timescale over which a trauma trace fades after activation; how long the echo persists.

How Early Is Early?

> "Before language, there is only the body."
>

LEARNING OBJECTIVES

By the end of this chapter, you will understand:

- Why the age at which trauma occurred matters to its character
- What happens to the structure of the soma-field when modification occurs before language develops
- Why "returning to the pre-trauma self" is a coherent goal for late trauma but not for pre-verbal trauma
- What "forward transformation" means as a mathematical and clinical concept

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Developmental Time

Children are not small adults. The nervous system develops in stages, and each stage has different capacities — for encoding, for integration, for language, for explicit memory. What a three-year-old can do with an overwhelming experience is not what a ten-year-old can do, and neither is what an adult can do.

This is relevant to trauma because the *character* of a traumatic modification depends on the developmental stage at which it occurs. Not the severity — severity is a separate question. The character. What structures are modified, how the modification is stored, and what it is even possible to change about it afterward.

The key developmental milestone for this model is the onset of reliable verbal encoding capacity — the ability to store experiences with a narrative, linguistic representation alongside the somatic one. This typically emerges between approximately 24 and 48 months of age, with considerable individual variation. We use $\tau_c \approx 36$ months as an approximate threshold.

The parameter τ_d — **developmental age at trauma** — is the age at which the primary modification occurred.

Below the Threshold: Pre-Verbal Trauma

For $\tau_d < \tau_c$ (pre-verbal trauma), several things are different from the late-trauma case.

The structure was formed under the modification. A nervous system that is being organised — that is still forming its basic coupling architecture — under conditions of unresolved physiological threat does not develop and then get modified. It develops *as* modified. The asymmetric couplings, the elevated vigilance attractor, the memory kernel coefficients — these are not perturbations on a pre-existing baseline. They are the baseline.

There is no prior self to recover. For trauma occurring after the baseline architecture is formed ($\tau_d > \tau_c$), there is a counterfactual: the person that would have developed without the traumatic modification. This counterfactual is partially encoded — in early memories, in narrative, in the patterns of functioning before the event. Therapeutic language of "returning to yourself" or "recovering the pre-trauma self" is coherent in this case: the target exists.

For pre-verbal trauma ($\tau_d < \tau_c$), the counterfactual does not exist as an encoded state. There was no formed nervous system that then got modified. The self-before-trauma never developed. There is nowhere to return to.

This is not a pessimistic statement. It is a precise one. And precision here matters because it changes the therapeutic question.

The Interpolation

The coupling matrix for a traumatised nervous system can be written as a function of developmental age:

$$W(\tau_d) = f(\tau_d) \cdot W_0 + (1 - f(\tau_d)) \cdot W_{\text{trauma}}$$

where f is a smooth interpolation function:

$$f(\tau_d) = \tanh\left(\frac{\tau_d}{\tau_c}\right)$$

```
STRUCTURAL FRACTION f(τ_d) = tanh(τ_d / τ_c)

┌──────────────────────────────────────────────────────────────────┐
│                                                                  │
│  f(τ_d) ▲  1.0 ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─ ─┌──────────      │
│  (how    │                              ┌─────┘                    │
│  much    │                        ┌────┘                         │
│  is W₀)  │  0.76 ─ ─ ─ ─ ─ ─ ─ ─ ┘   ← f(τ_c) = tanh(1)      │
│          │                        ↑                              │
│          │  0.5 ─ ─ ─ ─ ─ ─ ─ ┌──┘                            │
```

```
│          │                  ╭─╯                         │
│          │                ╭─╯                           │
│          │              ╭─╯                             │
│          │  0.0 ───╯                                    │
│          └──────────────────────────────────────────→   │
│           0    τ_c/2  τ_c    2τ_c    3τ_c      τ_d (months) │
│                       (36)                               │
│                                                          │
│  Left of τ_c:  W is mostly W_trauma — structural        │
│  Right of τ_c: W is mostly W₀ — perturbative            │
└──────────────────────────────────────────────────────────┘
```

```
Figure 6.1. The structural fraction f(τ_d). This function describes what proportion
of the coupling matrix is neurotypical baseline (W₀) versus trauma-formed (W_trauma),
as a function of developmental age at trauma. At τ_d = 0 (birth or in utero), the
coupling is entirely trauma-formed: f = 0. At τ_d = τ_c ≈ 36 months, f ≈ 0.76:
the baseline accounts for about three-quarters of the coupling. The interpolation is
smooth: there is no sharp cutoff, just a continuous change in character.
```

At $\tau_d = 0$: $f = 0$ and $W = W_{\text{trauma}}$. There is no baseline component.

At $\tau_d = \tau_c$: $f = \tanh(1) \approx 0.76$. The baseline accounts for 76% of the coupling; the modification is 24%.

At large τ_d: $f \to 1$ and $W \approx W_0$. The modification is a small perturbation on a fully formed baseline.

The therapeutic implication of this formula is significant. For $\tau_d \ll \tau_c$: the operation $W \to W_0$ — extracting the baseline from the current coupling — is not defined. The W_0 was never the dominant component. It cannot be recovered because it was not formed.

Forward Transformation

What *is* possible, for pre-verbal trauma, is a **forward transformation**: the construction of a new coupling matrix W' that has desirable properties — wider window of tolerance, shallower hypervigilance attractor, lower memory kernel amplitudes, greater capacity for social engagement — without that new matrix being a recovery of a prior state.

This is a different target, and it requires a different process:

- Not excavating the past for the lost self, but building forward
- Not reducing to a baseline that didn't form, but constructing a landscape that works
- Not recovery ($W \to W_0$, undefined), but transformation ($W \to W'$, unconstrained)

The route to W' uses the same therapeutic tools — somatic therapy, relational repair, interoceptive training, bodywork — but with a different intention. The intention is not to return somewhere but to arrive somewhere for the first time.

AUTHOR'S NOTE: τ_d = 18 Months

My developmental age at trauma: $\tau_d \approx 18$ months. Approximately half of τ_c.

At that age, the structural fraction is approximately $f(18/36) = \tanh(0.5) \approx 0.46$. Slightly less than half of the coupling matrix was neurotypical baseline at the time. More than half was trauma-formed. As the trauma continued over three months of hospitalisation — developmental ages 18 to 21 months — the modification was present throughout the period when the coupling architecture was being most actively organised.

There is no version of me that existed before this modification and then got modified. The preVerbalIsStructural theorem, which is in Appendix B, is a formal proof of the clinical fact that has taken decades of therapy to find words for: *there is nowhere to return to, and that is not a tragedy, it is simply the correct topography*.

The voyage is forward. This book is part of it.

GOING DEEPER: The preVerbalIsStructural Theorem

The following is a proof sketch in Lean 4, a proof assistant that requires mathematical arguments to be written in a form that a computer can verify. A `sorry` marks a step that is stated but not fully proved — an open obligation.

```
-- Key theorem: for pre-verbal trauma, no neurotypical W₀ can be
-- recovered by subtraction from the current coupling matrix
theorem preVerbalIsStructural {n : ⍰} (profile : TraumaProfile n)
    (h : profile.τ_d < τ_c) :
    structuralFraction profile.τ_d < Real.tanh 1 := by
  unfold structuralFraction
  apply Real.tanh_lt_tanh
  exact div_lt_one_of_lt h (by norm_num)
```

This theorem states: for any TraumaProfile with developmental age below τ_c, the neurotypical structural fraction is below $\tanh(1) \approx 0.76$. More than 24% of the coupling matrix is trauma-formed, not baseline-formed. At $\tau_d = 0$, 100% is trauma-formed.

Corollary (commented in the code): the therapeutic operation for pre-verbal trauma is forward transformation ($W \to W'$), not recovery ($W \to W_0$). The second operation is undefined because W_0 was never the dominant component.

KEY TERMS

Developmental age at trauma (τ_d) — the age, in months, at which the primary traumatic modification occurred.

Verbal encoding threshold (τ_c) — the approximate developmental age (≈36 months) at which reliable narrative memory and verbal encoding capacity emerges.

Structural fraction $f(\tau_d)$ — the proportion of the coupling matrix attributable to neurotypical baseline development; interpolated smoothly from 0 (purely structural modification) to 1 (purely perturbative modification).

Forward transformation — the therapeutic goal for pre-verbal trauma: constructing a new coupling matrix W' with wider attractor topology, rather than recovering a baseline that was not fully formed.

Interlude: A Voyage to the Alps

```
|                                                                  |
|   "Everything floats: the universe, the mountains, the body.   |
|    The question is only what it is floating in."               |
|                                                                  |
```

There is a campsite in the Klöntal valley, in the canton of Glarus, Switzerland, that I have been returning to for many years. I promised to write a book about it. This is the closest I have managed — and as it turns out, the campsite book and the soma-field book are the same book.

The Klöntal sits in a glacially carved valley a few kilometres from the town of Glarus, adjacent to the Swiss Tectonic Arena Sardona — a UNESCO World Heritage Site containing some of the most famous and legible tectonic structures in the world. The valley walls are parabolic: shaped by ice over millions of years into the form that an engineer would choose if they wanted to focus sound. Stand at one end and speak quietly, and the words arrive at the other end with startling clarity. The valley is a natural resonator: limestone and dolomite walls, near-perfect parabolic geometry, and an acoustic character that makes sound oscillate long after the source has fallen silent.

```
PARABOLIC VALLEY CROSS-SECTION

  valley rim                     valley rim
  (limestone)                    (limestone)
        \    ~   ~   ~   ~   ~   /
         \  ~               ~  /   ← sound reflects from walls
          \ ~  → source ←  ~ /
           \~               ~/
            \ ~  converge  ~/
             ________________
                valley floor

A parabolic cross-section focuses incoming sound to the focal region.
The same geometry governs satellite dishes, reflector telescopes, and
the resonant cavities of musical instruments. Mountain valleys with this
profile produce exceptional acoustics — sound oscillates long after the
source goes quiet.
```

The valley's acoustic behaviour is the physical intuition behind the soma-field wave description. The emotional field has modes — preferred patterns of activation, like standing waves in a resonant cavity — that continue to oscillate after the activating event has passed. The memory kernel $K(\tau)$ is the body's version of the valley's echo: not a recording, but a resonance that continues to shape what comes next.

Everything Floats

Geology teaches, and physics confirms, that everything floats.

At the **cosmological scale**: galaxies float in the curved spacetime that mass creates. The Milky Way is moving toward the Virgo Supercluster at approximately one million kilometres per hour — not through a fixed background, but on the spacetime manifold itself. There is no fixed frame. The background is the field.

At the **geological scale**: continents float on the asthenosphere, the semi-molten layer beneath the rigid lithosphere. The Alps exist because the African plate has been moving north at 2–3 centimetres per year for approximately 50 million years, crumpling the sediments of the ancient Tethys Sea into the mountains visible from the valley floor. The same forces are operating now, invisibly, at the speed of growing fingernails.

At the **somatic scale**: the emotional field floats in the Hamiltonian landscape — moving toward attractors, drawn by the energy gradient, oscillating around stable states, occasionally crossing a phase boundary into a new basin.

One equation governs all three:

$$\ddot{x} = -\nabla V(x) + F_{\text{ext}}$$

A galaxy, a tectonic plate, a nervous system: all governed by gradient descent on a potential with external forcing. The scales span 25 orders of magnitude. The structure does not vary.

Reading the Mountain

The Glarus Thrust (Glarner Hauptüberschiebung) is the tectonic feature that makes this region a UNESCO World Heritage Site. It is a thrust fault on which an enormous slab of Verrucano sandstone (Permian, approximately 250 million years old) was transported roughly 35 kilometres northward over much younger Flysch sediment (Eocene, approximately 40 million years old). The old sits on top of the new. The contact is visible across many mountain faces as a near-horizontal line: above it, ancient red sandstone; below it, young grey sediment.

```
GLARUS THRUST: SCHEMATIC CROSS-SECTION (not to scale)

Surface ════════════════════════════════════════════════════
```

```
          │  VERRUCANO  (~250 Ma, Permian)                │
          │  Ancient red sandstone                        │
          │  Formed long before the Alps existed          │
- - - - ├══════════════ THRUST CONTACT ═══════════════╡ ← THE LINE
          │  FLYSCH  (~40 Ma, Eocene)                    │ │
          │  Young grey marine sediment                  │ │
          │  Floor of the ancient Tethys Sea             │ │
Base      ════════════════════════════════════════════════╧══

  Direction of transport: ~35 km northward.
  The ancient slab (~250 Ma) was carried over the young sediment (~40 Ma).
  Read a single mountain face: 210 million years of geological history,
  visible in one glance. This is 4D geology — space encodes time.
```

A geological cross-section is four-dimensional: horizontal position records geography, but vertical position records time. Deep is old; shallow is recent. To read a mountain face is to read the history of the forces that shaped it — compression, burial, metamorphism, uplift, erosion — all preserved in the mineral record.

The soma-field coupling matrix W is four-dimensional in the same sense. The current configuration encodes the accumulated history of all the forces that shaped it. The asymmetries in W are the thrust faults of the emotional landscape: places where an ancient force has pushed its structure over something newer, and the contact is still legible if you know how to read it.

For pre-verbal trauma at $\tau_d \approx 18$ months: the Verrucano is very old, very deep in the developmental history, and emphatically on top.

M-Theory: Everything Floats in More Dimensions

M-theory, the current best candidate for a unified theory of physics, proposes that the universe is a *brane* — a membrane — floating in an 11-dimensional space. Our familiar four dimensions are a surface in a higher-dimensional structure. The other seven dimensions are curled too small to observe directly, but they leave measurable signatures in the physics of the accessible four.

The soma-field is not M-theoretic in any technical sense. But the intuition scales: the emotional field is a field on the brane of the body, and what we observe — threshold crossings, attractor dynamics, memory kernel echoes — are projections of a structure that extends into dimensions not directly accessible to ordinary awareness.

The pre-verbal, the sub-threshold, the procedural — somatic content that drives behaviour without entering conscious experience — is the body's version of the curled dimensions: real, causally active, not directly observable. Interoceptive practice is the project of unfolding them: making accessible what was previously curled below T.

The Valley at Dusk

I use Phase Plant, a modular synthesizer, to work with acoustic field recordings — routing them through resonant filter banks, mapping the frequencies that a resonant space prefers, listening for the modes that survive decay while others fall away. It is an unconventional approach to acoustics. But it is physics: finding the eigenfrequencies of a resonant cavity by attending to what persists.

The Klöntal valley has such frequencies. When the sun drops behind the peaks and the daytime noise subsides, what remains is the valley's own voice: a low, slow resonance in the limestone, carrying the frequencies that the parabolic geometry selects.

The emotional field has equivalent preferred frequencies. The trauma memory kernel $K(\tau) = \sum_k A_k e^{-|\tau|/\tau_k}$ encodes them: the values $1/\tau_k$ are the field's natural resonance rates, the A_k their amplitudes. Therapeutic work — reducing A_k, lengthening τ_k — is the project of quieting the modes excited by the original event until the field returns to its ground state.

In the valley at dusk, this is not a metaphor. It is audible.

PART III: THE PHYSICS UNDERNEATH

The Same Equation, Three Times

> "The unreasonable effectiveness of mathematics in the
> natural sciences."
> — Eugene Wigner, 1960
>

LEARNING OBJECTIVES

By the end of this chapter, you will understand:

- Why the same Hamiltonian appears in condensed matter physics, neural network theory, and the soma-field model
- What the Wick rotation is and why it connects quantum oscillations to trauma memory
- What string diagrams and Feynman diagrams are and what they say about emotional interaction
- The meaning of "the same mathematical structure" as evidence of structural reality

---

The Moment of Recognition

The Soma-Field Model did not begin with a plan to connect it to quantum field theory. It began with a neuroscience question: what is the simplest mathematical model of an emotional field that has stable states, dynamic transitions, and the capacity to be modified by experience?

The answer that emerged — a Hamiltonian field with a coupling matrix, evolving under Langevin dynamics — turned out to be an equation that physicists had seen before.

It is the Hopfield network Hamiltonian. Which is the Ising model Hamiltonian. Which is the classical limit of a quantum field theory in imaginary time.

This is not a coincidence crafted after the fact. It is the signature of something: when you write down "the simplest model of a field with stable states," you land on an equation that appears in three separate disciplines because three separate disciplines have independently answered the same mathematical question.

The Same Hamiltonian

The Ising model (condensed matter physics, early 20th century) describes a lattice of interacting spins — magnetic moments that can point up or down:

$$H_{\text{Ising}} = -\frac{1}{2}\sum_{i,j} J_{ij}\,\sigma_i\,\sigma_j - \sum_i h_i\,\sigma_i$$

The Hopfield network (computational neuroscience, Hopfield 1982 — Nobel Prize 2024) describes a network of interacting neurons that stores memories as stable states:

$$H_{\text{Hopfield}} = -\frac{1}{2}\sum_{i,j} W_{ij}\,x_i\,x_j - \sum_i \theta_i\,x_i$$

The Soma-Field Model describes the energy landscape of the emotional field:

$$H_{\text{soma}} = -\frac{1}{2}\sum_{i,j} W_{ij}\,e_i\,e_j - \sum_i \theta_i\,e_i$$

Replace $J_{ij} \to W_{ij}$, $\sigma_i \to e_i$, $h_i \to \theta_i$: these are the same equation written with different letters. The same mathematics describes magnetic spins in a crystal, memories in a neural network, and emotional states in a body.

This is the Hopfield equivalence — the observation for which Hopfield received the Nobel Prize: that the Ising spin model and a neural memory network are computing the same energy function. The Soma-Field Model extends that equivalence one step further: the same computation also describes the attractor structure of emotional dynamics.

Placed in the longer history of neural network modelling, the position of the Soma-Field Model is more precise than *an extension of the Hopfield framework*. Every artificial neural network built since McCulloch and Pitts (1943) — perceptrons, backpropagation networks, LSTMs, transformers — is a formal model of the neocortex. These systems learn to recognise patterns and minimise prediction error with increasing sophistication. None of them possess a limbic system: no internal valuation, no threat-detection architecture, no arousal modulation, no interoceptive loop from the body back to the field.

Hopfield's energy network is the most elegant of the neocortical models. It describes associative pattern-completion — exactly what the hippocampal-cortical system does for declarative memory. The Soma-Field Model is not a better cortex. It is the model of the system underneath the cortex that has been waiting, since 1943, to be written down.

Hopfield later described a wish that he had incorporated something analogous to 'maternal instincts' into the energy function. In the light of the Soma-Field Model, that wish was not a desire for a better neocortical model. It was an intuition pointing at the absent layer — the limbic system — for which he had no formal language at the time.

GOING DEEPER: The Missing Half of the Brain

Every artificial neural network ever built — from the perceptron in 1943 to the large language models of today — is a formal model of the neocortex. The neocortex recognises patterns, predicts sequences, and minimises error. It has been formally described, trained, and deployed at extraordinary scale.

The limbic system has not.

The limbic system is the older, deeper structure: amygdala, hippocampus, hypothalamus, cingulate cortex. It assigns value. It detects threat before the cortex has finished processing. It reinstates whole body states in response to a partial cue — a smell, a texture, a tone of voice. It holds trauma. It is the system that makes things *matter*.

Artificial intelligence has very effective cortex. It has no limbic system. It can tell you that fire is hot. It cannot be burned.

The Soma-Field Model provides the first formal field-theoretic architecture for the limbic system. Together with the Hopfield framework it describes — for the first time — both principal computational substrates of the vertebrate brain. The architecture is, formally, complete.

The Wick Rotation: One Substitution

The deepest correspondence in the model is the one that connects quantum mechanics to trauma memory. It requires a single substitution.

In quantum mechanics, the state of a system evolves in time via the time evolution operator:

$$U(t) = e^{-i\hat{H}t/\hbar}$$

The key feature is the i — the imaginary unit. This makes the exponential oscillatory: $e^{-i\omega t} = \cos(\omega t) - i\sin(\omega t)$. A quantum state oscillates in time rather than decaying.

Now make the substitution $t \to -i\tau$ — replacing real time with imaginary time. This is the **Wick rotation**, named after Gian-Carlo Wick (1954):

$$e^{-i\hat{H}(-i\tau)/\hbar} = e^{-\hat{H}\tau/\hbar}$$

The oscillatory phase has become a real decaying exponential. This is the Boltzmann weight $e^{-\beta\hat{H}}$ from statistical mechanics (at inverse temperature $\beta = \tau/\hbar$). The Wick rotation is the bridge between quantum mechanics and thermal physics.

```
╔══════════════════════════════════════════════════════════════════╗
║                    THE WICK ROTATION                             ║
╠══════════════════════════════════════════════════════════════════╣
║                                                                  ║
║  QUANTUM MECHANICS              THERMAL / SOMATIC PHYSICS        ║
║  (real time t)                  (imaginary time τ = it)          ║
║                                                                  ║
║  e^{-iHt/□}    ───────────────→ e^{-Hτ/□}                        ║
║                    t → -iτ                                       ║
║                                                                  ║
║  oscillates:                    decays:                          ║
║                                                                  ║
║       ∩   ∩   ∩                   |\                             ║
║   ────╯ ╰─╯ ╰─╯ ╰──               |  \                           ║
║                                   |    \___                      ║
║  Quantum wave                     |        _________             ║
║  function: oscillates              Thermal weight: decays        ║
║                                                                  ║
║  The i is the only difference between these two functions.       ║
║  Remove i → quantum oscillation becomes exponential decay.       ║
╚══════════════════════════════════════════════════════════════════╝
```

Figure 7.1. The Wick rotation. A single substitution (t → -iτ) transforms the oscillatory quantum phase factor into the real decaying exponential of thermal physics. The memory kernel K(τ) = Σ A□ e^{-|τ|/τ□} has exactly this form. The i in the quantum exponent is the only mathematical difference between a quantum field that oscillates and a trauma trace that decays.

And the memory kernel for C-PTSD trauma?

$$K_{\text{trauma}}(\tau) = \sum_k A_k \, e^{-|\tau|/\tau_k}$$

This is the Wick-rotated propagator. The QFT field mass m corresponds to $1/\tau_k$. The propagator amplitude $1/2m$ corresponds to A_k. These are not analogous. They are the same mathematical object with different domain-specific names.

Feynman Diagrams for Emotions

Feynman diagrams were developed in the 1940s as a way of computing interactions in quantum field theory. They represent particles as lines and interactions (couplings) as vertices. A photon and an electron meeting at a vertex and scattering is a Feynman diagram. The rules for computing physical quantities from these diagrams are exact — each diagram corresponds to a specific integral.

In the 1990s and 2000s, it was established (Penrose 1971, Baez and Lauda 2011, Selinger 2010) that Feynman diagrams are a special case of a more general mathematical language: **string diagrams** — diagrams for morphisms in symmetric monoidal categories. This is not a simplification. It is a theorem. String diagrams, Feynman diagrams, and morphisms in symmetric monoidal categories are the same mathematical object in three notations.

The soma-field operations — coupling of emotional modes, composition of field operators, tensor products of states — are morphisms in exactly this sense. The following diagram represents two emotional modes combining at an interaction vertex:

```
EMOTIONAL INTERACTION AS FEYNMAN VERTEX

Fear ───────┐
            ├───────── Freeze
Shame ──────┘
(coupling W_{fear,shame → freeze})

This is identical in structure to a Feynman vertex:

electron ───────┐
                ├───────── electron (scattered)
photon ─────────┘

Both are morphisms:  A ⊗ B → C
in a symmetric monoidal category.
Fear ⊗ Shame → Freeze  is a valid morphism in the soma-field category.
```

The clinical relevance: the Feynman diagram language gives us a way to represent and compute emotional interactions combinatorially — to ask what the "Feynman rules" for emotional coupling are, and what composite interactions are possible.

The Correspondence Table

QFT quantity	Soma-Field analogue
Field mode ϕ[illegible]	Emotional mode e_i
Coupling constant J_{ij}	Coupling matrix entry W_{ij}
Field mass m	Inverse decay time 1/τ[illegible]
Propagator amplitude 1/2m	Trauma trace amplitude A[illegible]
Euclidean propagator G_E	Memory kernel K(τ)
Vacuum energy [illegible]H[illegible]$_0$	Resting field energy H(e_calm)
Thermal fluctuation k_BT	Noise amplitude σ_0
Wick rotation t → −iτ	Real-time Langevin dynamics
Feynman vertex	Emotional mode interaction
Morphism A[illegible]B → C	Field coupling operation

Table 7.1. Formal correspondence between QFT quantities and Soma-Field analogues. Each row is a single mathematical entity in two different notation systems. The correspondences are not approximate analogies — they are exact identifications under the Wick rotation and the Hopfield equivalence.

GOING DEEPER: The Baez–Lauda Coherence Theorem

In 2011, John Baez and Aaron Lauda proved a coherence theorem establishing that string diagrams are a complete and sound notation for morphisms in symmetric monoidal categories. This means: anything you can write as a morphism in a symmetric monoidal category, you can draw as a string diagram, and vice versa, with perfect fidelity.

Feynman diagrams are string diagrams for the symmetric monoidal category of representations of the Poincaré group (the symmetry group of spacetime). Tensor network diagrams (used in quantum information and condensed matter) are string diagrams for the same structure.

The soma-field operations — emotional mode coupling, field composition, state tensor products — are morphisms in a symmetric monoidal category. Therefore, they can be drawn as string diagrams. Therefore, they can be computed with the same diagrammatic calculus as Feynman diagrams.

> This is not the claim that emotions are quantum mechanical. It is the claim that the mathematics of composition and coupling is universal — it appears wherever things interact, regardless of what the things are.

> **KEY TERMS**
>
> **Wick rotation** — the substitution $t \to -i\tau$ that transforms oscillatory quantum dynamics into real-time thermal/stochastic dynamics.
>
> **Feynman diagram** — a diagrammatic notation for computing interaction amplitudes in quantum field theory; each diagram represents a specific integral contribution to a physical quantity.
>
> **String diagram** — a diagrammatic notation for morphisms in a symmetric monoidal category; identical in structure to Feynman diagrams under the Baez–Lauda theorem.
>
> **Morphism** — a structure-preserving map between objects in a category; the general notion that subsumes functions, linear maps, and physical interactions.

The Nervous System as Phase Diagram

LEARNING OBJECTIVES

By the end of this chapter, you will understand:

- What phase transitions are and why they apply to the nervous system
- How the three polyvagal states correspond to different phases
- Why state changes in trauma feel sudden rather than gradual
- What ADHD represents in thermodynamic terms

Phase Transitions

Water can exist as ice, liquid, or steam. At atmospheric pressure, it transitions between these phases at specific temperatures: 0°C and 100°C. The transitions are dramatic: adding energy to ice below 0°C changes its temperature gradually; adding energy at exactly 0°C produces no temperature change — the energy goes entirely into breaking the crystal lattice, reorganising water molecules from a rigid ordered structure into a fluid disordered one. This is a **phase transition**: a qualitative reorganisation of the system's structure at a critical point, rather than a smooth gradual change.

Phase transitions appear wherever there is an energy landscape with multiple stable phases, and a parameter (temperature, pressure, magnetic field) that shifts the relative stability of those phases. They are universal.

The Three Phases of the Nervous System

The polyvagal hierarchy describes three functional states of the autonomic nervous system. In the Soma-Field Model, these correspond to three distinct phases of the field:

```
╔══════════════════════════════════════════════════════════════════════╗
║              SOMA-FIELD PHASE DIAGRAM                                ║
╠══════════════════════════════════════════════════════════════════════╣
║                                                                      ║
║  Arousal ▲  HIGH                                                     ║
║  level   │  ╔════════════════════════╗                               ║
║          │  ║  SYMPATHETIC PHASE     ║ Fight / Flight                ║
║          │  ║  Large oscillations    ║ High noise                    ║
║          │  ║  Fast transitions      ║ Mobilisation                  ║
║          │  ╚════════════════════════╝                               ║
║          │                ↕ phase boundary (T_upper)                 ║
```

```
║   MEDIUM │   ╔═══════════════════════════╗                              ║
║          │   ║  VENTRAL VAGAL PHASE      ║ Social engagement            ║
║          │   ║  Stable oscillations      ║ Regulated noise              ║
║          │   ║  Social capacity          ║ Window of Tolerance          ║
║          │   ╚═══════════════════════════╝                              ║
║          │              ↕ phase boundary (T_lower)                      ║
║     LOW  │   ╔═══════════════════════════╗                              ║
║          │   ║  DORSAL VAGAL PHASE       ║ Freeze / Shutdown            ║
║          │   ║  Minimal oscillations     ║ Very low noise               ║
║          │   ║  Disconnection            ║ Immobilisation               ║
║          │   ╚═══════════════════════════╝                              ║
║          └──────────────────────────────────────────────────────────    ║
║               perceived threat level →                                  ║
╚═════════════════════════════════════════════════════════════════════════╝
```

Figure 8.1. The nervous system as a phase diagram. Three distinct phases correspond to the three polyvagal states. Phase boundaries (T_upper and T_lower) mark the transitions. For a regulated nervous system, most experience occurs in the ventral vagal phase. For a trauma-modified system, the lower boundary T_lower may be close to the ventral vagal resting state, making the transition to freeze easier to trigger.

The critical feature of a phase transition — as opposed to a smooth change in arousal level — is that it happens *all at once*. Below the phase boundary, adding arousal increases activation level. At the phase boundary, the system tips: a qualitatively different organisation takes over. This is why the freeze response (dorsal vagal) is not "very very calm": it is a different phase with different physical properties, entered through a phase transition, not reached by gradual reduction.

This also explains why clients in therapy sometimes describe state changes as happening without warning: from their perspective, they were fine, and then suddenly they were not. From the model's perspective, they were gradually approaching a phase boundary, and the transition happened when they crossed it. The discontinuity is real — it is a property of the phase diagram, not a failure of self-awareness.

ADHD: A Thermodynamic Framing

Attention Deficit Hyperactivity Disorder (ADHD) presents quite differently from C-PTSD in the soma-field model. Rather than a modification of the coupling matrix structure, ADHD corresponds primarily to an increase in the **effective noise amplitude** σ_0 and a reduction in **damping** γ of the field dynamics.

The Langevin equation with these parameters:

$$\dot{\mathbf{e}} = -\gamma \, \nabla H(\mathbf{e}) + \sigma_0 \, \eta(t)$$

In the ADHD regime, σ_0 is large and γ is small. The implications:

- The field moves around the landscape quickly (high noise, low damping)
- It spends less time in any single attractor (shallow dwell time in all basins)
- Transitions between states are frequent and sometimes erratic
- The effective "temperature" of the system is high: many states are thermally accessible

NEUROTYPICAL (moderate σ_0, moderate γ):

——— e(t): settles at attractor, brief excursions, returns

>

ADHD (high σ_0, low γ):

——— e(t): fast, wide excursions, brief attractor dwell

>

Figure 8.2. Field dynamics in neurotypical (top) and ADHD (bottom) regimes.
ADHD is not a broken attractor structure – the landscape may be quite normal.
It is a high-temperature, low-damping dynamical regime in which the field moves
through the landscape rapidly and does not settle.

The clinical significance: ADHD is not a motivation or character failure. It is a nervous system running at a thermodynamic setting different from typical, with specific performance characteristics — excellent rapid exploration of large state spaces, poor sustained dwell in narrow regions. "Focus" difficulties arise not because the attractor is absent, but because the effective temperature is too high for the system to remain in it.

The co-occurrence of ADHD and C-PTSD — which is common, and is well-documented — creates a particularly complex landscape: the coupling matrix is asymmetrically modified (C-PTSD effect) *and* the field is running at high temperature (ADHD effect). The practical consequence is a system that has a large, deep hypervigilance attractor and the thermal energy to reach it from almost anywhere.

KEY TERMS

Phase transition — a qualitative reorganisation of a system's structure at a critical parameter value; not a gradual change but a discontinuous one.

Noise amplitude σ_0 — the magnitude of random fluctuations in the field dynamics; controls the effective temperature of the system.

Damping γ — the rate at which the field returns toward attractor states after perturbation; low damping means slow return.

Effective temperature — the ratio σ_0^2/γ; determines how widely the field explores the landscape relative to the depth of the attractors.

PART IV: WHAT CHANGES

The Instrument

LEARNING OBJECTIVES

By the end of this chapter, you will understand:

- What the Soma-Field Instrument is designed to measure
- The seven dimensions the instrument tracks
- What the ABCD operator circuit does
- How the instrument relates to clinical practice

The Map Is Not the Territory

The Soma-Field Model is a mathematical description. Like all mathematical descriptions of physical or biological systems, it simplifies. The soma-field is not the body; it is a model of the body, selected for the properties it can illuminate while necessarily omitting others. This is not a failure of the model. A map that included every detail of the territory would be the territory.

The **Soma-Field Instrument** is a clinical tool built on this model: a structured means of tracking the parameters of the soma-field over time — the coupling structure, the attractor positions, the threshold, the noise level, the memory kernel amplitudes — so that changes can be measured rather than merely described.

The instrument is not a questionnaire. It does not ask about narrative or history. It asks about the body: current activation levels across the emotional modes, attractor dwell times, threshold accessibility, interoceptive accuracy. The goal is to make the model's parameters observable.

The Seven Dimensions

The instrument tracks seven primary dimensions of soma-field state:

```
╔══════════════════════════════════════════════════════════════════╗
║          THE SEVEN DIMENSIONS OF THE SOMA-FIELD                 ║
╠══════════════════════════════════════════════════════════════════╣
║                                                                  ║
║  1. ACTIVATION LEVEL           How strongly are the modes       ║
║     e = (e₁,...,e□)            currently firing?                ║
║                                                                  ║
║  2. ATTRACTOR POSITION         Which state is the field         ║
║     e* = argmin H(e)           currently resting in?            ║
```

Figure 9.2. The Soma-Field instrument pipeline. Biofeedback sensors (HRV, EDA, EMG) feed the soma-field model, which produces a real-time emotion vector $\mathbf{e}(t) \in \square^{11}$. This drives The Tensor (the emotional score specification), which controls a synthesis engine (Phase Plant). A feedback loop via therapeutic intervention δW allows the practitioner to modify the coupling matrix directly — closing the loop between measurement and treatment. *Author's original figure.*

Figure 8: Figure 9.2. The Soma-Field instrument pipeline. Biofeedback sensors (HRV, EDA, EMG) feed the soma-field model, which produces a real-time emotion vector $\mathbf{e}(t) \in \square^{11}$. This drives The Tensor (the emotional score specification), which controls a synthesis engine (Phase Plant). A feedback loop via therapeutic intervention δW allows the practitioner to modify the coupling matrix directly — closing the loop between measurement and treatment. *Author's original figure.*

```
║                                                                  ║
║  3. THRESHOLD                 At what activation level does      ║
║     T                         the field become conscious?        ║
║                                                                  ║
║  4. WINDOW OF TOLERANCE       How wide is the basin around       ║
║     ΔT = T_upper - T_lower    the current attractor?             ║
║                                                                  ║
║  5. NOISE LEVEL               How much thermal fluctuation       ║
║     σ_0                       is present? (ADHD component)       ║
║                                                                  ║
║  6. MEMORY KERNEL AMPLITUDE   How strongly are past              ║
║     A = (A_1, A_2, ...)       activations currently echoing?     ║
║                                                                  ║
║  7. INTEROCEPTIVE ACCURACY    How reliably can the person        ║
║     α □ [0,1]                 read their own field state?        ║
║                                                                  ║
╚══════════════════════════════════════════════════════════════════╝
```

Figure 9.1. The seven dimensions of the Soma-Field Instrument. Each dimension corresponds to a parameter or derived quantity of the mathematical model. Clinical progress is tracked as change across these dimensions over time, rather than as narrative self-report alone.

The ABCD Operator Circuit

The instrument is organised around four operators that act on the soma-field:

A — Attention: the operation of directing conscious attention to a body region or emotional mode. Attention modulates the threshold T locally: attended regions have their activation brought closer to or above the threshold. Formally: a projection operator that selects a subspace of the field.

B — Body: the somatic grounding operations — breath, posture, movement, temperature. These directly influence the coupling matrix (changing which modes are activated together) and the noise amplitude (breath regulation reduces σ_0). Formally: a modification of the W and σ_0 parameters.

C — Coupling: the explicit work of mapping which emotional modes are coupled, how strongly, and in what direction. This is the diagnostic function of the instrument: identifying the current coupling structure so that modifications can be targeted. Formally: an estimation of W from observed field dynamics.

D — Dynamics: tracking field evolution over time — how the state moves, which attractors it visits, how long it dwells, what triggers transitions. This is the longitudinal function: measuring change across sessions.

```
THE ABCD CIRCUIT

┌───────────────────────────────────────────────────────────────┐
│                                                               │
│      A (Attention)    B (Body)                                │
│          │                │                                   │
│          ▼                ▼                                   │
│      ┌───────┐       ┌────────┐                               │
│      │ lower │       │ modify │                               │
│      │   T   │       │ W, σ   │                               │
│      └───┬───┘       └────┬───┘                               │
│          │                │                                   │
│          └────────┬───────┘                                   │
│                   │                                           │
│              ┌────▼────┐                                      │
│              │  FIELD  │ e(t)                                 │
│              │  STATE  │                                      │
│              └────┬────┘                                      │
│                   │                                           │
│          ┌────────┴───────┐                                   │
│          │                │                                   │
```

```
│      ┌───▼───┐      ┌────▼───┐                              │
│      │ map W │      │ track  │                              │
│      │       │      │  e(t)  │                              │
│      └───────┘      └────────┘                              │
│     C (Coupling)   D (Dynamics)                             │
│                                                              │
└──────────────────────────────────────────────────────────────┘
```

Figure 9.2. The ABCD operator circuit. Attention (A) and Body (B) are input operators that act on the field. Coupling (C) and Dynamics (D) are measurement operators that read from the field. Together they form a closed loop: the measurement informs the input, which modifies the field, which is measured again.

Forward Transformation

```
> "The opposite of trauma is not safety.
> It is a nervous system that can find safety."
>
```

LEARNING OBJECTIVES

By the end of this chapter, you will understand:

- Why "healing" in the traditional sense is not the right goal for all trauma
- What forward transformation means in the language of the model
- What therapy "does" when it works, in terms of field parameters
- What the new landscape looks like

The Wrong Goal

The dominant model of trauma recovery involves, in some form, a return. Processing the memory until it no longer carries charge. Resolving the dissociated parts. Finding the self that existed before. Returning to baseline.

For late trauma — modification occurring after the baseline is formed — this model is coherent. A baseline exists. The modification can, in principle, be subtracted from the current coupling matrix to recover something close to it. The therapeutic work, however difficult, is working toward a target that is real.

For pre-verbal trauma, this model generates a problem. The baseline was never fully formed. The target of recovery — the self before the modification — is a mathematical object that does not exist. Attempting to drive the field toward it is attempting to converge on an undefined value.

Clinically, this manifests as therapy that helps, and helps, and helps — and never arrives. Each session improves things. The client gets better at regulation, more tolerant of activation, more able to function. But the destination remains unreachable. The gap persists. The sense of having "a self before all this" that the therapy is trying to restore — never narrows to nothing.

This is not a failure of the therapy or the therapist. It is a consequence of using the wrong map. The destination does not exist; the voyage toward it cannot terminate.

The Right Goal

Forward transformation changes the question.

Instead of: *how do we remove the modification to recover what was there before?*

We ask: *what kind of coupling matrix W' would give this nervous system the widest possible window of tolerance, the deepest possible calm attractor, and the lowest possible memory kernel amplitudes — starting from where it is now?*

This is a well-posed optimisation problem. W' does not have to be W_0. It does not have to resemble a neurotypical baseline. It has to have desirable dynamical properties as specified by the clinical goals of this person.

The voyage is not back. It is forward into a landscape that has never existed — a landscape being constructed, not recovered.

```
THERAPEUTIC TRAJECTORY: FORWARD TRANSFORMATION

┌──────────────────────────────────────────────────────────────────┐
│                                                                  │
│  CURRENT LANDSCAPE (W)         TARGET LANDSCAPE (W')            │
│                                                                  │
│  Energy H ▲                    Energy H ▲                       │
>
│            │  │  └──┘  │                  │└ calm *             │
│            │  │calm *  │  hyper*          │   wide basin         │
│            │  │(narrow)│  (deep)          │                     │
│            └──┴────────┴───────           └───────────────      │
│                                                                  │
│  W → W': calm basin widens, hypervigilance basin shallows,      │
│          memory kernel amplitudes reduce.                        │
│          The new landscape has never existed before.             │
│          It is being built, not recovered.                       │
│                                                                  │
└──────────────────────────────────────────────────────────────────┘
```

Figure 10.1. Forward transformation. The target W' is not a reconstruction of a prior baseline (which may not have existed). It is a new configuration with desired dynamical properties: a wide calm basin, shallow hypervigilance attractor, and reduced memory kernel amplitudes. The path from W to W' uses therapeutic tools as the mechanism of landscape modification.

What Therapy Does

In the language of the model, effective somatic therapy for pre-verbal trauma does the following, measurable in terms of the model's parameters:

1. **Widens the window of tolerance** ($T_{\text{upper}} - T_{\text{lower}}$ increases): more activation is tolerable without triggering a phase transition.
2. **Reduces memory kernel amplitudes** (A_k decrease): past activations exert less pull on the current field state. The echoes get quieter.
3. **Increases memory kernel decay times** (τ_k increase): the echoes that remain fade more quickly. The field returns to rest between episodes.
4. **Symmetrises the coupling partially** (W becomes more symmetric): the asymmetric directional flows decrease. Getting from hypervigilance to calm becomes less difficult relative to the reverse journey.
5. **Deepens the calm attractor** (calm basin gets deeper and wider): the field can be perturbed further from rest and still return there.
6. **Improves interoceptive accuracy** (α increases): the person gets better at reading their own field state, which improves the precision of all the above.

None of these changes brings the field to W_0. All of them make the field W' more functional, more flexible, and more capable of safety. The model does not specify how these changes are achieved — that is the domain of clinical practice. It specifies what is changing when they are achieved.

The Therapeutic Relationship as Field Coupling

A note on the relational dimension, which the model's formalism can sometimes obscure.

The coupling matrix W is not static. It is updated by experience. The experience of being in a regulated relationship — of having an other whose field is predominantly ventral vagal, engaged, and non-threatening — is itself field-modifying. The nervous system learns from co-regulation.

In field language: the therapist's soma-field is coupled to the client's soma-field during a session. This coupling is weak (they are separate bodies) but not zero. Repeated experiences of this coupling — of another field that is stable and available — gradually shift the client's attractor structure. The calm that is borrowed from the relational field slowly becomes encoded in the client's own coupling matrix.

This is why relational therapy works even in the absence of explicit body-focused techniques. The relationship is the technique. The therapist's regulated nervous system is the instrument.

AUTHOR'S NOTE: The Voyage Forward

I wrote this model in part because I needed a description of my own landscape that was precise enough to work with.

The traditional therapeutic story — you process the trauma, you return to yourself, you heal — did not fit. I got better, session by session, year by year. The regulation improved. The activation windows widened. The freeze responses got shorter. But there was nowhere I was arriving at, no self I was returning to, because the modification had not been added to a prior self. It was the self.

What the model gave me was a different story: not a return, but a construction. Not going back to something, but going forward to something that has never existed. And because the target is W' rather than W_0, the voyage does not need to end.

There is no failure in that. There is, in fact, considerable freedom.

KEY TERMS

Forward transformation — the construction of a new coupling matrix W' with desired dynamical properties, as opposed to the recovery of a prior baseline W_0.

Co-regulation — the process by which the soma-field of one person influences the soma-field of another through relational coupling; the mechanism by which the therapeutic relationship modifies the landscape.

PART V: APPLICATIONS

A Voyage into the Field

LEARNING OBJECTIVES

By the end of this chapter you should be able to:

- Explain what it means to *navigate* an emotional field rather than merely observe it
- Describe the two sectors of soma-field dynamics: perturbative (within a basin) and non-perturbative (threshold crossings)
- Explain what a Feynman diagram represents and apply the concept informally to emotional coupling
- Define the *emotional score* of a narrative and distinguish it from its container
- Explain the holographic principle as applied to clinical assessment
- Describe what EmotionML captures and what it omits

Two films. One from 1966. One never made.

In *Fantastic Voyage* (Fleischer, 1966), a submarine called the *Proteus* is miniaturised to microscopic scale and injected into the bloodstream of a critically injured scientist. The crew has sixty minutes to navigate from the carotid artery to a blood clot in the brain, dissolve it, and exit before the miniaturisation reverses. The body is the territory. The voyage is literal.

In a therapy session, something similar happens. Attention — the therapist's, and eventually the patient's own — is directed inward. It navigates through layers of somatic sensation, emotional activation, and memory echo. It encounters resistance: the field's own defence against being observed. It approaches regions of high activation — the deep attractors — and, if conditions permit, crosses the threshold into them rather than turning back.

The body is the territory in both cases. The voyage is into an interior that has its own geography, its own currents, its own immune responses. The *Proteus* crew is attacked by white blood cells — the body's machinery for destroying foreign objects. The therapeutic attention is resisted by the field's own homeostatic mechanisms: avoidance, dissociation, intellectualisation, the body's insistence that some regions remain unvisited.

It was always the same film.

The Soma-Field Model provides the mathematics of that film: the Hamiltonian landscape the crew must navigate, the attractor basins where the submarine drifts without effort, the energy barriers that require thrust to cross, the memory kernel that makes past routes echo in present navigation. This chapter develops the geometry of the voyage.

The Navigable Landscape

In Chapter 4 we introduced the Hamiltonian $H(\mathbf{e})$ as the energy function of the emotional field. The state of the field is a point in the high-dimensional space of all possible emotional-somatic configurations. The dynamics move the field downhill toward local minima — the attractor basins.

Think of this as terrain. The attractor basins are valleys. The field settles naturally into whichever valley it is closest to, and tends to stay there unless external energy (a trigger, a somatic cue, a therapeutic intervention) pushes it uphill toward a ridge and over into another valley.

The therapeutic voyage is a navigation across this terrain: starting in one valley (the presenting state), moving toward another (the target state — safety, integration, the capacity for contact), crossing the ridges in between. The ridges are the thresholds. The crossing is the therapeutic event.

What the terrain looks like. For a field with two strongly coupled modes — call them *fear* and *shame* — the landscape is a surface in three dimensions: fear on one axis, shame on another, energy on the vertical axis. The attractor is a bowl. The trauma state may have two bowls: a "fear-then-shame" sequence, and a "freeze" attractor from which shame and fear are both absent but unreachable.

For n modes, the terrain is n-dimensional. Visualisation requires projection, but the mathematics is the same regardless of dimension.

Fractal basin boundaries. When the coupling matrix W is asymmetric — when fear drives shame more strongly than shame drives fear, as is common in post-traumatic presentations — the boundary between attractor basins is not a smooth curve. It is fractal: the boundary between the "hypervigilance" basin and the "collapse" basin in a traumatised field has infinitely complex interdigitation at every scale of magnification.

The Mandelbrot set is the mathematical archetype of a fractal basin boundary: the boundary between the set and its complement is a Julia set, infinitely detailed at every scale. The fractal basin boundaries of the soma-field are of the same class. The visualisation is not merely aesthetic — the mathematics is the same.

> **GOING DEEPER: Fractals, Julia Sets, and Attractor Boundaries**
>
> The iteration $z \mapsto z^2 + c$ that generates the Mandelbrot set is a discrete dynamical system on the complex plane. Its attractor is the origin (the sequence converges to 0), or infinity (the sequence escapes). The boundary between the two basins is the Julia set J_c — a fractal object that, for most values of c, has non-integer (Hausdorff) dimension strictly between 1 and 2.
>
> The soma-field is a continuous dynamical system on $\mathbb{R}^n$, not a discrete iteration on $\mathbb{C}$.

> But the mechanism is the same: nonlinear coupling between modes (the W_{ij} terms + the threshold nonlinearity) generates sensitivity at the boundary that propagates across scales. The Hausdorff dimension of the boundary is a direct function of the asymmetry of W and the steepness of the threshold nonlinearity. In a severely traumatised field with highly asymmetric coupling, the boundary dimension approaches 2: the boundary is space-filling. There is, in the formal sense, no clean edge between hypervigilance and collapse. Just increasingly complex interdigitation.

Clinical implication. The fractal character of the basin boundary means that small perturbations near the threshold have disproportionate effects — the butterfly effect is concentrated at the boundary. A session conducted near a threshold crossing is qualitatively different from a session conducted well inside a basin. The geometry predicts this before any clinical experience confirms it.

Emotions Looking for Each Other

In particle physics, interactions are drawn as Feynman diagrams: lines representing particles moving through space and time, meeting at vertices where something happens. An electron emits a photon. A quark changes flavour. Two particles scatter.

The same formalism applies to the soma-field, and the interpretation is immediate.

The propagator. A single emotional mode — call it *fear* — traveling through time without interacting with anything else is drawn as a single line, moving from left to right (earlier to later). The line gets fainter as time increases: the mode decays toward its equilibrium unless maintained by coupling or stimulus. This is the *propagator* — the Green's function of the free dynamics.

```
Single-mode propagator:

fear ─────────────────────────────> (decays at rate |W_fear,fear|)
     t'                           t
```

The coupling vertex. When fear and shame are coupled — $W_{\text{fear, shame}} \neq 0$ — they can scatter: fear activates shame, shame amplifies fear. This is drawn as two lines meeting at a vertex, with the coupling strength W_{ij} labeling the junction.

```
Fear-shame coupling vertex:

fear  ────────────● ──────────── fear
                  │  W_fs
shame ────────────● ──────────── shame
```

If $W_{\text{fear, shame}} > 0$, shame excites fear. If $W_{\text{fear, shame}} < 0$, shame suppresses fear. For the asymmetric case $W_{\text{fear, shame}} \neq W_{\text{shame, fear}}$ — one emotion drives the other more than it is driven in return — the vertex is directional. Fear leads shame in a post-traumatic field. Shame may or may not respond in kind.

Feedback loops. When fear excites shame and shame excites fear in a closed cycle, the diagram is a loop. The loop is not merely a metaphor: it is a precise mathematical object whose value — computed by integrating over the intermediate times — gives a correction to the effective coupling at the loop's characteristic timescale.

```
Fear-shame feedback loop:


FEAR ────────────●─────────────────────●──── FEAR
                 │  W_fs               │
                 └────── SHAME ────────┘
                           W_sf
                 Loop correction: loop runs faster,
                 effective W_fear,fear increases.
                 This is sensitisation.
```

Repeated co-activation of fear and shame — as occurs in a trauma where shame was the response to terror — consolidates the loop: the effective coupling grows. The Feynman diagram is a picture of how shame becomes a reliable trigger for fear across sessions and years.

The memory vertex. The trauma memory kernel introduces a vertex that is non-local in time: mode j at some earlier time t' contributes to mode i now, at time t, with weight $K(t - t')$. The diagram has an internal arrow going backward in time — not acausally, but in the sense that the *past state* of the field is still driving the *present state*.

```
Memory kernel vertex:


shame(t') ───┐  K(t-t') J_fs
             └────────────────────── fear(t)


The shame at time t' is still driving fear now,
weighted by how much the memory kernel retains it.
```

For a field with a slow memory kernel (large τ_k), past activations echo far into the future. A traumatic incident twenty years ago is still driving present fear via the memory vertex — not as a belief or a narrative, but as a dynamical coupling with a specific timescale.

The instanton: the pivot. No finite collection of Feynman diagrams — no sum of scattering and loop corrections — describes a threshold crossing. The topological change from one basin to another is a *non-perturbative* event: an instanton. It is not a series of small steps; it is a qualitative transition, a jump between attractors. In physics, instantons are the events that perturbation theory cannot see. In the therapy room, they are the sessions that change something permanently.

GOING DEEPER: The Two Sectors

Every field theory divides into a *perturbative* sector (small fluctuations, describable by Feynman diagrams) and a *non-perturbative* sector (large topological events, described by instantons, solitons, and other saddle-point solutions).

The soma-field has the same division:

Sector	Events	Mathematical description
Perturbative	Emotional coupling, sensitisation, habituation, day-to-day activation	Feynman diagrams: propagators, vertices, loops
Non-perturbative	Threshold crossings, basin transitions, pivotal sessions	Instantons: minimal-action paths between basins

The perturbative sector is accessible to standard talk therapy (changing W_{ij} by desensitisation; damping memory kernel amplitudes A_k; adjusting thresholds). The non-perturbative sector requires conditions for threshold crossing: sufficient activation energy, a safe enough container, and — often — direct somatic engagement. You cannot reach an instanton by accumulating small perturbative steps. That is the formal reason why some therapeutic approaches reach a ceiling.

The Emotional Score

A musical score is not a performance. It is the abstract structure that can be performed in many ways — by different orchestras, in different halls, at different tempos — while remaining recognisably itself. The *notes* are the invariant; the *sound* is the realisation.

A film has an emotional score. Not the music (though the music is part of its expression), but the trajectory of the emotional field that the film traces over its duration: how activation rises and falls, which modes are coupled, where the thresholds are crossed, what the final attractor state is.

This emotional score is independent of the narrative container — the specific story in which it is realised. The same score can be realised in a river journey, a war, a marriage, a career, a therapy, or a voyage through a bloodstream.

Formally. The emotional score is a trajectory $\mathbf{e}^*(t)$ in the emotional field space, parameterised by story-time $t \in [0, 1]$ (opening to closing). A film, novel, or therapy session is:

$$\text{Realisation} = (\mathbf{e}^*(t),\ \text{Container})$$

The container provides the narrative surface: characters, setting, imagery, plot. The emotional score provides the dynamics: which modes activate, in what sequence, at what coupling strength.

The Conrad example. *Heart of Darkness* and its film realisation *Apocalypse Now* (Coppola, 1979) share a score: an upstream journey toward something pre-verbal, toward a figure (*Kurtz*) who represents the field's deepest attractor — the place where normal threshold regulation has dissolved. The journey upstream is a journey toward decreasing τ_d (shorter developmental time), toward earlier, more diffuse, less differentiated emotional modes. The field becomes less structured as the journey continues. Kurtz is the attractor at the bottom of the developmental basin — not a monster, but the deepest attractor, the one with no threshold above it.

The score is: *progressive reduction of threshold distance, increasing weight of pre-verbal modes, final approach to a basin from which ordinary return is blocked.* The container (Congo river / Vietnam river) is a surface over which this score is played.

Multi-scale structure. The score has fractal structure: the same emotional pattern recurs at the level of the full film, the act, the scene, and the moment. A scene in which a character approaches and retreats from a threshold is a micro-version of the film's macro-structure. This is not a metaphor — the soma-field dynamics are scale-invariant near a critical point, so the same Hamiltonian structure repeats across timescales. A good filmmaker composes at all scales simultaneously.

The viewer's field. The viewer has their own emotional field $\mathbf{e}_V(t)$ which couples to the screen signal $S(t)$:

$$\dot{\mathbf{e}}_V = -\nabla H_V(\mathbf{e}_V) + \lambda \cdot S(t) + \eta_V$$

The director controls $S(t)$ — the screen signal — but not H_V (the viewer's own landscape). A viewer whose own field has a deep shame attractor will have a different response to the same $S(t)$ than a viewer without it. The film is the same; the voyage is different. This is the formal account of why films affect different people differently, and why re-watching a film after therapeutic work can produce a qualitatively different emotional experience: H_V has changed.

The Holographic Clinic

In theoretical physics, the holographic principle (Susskind, 1995; Bousso, 2002) states that the complete description of a volume of space can be encoded on its boundary surface, with no loss of information. A three-dimensional object is fully represented by a two-dimensional hologram. The interior is encoded in the edge.

The soma-field has a holographic structure that is clinically actionable.

The boundary. The observable boundary of the soma-field is what can be seen from outside: behaviour, posture, facial expression, reported affect, the pattern of threshold crossings in session, the rate of escalation and de-escalation, the latency between stimulus and response. This is the boundary data — the hologram.

The bulk. The interior of the soma-field is inaccessible to direct observation: the weight matrix W, the memory kernel $K(\tau)$, the effective thresholds T_i, the attractor topology. These are the bulk fields.

The reconstruction theorem. If the boundary data is sufficiently rich — if we observe enough threshold crossings, enough coupling patterns, enough temporal correlations — the bulk fields can be reconstructed. The weight matrix W_{ij} can be estimated from the co-activation statistics of observed modes. The memory kernel time constants τ_k can be estimated from the delay between stimulus and response at different frequencies. The attractor topology can be inferred from which basins the field visits and how long it dwells in each.

The body tells you everything. This is not a therapeutic truism. It is a measurement theorem: given sufficiently rich boundary data, the full soma-field is recoverable from external observation. The body is a hologram of its own interior.

Clinical implication. A thorough intake assessment — one that tracks not just presented symptoms but response latencies, co-occurrence statistics, threshold distances, and interoceptive access — is a holographic measurement. It gives access to the bulk fields without requiring the patient to verbally report what they do not have words for. The body has been keeping a precise record. The therapist's task is to read it.

EmotionML: Labels Without Dynamics

The W3C EmotionML standard (Schröder et al., 2011) provides a formal vocabulary for annotating emotional states in human-computer interaction. It specifies representation formats for emotion categories (anger, fear, joy, sadness...), dimensions (valence, arousal, dominance), and appraisals (novelty, intrinsic pleasantness, goal congruence). It is a well-engineered taxonomy.

It is not a dynamical theory.

EmotionML says what emotional state a system is in at time t. It does not say how that state changes, what coupling it has to other states, what its threshold distance is, how its memory kernel drives its future evolution, or what basin transition conditions apply. It provides a label; the Soma-Field Model provides the dynamics.

The relationship is analogous to the relationship between a chemical nomenclature (naming compounds) and a rate equation (describing how compounds react). The nomenclature is necessary but not sufficient. Knowing that a patient presents as "fearful" is the EmotionML layer. Knowing the coupling $W_{\text{fear, shame}}$, the threshold T_{fear}, the memory kernel time constants, and the attractor depth is the soma-field layer. The second layer strictly includes the first.

> **AUTHOR'S NOTE: Why Taxonomy Is Not Enough**
>
> The history of psychiatry is largely a history of improving the taxonomy: from humours to syndromes to DSM categories to dimensional models. Each generation's taxonomy is more precise than the previous. Each is still a taxonomy.
>
> The shift from taxonomy to dynamics is not a refinement. It is a change of mathematical structure: from a set of labels to a vector field on a state space, with a Hamiltonian, a noise term, and a coupling matrix. The prediction capability is qualitatively different. A taxonomy tells you what something is called. A dynamical model tells you what it will do next and what it would take to change it.
>
> EmotionML is a very good taxonomy. The Soma-Field Model is the next layer.

> **KEY TERMS**
>
> **Navigable landscape** — the Hamiltonian $H(\mathbf{e})$ understood as terrain, with attractor basins as valleys and thresholds as ridges that must be crossed.

Fractal basin boundary — the boundary between attractor basins when the coupling matrix W is asymmetric; has non-integer Hausdorff dimension and is sensitive to perturbation at all scales.

Feynman diagram — a graphical notation for the terms in a perturbative expansion; in the soma-field, lines represent propagating modes, vertices represent couplings, and loops represent feedback cycles.

Perturbative sector — dynamics within an attractor basin, describable by Feynman diagrams; accessible to standard desensitisation and coupling-modification approaches.

Non-perturbative sector — threshold crossings and basin transitions; described by instantons; requires conditions for the full threshold-crossing event.

Instanton — a non-perturbative saddle-point solution connecting two attractor basins; in the therapy room, the pivot moment that changes the field topology.

Emotional score — the trajectory $\mathbf{e}^*(t)$ that defines a narrative's emotional structure independently of its container; formally Realisation $= (\mathbf{e}^*(t), \text{Container})$.

Holographic principle — the claim that boundary observables (behaviour, symptoms, threshold patterns) encode the full bulk fields (W, K, attractor topology); the basis for clinical assessment as measurement.

EmotionML — W3C standard for emotion annotation; provides taxonomy (labels) but not dynamics (evolution equations); a necessary but not sufficient layer.

CHAPTER SUMMARY

The Soma-Field Model provides the geometry of a voyage. The Hamiltonian landscape is the territory: attractor basins are valleys, thresholds are ridges, and the field navigates this terrain continuously. For asymmetric coupling matrices, the basin boundaries are fractal — infinitely complex at every scale, sensitive to perturbation everywhere along the edge.

The dynamics divide into two sectors. The perturbative sector — small fluctuations within a basin — is organised by Feynman diagrams: propagators carry modes through time, coupling vertices describe interactions between modes, feedback loops describe sensitisation, and memory kernel vertices describe the echo of past activations into the present. The non-perturbative sector — threshold crossings — is described by instantons: the minimal-action paths between basins that no perturbative sum can reach.

> Narratives have emotional scores: trajectories $\mathbf{e}^*(t)$ that define a film or story independently of its narrative container. The same score can be realised in a river journey, a war, or a voyage through a bloodstream. The viewer's own soma-field couples to the screen signal; what they experience depends on their own Hamiltonian.
>
> The boundary of the soma-field encodes the bulk: behavioural observation, response latency, co-activation statistics, and threshold patterns give access to the full weight matrix, memory kernel, and attractor topology without requiring verbal report of what has no words. The body is a hologram of its own interior.
>
> EmotionML provides the taxonomy. The Soma-Field Model provides the dynamics. Both are needed; the second strictly extends the first.

Epilogue: The T's

There are four T's in this book, and they are not accidental.

Theory — the formal structure that makes prediction possible. A theory is not a guess or an opinion. It is a precise description that can be tested, that makes specific predictions, and that says exactly what evidence would falsify it. The Soma-Field Model is a theory of emotional dynamics: it makes predictions about attractor structure, about the character of pre-verbal versus late trauma, about what parameters change in effective therapy. Whether those predictions survive contact with data is an empirical question, and the empirical work is needed.

Threshold — the parameter T, which appears in the model as the boundary between sub-threshold somatic activity and conscious emotional experience. The threshold is not a switch. It is a continuous parameter, differently set in different nervous systems, modifiable through practice and therapy. The difference between a body that feels everything and a body that feels nothing is, in formal terms, a difference in T. The therapeutic expansion of the window of tolerance is, in formal terms, a widening of the range around T within which the field can move without triggering a phase transition.

Time — the developmental time τ_d, which changes the character of a modification from perturbative (late trauma: $W = W_0 + \delta W$, a baseline plus a modification) to structural (pre-verbal trauma: $W = W_{\text{trauma}}$, the modification is the structure). Time also appears in τ_k — the decay time of the memory kernel, how long an echo persists. Therapy changes τ_k. The passage of time, in the absence of intervention, does not reliably change τ_k for pre-verbal somatic traces.

Transformation — the fourth T, the one that this book is about, finally. Not recovery. Not return. Not the restoration of a prior state. The construction of a new landscape: wider, more flexible, with deeper calm and shallower hypervigilance, arrived at from where the system is, going somewhere it has not been before.

There is a fifth T, which gave this book its title: **Trance** — in two senses. The first: the altered state at the threshold, the phase transition, the field crossing a boundary and remaining in the other phase. Trance is not a malfunction; it is a dynamic state, momentarily ungoverned by the usual attractors. In those moments, something is possible that is not possible from within a stable phase.

The second sense is the title itself. *A Voyage into Trance* (1995) is a Goa trance compilation by Paul Oakenfold. The trance state produced by extended rhythmic music and the freeze response of a traumatised nervous system are not the same experience. They are governed by the same mathematics. Both drive an arousal variable across a threshold and sustain it there. Music does this deliberately; trauma does it involuntarily. The mathematics does not distinguish.

The voyage into trauma is not a straight line. It passes through all five T's, sometimes in order, sometimes not.

The model is the map. The body is the territory. The voyage is yours.

Appendices

Appendix A: The Mathematics in Full

The following is a condensed version of the academic paper Soma-Field Model of Emotional Dynamics and C-PTSD *for readers who want the formal derivations. Full derivations, Lean 4 type sketches, and bibliography are in the companion document* `soma-field-paper.md`*.*

A.1 The Hamiltonian

$$H(\mathbf{e}) = -\frac{1}{2}\,\mathbf{e}^\top W\,\mathbf{e} - \theta^\top\mathbf{e}$$

where $W \in \mathbb{R}^{n\times n}$ is the coupling matrix and $\theta \in \mathbb{R}^n$ is the threshold bias vector.

A.2 The Dynamics

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}) + \sigma_0\,\eta(t) = W\mathbf{e} + \theta + \sigma_0\,\eta(t)$$

where $\eta(t)$ is white noise with $\langle \eta_i(t)\eta_j(s)\rangle = \delta_{ij}\delta(t-s)$.

A.3 The C-PTSD Modification

$$W_{\text{C-PTSD}} = W_0 + \Delta W, \qquad \Delta W_{ij} \neq \Delta W_{ji}$$

The asymmetry of ΔW breaks the gradient flow property and introduces directional cycles in the landscape.

A.4 The Memory Kernel

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}) + \int_0^t K(t-s)\,\mathbf{e}(s)\,ds + \sigma_0\,\eta(t)$$

$$K(\tau) = \sum_k A_k\,e^{-\tau/\tau_k}$$

A.5 Developmental Time Parameterisation

$$W(\tau_d) = f(\tau_d)\cdot W_0 + (1 - f(\tau_d))\cdot W_{\text{trauma}}$$

$$f(\tau_d) = \tanh\left(\frac{\tau_d}{\tau_c}\right), \qquad \tau_c \approx 36 \text{ months}$$

A.6 The QFT Correspondence

Under the Wick rotation $t \to -i\tau$:

QFT	Soma-Field
$G_E(\tau) = \frac{1}{2m} e^{-m\|\tau\|}$	$K(\tau) = \sum_k A_k e^{-\|\tau\|/\tau_k}$
Field mass m	Inverse decay time $1/\tau_k$
H_{Ising}	H_{soma}

Appendix B: Lean 4 Type Sketches

The following are proof sketches in Lean 4. `sorry` marks open proof obligations.

```
-- Core soma-field structures
structure CouplingMatrix (n : □) where
  W : Matrix (Fin n) (Fin n) □
  θ : Fin n → □

structure SomaField (n : □) where
  e     : Fin n → □      -- current activation vector
  W     : CouplingMatrix n
  T     : □              -- threshold parameter
  sigma : □              -- noise amplitude

-- The Hamiltonian
noncomputable def hamiltonian {n : □} (W : CouplingMatrix n) (e : Fin n → □) : □ :=
  -0.5 * Matrix.dotProduct (Matrix.mulVec W.W e) e
  - Matrix.dotProduct W.θ e

-- C-PTSD modification: asymmetric coupling
def isCPTSDModified {n : □} (W : CouplingMatrix n) : Prop :=
  □ i j, W.W i j ≠ W.W j i

-- Developmental time parameterisation
structure TraumaProfile (n : □) where
  τ_d        : □
  asymmetry  : Matrix (Fin n) (Fin n) □
  amplitudes : List (Fin n → □)
  decayTimes : List (Fin n → □)

def τ_c : □ := 36

noncomputable def structuralFraction (τ_d : □) : □ :=
```

```
  Real.tanh (τ_d / τ_c)


-- For pre-verbal trauma: structural fraction < tanh(1) ≈ 0.76
theorem preVerbalIsStructural {n : □} (profile : TraumaProfile n)
    (h : profile.τ_d < τ_c) :
    structuralFraction profile.τ_d < Real.tanh 1 := by
  unfold structuralFraction
  apply Real.tanh_lt_tanh
  exact div_lt_one_of_lt h (by norm_num)
```

Appendix C: The Cross-Language Correspondence Table

Mathematical language	Emotional dynamics
Symmetric monoidal category $\mathcal{C}$	Soma-field operator algebra
Object $A \in \mathcal{C}$	Emotional mode type
Morphism $f : A \to B$	Field operator (maps one mode to another)
Tensor product $A \otimes B$	Simultaneous activation of modes A and B
Composition $g \circ f$	Sequential field operations
Identity morphism id_A	Identity (mode persists unchanged)
Braiding $\sigma : A \otimes B \cong B \otimes A$	Mode-order independence of simultaneous states
Feynman vertex	Emotional interaction (coupling W_{ij})
Loop diagram	Memory kernel (self-coupling over time)
Feynman propagator	Memory trace decay $e^{-\lvert\tau\rvert/\tau_k}$
Vacuum state	Resting soma-field (minimal activation)
Partition function Z	Field normalisation (probability distribution over states)
Renormalisation group flow	Therapeutic modification of coupling constants
Phase transition	Polyvagal state transition (ventral/sympathetic/dorsal)
Symmetry breaking	Asymmetric W (C-PTSD modification)

The correspondences in this table are not analogies. They are identifications of the same mathematical object in two notation systems, established by the Baez–Lauda coherence theorem (2011) for the categorical column and the Wick rotation for the QFT column.

Appendix D: Glossary

Amplitude A_k — The strength of a trauma memory trace's influence on the current soma-field. Reduced by effective somatic therapy.

Attractor — A stable state in the energy landscape; a position toward which the soma-field naturally moves from nearby states.

Basin of attraction — The region of state space from which the field flows toward a given attractor.

C-PTSD operator — The modification ΔW to the coupling matrix that represents the effect of complex developmental trauma on the soma-field landscape.

Co-regulation — The process by which the soma-field of one person influences another through relational coupling; the somatic mechanism of relational healing.

Coupling matrix W — The matrix encoding the interactions between emotional modes; determines the shape of the energy landscape. Symmetry of W guarantees stable attractors.

Damping γ — The rate at which the field returns toward attractors after perturbation. Low damping (ADHD) produces rapid, wide-ranging field dynamics.

Decay time τ_k — How long a trauma memory trace persists before fading. Pre-verbal traces typically have longer decay times.

Developmental age at trauma τ_d — The age in months at which the primary traumatic modification occurred. Determines whether the modification is perturbative (late trauma) or structural (pre-verbal trauma).

Effective temperature — The ratio σ_0^2/γ; determines how widely the soma-field explores the landscape relative to the depth of the attractors.

Forward transformation — The construction of a new coupling matrix W' with desired dynamical properties, as the correct therapeutic goal for pre-verbal trauma (as opposed to recovery of a prior baseline).

Hamiltonian H — The energy function that assigns a value to every possible soma-field state; determines the landscape's hills and valleys and thus the dynamics.

Interoception — The nervous system's process of sensing the body's internal state.

Interoceptive accuracy α — The precision with which a person can read their own soma-field state. Disrupted by trauma; improvable through training and therapy.

Memory kernel $K(\tau)$ — The function describing how past field activations continue to influence the current state. In C-PTSD: a sum of decaying exponentials.

Noise amplitude σ_0 — The magnitude of random fluctuations in field dynamics. Elevated in ADHD; reduced by breath and autonomic regulation.

Phase transition — A qualitative reorganisation of the field's state at a critical parameter value. Polyvagal state changes (e.g., ventral vagal to freeze) are phase transitions, not gradual changes.

Soma — The body as experienced from the inside; the totality of interoceptive signals.

Soma-field — The vector **e** of somatic activation levels across emotional modes; the state of the body's emotional field at a given moment.

Structural fraction $f(\tau_d)$ — The proportion of the coupling matrix attributable to neurotypical baseline development versus trauma-formed modification.

Threshold T — The activation level above which a soma-field mode becomes conscious experience. The boundary between felt emotion and sub-threshold somatic activation.

Verbal encoding threshold τ_c — The approximate developmental age (≈36 months) at which narrative memory and verbal encoding capacity reliably emerges.

Wick rotation — The substitution $t \to -i\tau$ that transforms oscillatory quantum dynamics into real-time thermal/stochastic dynamics; the bridge connecting QFT propagators to soma-field memory kernels.

Window of Tolerance — The range of arousal within which the nervous system can function flexibly, process information, and engage socially.

Bibliography

The following references are cited in this book. Full academic citation details are in the companion document `bibliography.bib`.

- Baez, J. C., & Lauda, A. D. (2011). A prehistory of n-categorical physics.
- Damasio, A. (1994). *Descartes' Error: Emotion, Reason, and the Human Brain*.
- Gendlin, E. T. (1978). *Focusing*.
- Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *PNAS, 79*(8), 2554–2558.
- Levine, P. A. (2010). *In an Unspoken Voice*.
- Ogden, P., Minton, K., & Pain, C. (2006). *Trauma and the Body*.
- Penrose, R. (1971). Applications of negative dimensional tensors.
- Porges, S. W. (2011). *The Polyvagal Theory*.
- Schore, A. N. (2001). The effects of early relational trauma on right brain development.
- Selinger, P. (2010). A survey of graphical languages for monoidal categories.
- van der Kolk, B. A. (2014). *The Body Keeps the Score*.

- Vitiello, G. (2001). *My Double Unveiled*.

A Voyage into Trauma: The Soma-Field Theory of Emotional Life First edition, 2026. Companion academic paper: soma-field-paper.md

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Listening Notes

This book was written in a single session on the night of 16–17 May 2026.

The development was set to *Silver Machine* by Hawkwind. It closed with *It's So Easy* by Hawkwind.

Both choices were correct.

> *AI has had a brain since 1943. Now it has a body.*

Introduction

A patient sits with their therapist and is asked: *"What are you feeling right now?"* The question is deceptively simple. They may say *anxious*, yet that word covers a vast and heterogeneous territory — a tightness in the chest, a running commentary of worry, a vague readiness to flee, a memory surfacing from childhood. Another patient, asked the same question, reports feeling nothing at all; and yet their posture, respiration, and the quality of their silence suggest otherwise. The emotion is there. It is simply not yet conscious.

This gap between emotional presence and emotional awareness is one of the most clinically significant phenomena in psychotherapy. Theories of affect regulation (Schore, 2001), somatic experiencing (Levine, 2010), sensorimotor psychotherapy (Ogden, Minton & Pain, 2006), and polyvagal theory (Porges, 2011) all grapple, in different ways, with the same observation: emotions exist in the body before — and often without — being named in the mind. Eugene Gendlin called the sub-verbal bodily sense of an emotional situation the *felt sense* (Gendlin, 1978): something that is there, whole and present, but not yet articulate.

The Soma-Field Model proposed here attempts to give this clinical observation a formal structure. It does so by borrowing a conceptual tool from physics: the field. In physics, a field is not a thing that exists at a point. It is a quantity that exists everywhere in a space, continuously, whether or not it is observed. Particles — the things we can measure — are not separate from the field; they are *excitations* of it, local concentrations of energy that arise when the field is perturbed above a certain threshold.

The central claim of this paper is that this structure accurately describes the phenomenology of emotion. The emotional field is always there, distributed across body and nervous system. What we call a conscious emotional experience is an excitation of that field — a local concentration that has crossed a perceptual threshold and entered awareness. The field continues below the threshold whether or not we attend to it, and its sub-perceptual activity shapes our behaviour, physiology, and cognition continuously.

The Soma-Field Model contributes the first formal field-theoretic architecture for the limbic system. Every artificial neural network since McCulloch and Pitts (1943) (McCulloch & Pitts, 1943) is a formal model of the neocortex — the pattern-recognition and prediction layer. The limbic system — responsible for emotional valuation, threat detection, and the somatic state reinstatement that underlies trauma — has never received a comparable formal treatment. The Soma-Field Model is that treatment. Together with the Hopfield framework, it constitutes the first complete formal description of the two principal computational substrates of the vertebrate brain.

The paper proceeds as follows. Section 2 reviews the relevant background in somatic clinical models, and introduces the two theoretical tools borrowed from physics and computer science: quantum field theory and Hopfield network energy functions. Section 3 develops the Soma-Field Model in detail. Section 4 describes the energy landscape, including the attractor states corresponding to fight, flight, freeze, and regulated calm. Section 5 discusses dissonance and resolution as mechanisms of emotional interaction. Section 6 describes the Soma-Field Instrument, a practical tool for therapeutic use. Section 7 addresses clinical implications.

Background

The Body-Mind Problem in Clinical Practice

Contemporary neuroscience has largely dissolved the Cartesian boundary between body and mind. Damasio (1994) demonstrated that emotion is inseparable from rational cognition: patients with damage to the ventromedial prefrontal cortex — preventing the normal generation of somatic signals — lose not only their emotional range but also their capacity for effective decision-making. Van der Kolk (2014) documented extensively how traumatic emotional states are encoded not merely in explicit memory but in posture, gesture, visceral sensation, and autonomic regulation. Porges' polyvagal theory (2011) provided a neurobiological account of how the autonomic nervous system generates three hierarchically organised states — ventral vagal (social engagement), sympathetic (mobilisation: fight/flight), and dorsal vagal (immobilisation: freeze) — each with characteristic phenomenological and behavioural signatures.

What these frameworks share is a conviction that emotional states are not located in the brain alone, nor in the body alone, but in a coupled system that is best understood as a single functional unit. The term *soma* — from the Greek for body — is used here to denote this unified body-mind system, following the tradition of somatic psychotherapy.

The Felt Sense and Sub-Perceptual Emotion

Gendlin's concept of the *felt sense* (1978) is of particular relevance. He described it as "a special kind of internal bodily awareness... a body sense of meaning." It is not an emotion in the ordinary sense — not a named feeling — but something more diffuse: a pre-articulate sense that *something is there*, present in the body, before it has been identified or named. Focussing, the therapeutic method Gendlin developed, works precisely by attending to this pre-threshold signal and allowing it to surface into conscious articulation.

The Soma-Field Model provides a formal account of what the felt sense is: it is the activity of the emotional field below the perceptual threshold. It is real, causal, and continuously present. It shapes cognition and behaviour even when it does not surface as a named feeling.

Quantum Field Theory: Structure, Not Metaphor

Quantum Field Theory (QFT) is the framework of modern particle physics. Its central departure from classical physics is the priority of the *field* over the *particle*. In QFT, what we call particles — electrons, photons — are not fundamental objects. They are *excitations* of an underlying field: local, stable configurations of energy that arise when the field receives a sufficient perturbation.

The quantum vacuum — the ground state of the field — is not empty. It is a seething background of virtual fluctuations: momentary excitations that do not have enough energy to persist as observable particles. The vacuum is active, but sub-threshold.

```
A SINGLE FIELD MODE — amplitude over time
(e.g. a mode of the electromagnetic field; or, later, a mode of the emotional field)


|                                    ╭──────────────────╮
|          ╭──╮              ╭──╮   /                    \             ╭──
|   ╭─╮   /    \    ╭─╮     /    \ /                     \    ╭──╮   /
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T /─────\/───────\╯─────\╯───────────────────────────────────\/────────── T
|         \───────╯       \─────╯                             \──────────
|
└──────────────────────────────────────────────────────────────────────► time


←── VIRTUAL: field fluctuates but stays sub-threshold ──────────→ ←REAL→
     present, active, causally real — but not locally detectable      ↑
     (the QUANTUM VACUUM: not empty; seething with activity)       particle
                                                                   created
```

Figure 0. A single field mode in quantum field theory. The field oscillates continuously. Below the detection threshold T, excitations are sub-threshold — real and causally active, but not detectable as particles. The

quantum vacuum is not empty; it is a field in constant motion that never quite crosses the threshold. When the amplitude does cross T, a particle exists: a locally observable excitation. The same structure — field always present, consciousness only when threshold crossed — is the core of the Soma-Field Model.

This paper does not claim that emotions are quantum phenomena in any literal sense: the soma-field is a classical field, not a quantised one. The claim is stronger and more specific than analogy: the mathematical object being constructed — the Green's function of a coupled field manifold — is formally of the same *type* as the objects that arise in QFT, differing only in the dimensionality of the manifold and the nature of the probe. What was previously described as a structural analogy is here identified as a formal correspondence: a particle is a pole in the propagator of its field; a conscious emotional percept is a pole in the propagator of the soma-field. Different physics. Same mathematics.

That correspondence gives the model precise vocabulary for the following set of ideas, which are central to the clinical observation of emotion:

- A quantity that exists everywhere, continuously, even when unobserved
- A background of sub-threshold activity that is real and causally effective
- The emergence of observable phenomena (conscious feelings) through threshold-crossing excitation of that background
- The possibility of multiple simultaneous excitations that interact with one another

Note (*May 2026*): A subsequent experiment (QUANT-EXP-1) demonstrates that the quantum extension of the Hopfield landscape used in this model — replacing the classical Langevin process with a transverse-field quantum annealer — produces a measurable *topological reachability advantage*: quantum annealing reaches attractor basins that cold classical dynamics cannot reach at any finite noise level. This upgrades the formal correspondence from a structural claim to a testable empirical prediction. See the companion paper *Quantum Soma and the Penrose Gap* (doi:10.5281/zenodo.20351230) for the full results and theoretical implications.

One further consequence follows. The clinical phenomena of alexithymia — difficulty identifying and naming feelings — and its apparent opposite, emotional flooding or hypervigilance, have always been treated as separate conditions requiring separate explanations. In the Green's function framing, they are the same structure at two extremes of the same parameter: the perception threshold T_i is too high (the bulk dynamics cannot cross into observable experience) or too low (bulk fluctuations flood the boundary without filtering). This is structurally identical to one of the deepest open problems in particle physics — the **hierarchy problem** — which asks why gravity is so much weaker than the other forces. The standard answer is that gravity propagates in the full higher-dimensional bulk while other forces are confined to a lower-dimensional brane; the coupling across the brane boundary determines the apparent weakness. The soma-field correspondence is exact: the threshold T_i *is* the brane. Perception is confined to the one-dimensional boundary of an eleven-dimensional dynamics. The hierarchy of emotional experience — why conscious feeling is so much weaker and more transient than the underlying field activity — has the same formal structure as the hierarchy of forces.

Neural Network Energy Functions and Hopfield Networks

In 1982, John Hopfield (awarded the Nobel Prize in Physics in 2024) proposed a model of associative memory based on a network of interconnected neurons (Hopfield, 1982). The critical insight was borrowed directly from statistical physics: the network could be assigned an **energy function** — a scalar quantity that decreases with each state update — such that the network would always evolve toward a local energy minimum. These minima are the stable states of the network: its memories, or more precisely, its *attractors*.

Hopfield observed that his neural network's dynamics were mathematically identical to those of an Ising spin-glass model from condensed matter physics — a system of interacting magnetic spins that minimises its total energy by aligning or anti-aligning with neighbours. The energy function he used is:

$$H(\mathbf{s}) = -\frac{1}{2}\sum_{i,j} W_{ij}\, s_i s_j - \sum_i \theta_i s_i$$

where $\mathbf{s}$ is the state of the network, W_{ij} is the coupling strength between units i and j, and θ_i is the activation threshold of unit i. The network always moves in the direction of decreasing H.

The Soma-Field Model applies this energy function directly to emotional dynamics. The *emotional coupling matrix* W encodes the relationships between emotional modes — which emotions amplify one another, which suppress one another — and the energy function determines the direction in which the emotional field naturally evolves.

Hopfield's network is a formal model of the *neocortex*: a system for storing cognitive patterns and retrieving them from partial cues by minimising an energy function. Every artificial neural network constructed since McCulloch and Pitts (1943) (McCulloch & Pitts, 1943) — from perceptrons to backpropagation networks to transformers — sits in this neocortical lineage. These systems recognise patterns, predict sequences, and minimise prediction error with increasing sophistication. None of them possess a limbic system. They have no internal valuation, no arousal modulation, no threat-detection architecture, no attachment structure, no interoception. They have very effective cortex.

The Soma-Field Model does not add to the neocortical lineage. It proposes the architectural layer that has never been formally built: *an artificial limbic system*.

Hopfield memory is associative and pattern-completing; somatic memory is state-reinstating. The field does not merely remember what happened. It re-lives it. *A body with a past.*

Hopfield's later-reported wish to have incorporated something analogous to 'maternal instincts' into the energy function was, in this reading, not a desire for a better cortex. It was an intuition pointing directly at the absent system — the layer beneath the cortex that assigns value, registers threat, and holds the body in a particular way of being long after the event that caused it.

This positions the Soma-Field Model not as a supplement to the neocortical lineage but as its completion. Artificial neural networks have, for eighty years, been increasingly sophisticated formal models of the neocortex: pattern recognition, sequence prediction, error minimisation. The cortex has been mapped in extraordinary detail. The limbic system — which assigns value, detects threat, modulates arousal, maintains attachment, and reinstates whole somatic states in response to partial cues — has had no comparable formal treatment. The architectural description of the vertebrate brain was, until this paper, half-built.

Four kinds of formal intelligence. This architectural gap can be situated within a wider taxonomy. Four quotients have been proposed to describe the landscape of biological intelligence across popular and scientific usage. They map onto the formal components of this model with an exactness that is not coincidental:

Quotient	What it measures	Biological substrate	Soma-Field status
IQ — cognitive	Pattern recognition, reasoning, prediction	Neocortex	Built (1943–): McCulloch & Pitts → Hopfield → transformers
EQ — emotional	Valuation, arousal, affect regulation	Limbic system	**Built here**: W, $K(\tau)$, $H(\mathbf{e})$, C_{HRV}, $\dot{H}$
AQ — adversity	Structural resilience under threat	PFC–limbic axis	**Built here**: S_{inst}, $\partial\|W\|/\partial t$, $C_{\text{HRV}}^{\text{recovery}}$
SQ — social	Attunement, theory of mind, relational navigation	Mirror system, TPJ	*Next paper*: κ_r, multi-field coupling

Table 3. Four dimensions of biological intelligence mapped onto the Soma-Field Model. The neocortical lineage (IQ) has been formally modelled for eighty years. Emotional intelligence (EQ) and adversity resilience (AQ) are formalised here for the first time. Social intelligence (SQ) is defined as the next extension of the framework.

AQ — adversity quotient — is formally the capacity to update W after adversity without the adversity permanently becoming W. Its mathematical definition appears in Section 3.4; its pathological lower bound is C-PTSD, in which all three components of AQ are simultaneously compromised (Appendix B.2).

The AI alignment implication follows directly. Current artificial systems have high IQ by construction and zero EQ, AQ, or SQ. The absence of internal valuation means that valuation must be injected externally — through reinforcement learning from human feedback (RLHF) and related techniques — which is structurally brittle for the same reason that a field with no limbic layer is brittle: the system has no internal stake in what it does. The Soma-Field formalisation specifies what that internal stake would look like, were it ever built.

A further lineage note is worth recording. Ramsauer et al. (2020) demonstrated that continuous-state modern Hopfield networks are mathematically equivalent to the self-attention mechanism in transformer language models. The softmax attention operation that drives contemporary large language models is a Hopfield retrieval step. The Soma-Field Model sits in this same energy-based lineage: the equations underlying associative memory, language understanding, and somatic trauma response are, at the appropriate level of abstraction, the same equations.

A historical irony completes the picture. String theory was not discovered as a theory of strings. In 1968, Gabriele Veneziano wrote down a scattering amplitude — a response function encoding how particles scatter — and only later did Nambu, Nielsen, and Susskind identify the string as whatever object produces that amplitude (Veneziano, 1968). The response function came before the thing. The Soma-Field Model recapitulates this historical order deliberately: the primary object is the eleven-dimensional coupling manifold; the string — the one-dimensional conscious percept — is what the manifold produces when probed. We retain Veneziano's discovery and decline to reify the string.

The Formal Correspondences: Where the Link Was Seen

The structural analogy between QFT and the Soma-Field Model is not merely conceptual. There are three places where equations from different disciplines become, after substituting the relevant quantities, literally the same functional form. The following sets them side by side. The point is not to impress with notation but to show exactly where the recognition happened — the moment when the same Greek letters appeared in the same positions in two fields that had no prior reason to be connected.

The same Hamiltonian: Ising spin model (condensed matter physics, 1920s) — Hopfield neural network (computational neuroscience, 1982) — Soma-Field Model:

$$H_{\text{Ising}}(\sigma) = -\frac{1}{2}\sum_{i,j} J_{ij}\,\sigma_i\,\sigma_j - \sum_i h_i\,\sigma_i$$

$$H_{\text{soma}}(\mathbf{e}) = -\frac{1}{2}\sum_{i,j} W_{ij}\,e_i\,e_j - \sum_i \theta_i\,e_i$$

Replace $J_{ij} \to W_{ij}$, $\sigma_i \to e_i$, $h_i \to \theta_i$: identical. The physicist, the neural network theorist, and the somatic clinician are computing the same energy function on different state spaces. The Hopfield 2024 Nobel Prize was awarded for discovering this identity between spin physics and neural computation; the Soma-Field Model extends the same identity one step further to emotional dynamics.

The Wick rotation — why the same exponential appears in QM and in memory:

In quantum mechanics, the time evolution operator is a complex phase:

$$U(t) = e^{-i\hat{H}t/\hbar}$$

Substitute $t \to -i\tau$ (the *Wick rotation* — replacing real time with imaginary time):

$$e^{-i\hat{H}(-i\tau)/\hbar} = e^{-\hat{H}\tau/\hbar}$$

The oscillating complex exponential becomes a real decaying exponential. This is the Boltzmann weight $e^{-\beta\hat{H}}$ at $\beta = \tau/\hbar$. The Langevin equation $\dot{\mathbf{e}} = -\nabla H + \eta$ is the classical limit of this Wick-rotated dynamics. Every simulation of the soma-field running this equation is, formally, a path integral in imaginary time.

The same propagator: Euclidean QFT (imaginary-time two-point correlator for a massive scalar field) — C-PTSD trauma memory kernel:

$$G_E(\tau) = \langle \phi(0)\, \phi(\tau) \rangle_{\text{QFT}} = \frac{1}{2m}\, e^{-m|\tau|}$$

$$K_{\text{trauma}}(\tau) = \sum_k A_k\, e^{-|\tau|/\tau_k}$$

Same form. The QFT field mass m corresponds to $1/\tau_k$ — the reciprocal of the trauma trace decay time. A heavier particle has a shorter-range propagator; a shorter-lived trauma trace decays faster. Therapeutic processing (reducing A_k, increasing τ_k) is, in the QFT language, changing the mass and amplitude of the propagator until the correlation function vanishes.

The specific visual moment: the quantum phase factor is $e^{-i\omega t}$. Remove the i (Wick rotation) and it becomes $e^{-\omega\tau}$. The memory kernel is $e^{-\tau/\tau_k}$. These are the same exponential. The i is the only difference between a quantum field that oscillates and a trauma trace that decays.

QFT quantity	Symbol	Soma-Field analogue	Symbol
Field mode	ϕ_k	Emotional mode	e_i
Coupling constant	J_{ij}	Coupling matrix entry	W_{ij}

QFT quantity	Symbol	Soma-Field analogue	Symbol
Field mass	m	Inverse decay time	$1/\tau_k$
Propagator amplitude	$1/2m$	Trauma trace amplitude	A_k
Euclidean propagator	$G_E(\tau) \propto e^{-m\tau}$	Memory kernel	$K(\tau) \propto e^{-\tau/\tau_k}$
Vacuum energy	$\langle H \rangle_0$	Resting field energy	$H(\mathbf{e}_{\text{calm}})$
Thermal fluctuation	$k_B T$	Noise amplitude	σ_0
Wick rotation	$t \to -i\tau$	Real-time Langevin	$\dot{\mathbf{e}} = -\nabla H + \eta$

Table 2. Formal correspondence between QFT quantities and Soma-Field analogues. Each row is a single mathematical entity in two notations. These correspondences were not constructed after the fact; they are the reason the QFT framework was recognised as relevant.

The central identification — particle and percept as poles in their respective propagators. All four correspondences above follow from one structural fact. In QFT, a particle is not a separate object from the field. It is a *pole* in the field's propagator — the Green's function evaluated in momentum space:

$$\tilde{G}_{\text{QFT}}(k^\mu) = \frac{i}{k^2 - m^2 + i\varepsilon}$$

The particle exists precisely when the four-momentum satisfies $k^2 = m^2$ — the *on-shell condition*. The particle is the singularity in the field's response to a point source: the field's Green's function, evaluated at its own resonance.

Diagonalise W with eigenvalues λ_i (the natural resonance frequencies of the emotional modes). The soma-field propagator — the two-point correlator $\langle e_i(t)\, e_i(t') \rangle$ in the frequency domain — is:

$$\tilde{G}_{ii}(\omega) = \frac{\sigma_{\text{eff}}^2}{\omega^2 + \lambda_i^2}$$

A conscious emotional percept in mode i exists precisely when the excitation frequency ω approaches $i\lambda_i$ — the mode's natural resonance. The percept is the singularity in the soma-field's response to a somatic probe.

Setting the two propagators side by side:

$$\frac{i}{\underbrace{k^2 - m^2 + i\varepsilon}_{\text{QFT: particle at mass-shell } k^2 = m^2}} \quad \longleftrightarrow \quad \frac{\sigma_{\text{eff}}^2}{\underbrace{\omega^2 + \lambda_i^2}_{\text{Soma-Field: percept at resonance } \omega = i\lambda_i}}$$

Both are poles in the propagator of their respective field manifold. A photon is not the electromagnetic field; it is the field's Green's function evaluated at a resonance. A flash of conscious emotion is not the soma-field; it is the field's Green's function evaluated at a threshold-crossing resonance. The manifolds differ — one is the four-dimensional spacetime vacuum, the other is the eleven-dimensional emotional coupling geometry. The mathematical type is the same. This is not analogy.

The Body Schema, Interoception, and Pain

A complete model of the emotional field must address a phenomenon that standard psychological accounts of emotion consistently underspecify: the field is not a model of the physical body. It is the nervous system's *predictive model* of the body — a continuously updated internal representation of what the soma should be experiencing, revised by incoming interoceptive signals.

The clinical proof of this distinction is phantom limb pain (Ramachandran & Hirstein, 1998). Patients who have undergone amputation routinely experience pain in the absent limb. The pain is real: it activates the same neural circuits, produces the same suffering, and responds to the same analgesics as pain from an intact limb. The limb is gone. The neural model of the limb persists. What hurts is the *brain's representation* of the foot, not the foot.

This is not an anomaly. It is the normal condition of all somatic experience. The brain does not receive raw signals from the body — it maintains a continuous predictive model of the body (the *body schema*) and generates somatic experience from that model. Interoception — the sense of the internal body state — is a prediction, not a direct readout (Seth, 2021). The brain predicts what the heart should be doing, what the gut should feel like, where tension should be. The felt body is the predicted body.

The formal consequence is direct: the soma-field's state vector $\mathbf{e}(t)$ must include **somatic modes** — pain states, regional tension, visceral sensation, proprioceptive activation — alongside emotional modes. These are modes of the same field, governed by the same coupling matrix W. The W_{ij} between fear modes and somatic pain modes is the formal account of why fear amplifies pain, why safety reduces it, and why chronic pain and C-PTSD are highly comorbid. They are not separate conditions sharing a correlation. They are the same attractor architecture operating across emotional and somatic modes simultaneously.

Phantom limb as attractor persistence. An amputated limb's somatic modes do not disappear from W when the limb is removed. The neural model persists. When movement- intention modes are activated — attempting to move the absent foot — foot-sensation modes are co-activated via W. If co-activation exceeds threshold, it is experienced as pain. Ramachandran's mirror box provides visual input that disconfirms the prediction error: new sensory evidence that the limb is moving, reducing coupling-driven co-activation, and therefore reducing the pain. This is $W \to W'$: therapy as structural rewriting of the field.

The load-bearing hyphen. The term *emotional-somatic* in clinical literature is not a stylistic compound. The hyphen marks an ontological claim: emotional states and somatic states are not two separate things that correlate. They are two aspects of the same field. The coupling matrix W is precisely the hyphen, made formal.

Therapeutic implication. Somatic therapies — body scanning, sensorimotor work, EMDR's bilateral stimulation — work not on the physical body but on the brain's model of the body. They provide new interoceptive evidence that updates the prediction. They change W. Therapy does not fix the tissue. It updates the model.

Correspondence with Existing Emotion Representations

A reasonable objection to any new framework is: *there is already a great deal of structure out here.* This is true. The emotion research literature contains several well-developed representational systems, and the Soma-Field Model must be positioned relative to them. The short answer is that every existing representation is *descriptive*; the Soma-Field Model is *dynamical*. The longer answer follows.

Categorical taxonomies (Ekman 1972; Plutchik 1980; Parrot 2001) assign names and hierarchical membership to emotional states. They are ontologies in the formal sense: a T-Box of classes and subclass relations. Plutchik's wheel additionally defines a *blend* operation — Love := Joy ⊓ Trust, Awe := Fear ⊓ Surprise — which is precisely the OWL2 `intersectionOf` construction. These systems tell you what to call a state. They do not tell you how a state evolves, or which attractor a system settles in when two mechanisms fire simultaneously.

Dimensional models (Russell 1980; Mehrabian and Russell 1974) embed emotions in a continuous space, canonically Valence × Arousal (the *circumplex*), sometimes extended to Pleasure × Arousal × Dominance. These models capture the *coordinates* of a state. The energy landscape of the Soma-Field Model — the function $H(\mathbf{e})$ over emotion-space — is the dynamical generalisation of the circumplex: the circumplex is a snapshot of positions; the energy landscape is the surface over which the field moves. The stable attractors of H are the emotion categories; their coordinates are the circumplex positions.

Process and appraisal models (Scherer 1999; Frijda 1986; the OCC model of Ortony, Clove and Collins 1988) describe the *sequence of evaluations* through which a stimulus becomes an emotion. They are closer to the Soma-Field dynamics — they include temporal stages — but they are deterministic and single-threaded: one appraisal chain, one output. The Soma-Field replaces this with a parallel field update: all modes evolve simultaneously, governed by the full W matrix.

Music-specific schemas (BRECVEMA, Juslin and Västfjäll 2008; Juslin *et al.* 2011; GEMS, Zentner *et al.* 2008) are the closest antecedents to the present model. The BRECVEMA framework identifies eight distinct psychological mechanisms through which music evokes emotion — Brain stem reflex, Rhythmic entrainment, Evaluative conditioning, Contagion, Visual imagery, Episodic memory, Musical expectancy, Aesthetic judgement — each with distinct evolutionary origins, processing speeds, and neural substrates. These mechanisms are the *object properties* of the emotion-induction ontology: they specify which musical features activate which emotional outputs. Juslin explicitly identifies the open problem: *"Exploring how various musical emotions come about through the interaction of multiple psychological mechanisms is an exciting endeavour that has just begun"* (Juslin & Sloboda, 2011, p. 638). The W coupling matrix is the formal answer to that open problem. Where BRECVEMA gives a list of mechanisms with characteristic outputs, the Soma-Field gives the interaction tensor W_{ij} that specifies, with numerical precision, what happens when mechanisms i and j fire concurrently.

The deeper connection is spectral. The *eigenmodes* of W — the directions in emotion-space that evolve independently — are the natural resonances of the soma-field: the patterns the field rings with when struck. BRECVEMA mechanisms are inputs: they excite specific rows of W. The eigenspectrum of W is the response: the set of frequencies the manifold can sustain. Where BRECVEMA is a taxonomy of *stimuli*, the eigenspectrum of W is a taxonomy of *responses*. Juslin's open problem — how mechanisms interact — is the question of how stimulus-space maps onto eigenmode-space through W. Section 3.3 develops this.

Body maps (Nummenmaa *et al.* 2014) map emotions to their somatic distribution — where in the body each emotion is felt. These are precisely the spatial support of the soma-field modes: the field configuration corresponding to an attractor state is the body map of that emotion. Body maps are measurements of the attractors; the Soma-Field is the dynamical system that generates them.

The formal correspondence table extends Table 2 to include these systems:

Existing representation	What it captures	Soma-Field equivalent
Ekman categories	Attractor labels (names)	Values of $\mathbf{e}$ at energy minima
Plutchik dyads ($A \sqcap B$)	Blend attractors	Metastable states between two energy minima
Russell circumplex	Coordinates (valence, arousal)	Projection of $H(\mathbf{e})$ onto two axes
OCC appraisal tree	Single-path sequential process	Single trajectory in the full field
BRECVEMA mechanisms	Object properties: stimulus → emotion	Rows of W: mechanism i activates mode j
Body maps (Nummenmaa)	Spatial support of each attractor	Modal structure of $\mathbf{e}$ at each minimum

None of these correspondences require modifying either the existing representations or the Soma-Field Model. They are consequences of the model's structure. The formal machinery for exploring these correspondences — typing BRECVEMA mechanisms as Lean inductive constructors, Plutchik blends as type intersections, mechanism profiles as decidable propositions — is developed in the companion file `src/EmotionOntology.lean`.

The Soma-Field Model

The field is primary. The felt emotion is secondary — it is what registers when the field is probed. This is the same ontological relationship as between a quantum field and a particle: the field exists continuously and everywhere; the particle is what you observe at the moment of measurement. The Soma-Field Model does not describe what emotions are *made of*. It describes the manifold whose impulse response *is* conscious emotional experience.

Emotions as a Persistent Wave Field

The foundational claim of the Soma-Field Model is simple: emotions are not events. They are a *field* — a distributed, continuous quantity defined over the entire soma (body-mind system) at all times.

This field has two coupled components:

1. **The somatic wave $\mathbf{E}_{\text{body}}(x,t)$**: distributed across the body as patterns of visceral sensation, muscle tone, proprioception, interoception, and autonomic state.
2. **The neural wave $\mathbf{E}_{\text{neural}}(x,t)$**: distributed across the nervous system as patterns of activation in cortical, subcortical, and peripheral neural circuits.

These two components are not separate systems. They are coupled — each continuously influencing the other. The total emotional field is their combined state:

$$\mathbf{E}(x,t) = \mathbf{E}_{\text{body}}(x,t) \otimes \mathbf{E}_{\text{neural}}(x,t)$$

The field is characterised by:

- **Multiplicity**: multiple emotional modes can be simultaneously active and interfering
- **Continuity**: it exists at all times, not only during episodes of conscious feeling
- **Spatial distribution**: different aspects of the field are localised in different regions of the soma (the familiar clinical observation that grief is felt in the chest, fear in the gut, anger in the jaw and fists)
- **Temporal dynamics**: the field evolves continuously, driven by the energy function

Figure 1. The Soma-Field. The body and brain are not separate containers of emotion but two coupled components of a single distributed wave field. Neither is primary; each continuously modifies the other. The ≈ symbols indicate that wave activity is always present in each region, not only during episodes of conscious feeling.

The Perception Threshold

Not all activity in the emotional field is consciously perceived. The field has a **perception threshold** T_i for each emotional mode i. Below this threshold, the emotional mode is sub-perceptual: it exists, it influences behaviour and physiology, but it does not surface as a named conscious feeling.

$$\text{Emotion } i \text{ is consciously perceived} \iff |\mathbf{E}_i(t)| > T_i$$

This threshold crossing corresponds precisely to the QFT excitation analogy: the emotional mode behaves like a virtual particle that has accumulated enough energy to become real — to emerge from the sub-threshold background and enter awareness.

This accounts for a range of clinically significant phenomena:

Clinical Observation	Soma-Field Account
Patient reports no feeling but shows physiological signs of distress	Sub-threshold field activity below T_i
Sudden unexpected flood of emotion in session	Rapid threshold crossing after gradual accumulation
Emotion felt somatically but not named	Threshold crossed in $\mathbf{E}_{\text{body}}$, not yet in $\mathbf{E}_{\text{neural}}$
Alexithymia (difficulty identifying feelings)	Elevated T_i — high threshold requiring more energy to cross
Hypervigilance / emotional flooding	Lowered T_i — reduced threshold, field crosses to conscious easily

Table 1. Clinical observations mapped onto the perception threshold model.

Figure 2. The perception threshold T_i for a single emotional mode. The field is active continuously (lower trace). Conscious experience arises only when amplitude exceeds T_i (upper trace). Everything below the line is still there — shaping body and behaviour before it can be named.

Figure 0. Continuous soma-field activity (blue) with a single threshold-crossing event. The field is always active; conscious experience (shaded) arises only when amplitude exceeds the perception threshold θ (red dashed). Below the threshold: real, causally active, but not yet conscious.

The Interaction of Emotional Modes

Multiple emotional modes are simultaneously active in the field at all times. They do not simply co-exist: they interact. The nature of these interactions is encoded in the **emotional coupling matrix** W, where W_{ij} represents the influence of emotional mode j on emotional mode i.

- If $W_{ij} > 0$: emotion j amplifies emotion i (e.g., fear can amplify shame)
- If $W_{ij} < 0$: emotion j suppresses emotion i (e.g., calm suppresses anxiety)
- If $W_{ij} = 0$: emotions i and j are independent

The field evolves according to the energy gradient:

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}) + \eta(t)$$

where $\eta(t)$ represents the continuous low-level fluctuations of the sub-perceptual field — the emotional equivalent of quantum vacuum noise. The field is always moving, always seeking lower energy, never at absolute rest.

The Three-Layer Architecture

The nervous system that implements the soma-field is not architecturally flat. Three hierarchically organised layers contribute to field dynamics, each corresponding to a distinct evolutionary substrate and a distinct role in the model. The clinical literature (Porges, 2011; van der Kolk, 2014; Ogden et al., 2006) converges on this stratification; what follows is its formal expression.

Layer 1 — Brainstem / autonomic baseline. The oldest structures: vagal nuclei, arousal systems, interoceptive machinery. In the model, this layer is represented by the noise term and, specifically, by the heart rate variability coherence C_{HRV}, which modulates effective noise amplitude across the whole field:

$$\sigma_{\text{eff}} = \frac{\sigma_0}{C_{\text{HRV}}}$$

High HRV coherence narrows effective noise, stabilising the field in its current attractor. This is the mechanism of HRV biofeedback as a regulatory intervention: it does not target any specific emotional mode but lowers the fluctuation floor of the entire field.

Layer 1 extension: cardiac acceleration and landscape tilt. The term C_{HRV} measures the *current state* of cardiac regularity — where the heart is. A complementary quantity is $\dot{H}(t)$, the first time-derivative of heart rate, in units of beats/s^2. This is the **cardiac acceleration**: not what the heart rate is, but where it is going.

The dimensional parallel with gravity is exact: gravitational acceleration g carries units m/s^2; cardiac acceleration $\dot{H}$ carries units beats/s^2. Both are accelerations; both describe a force field rather than a position. Gravity does not tell you where a test mass is — it tells you how it will move next. Cardiac acceleration tells you not the current BPM but the direction of the next one: the N+1 state.

In the soma-field, $\dot{H}(t)$ enters the dynamics not as noise modulation but as a **landscape tilt** — a time-varying bias added to the Hamiltonian that tips the energy function toward activation or rest attractors:

$$H(\mathbf{e}, t) = H_0(\mathbf{e}) - \alpha\,\dot{H}(t)\,\beta \cdot \mathbf{e}$$

where $\alpha > 0$ is the cardiac-somatic coupling constant and β is a mode-coupling vector (at leading order, $\beta = \mathbf{1}$: the tilt acts uniformly across all modes). When $\dot{H}(t) > 0$ (heart accelerating), the landscape tilts toward higher activation states before any cognitive or affective threshold is crossed. When $\dot{H}(t) < 0$ (heart decelerating), it tilts toward rest. The full three-layer equation including the cardiac acceleration term is:

$$\dot{\mathbf{e}}(t) = -\nabla H_0(\mathbf{e}) + \alpha\,\dot{H}(t)\,\beta + R_{\text{PFC}}(\mathbf{e}, t) + \frac{\sigma_0}{C_{\text{HRV}}}\,\xi(t)$$

The two cardiac terms serve distinct functions: C_{HRV} (state) modulates the noise floor; $\dot{H}$ (acceleration) tilts the deterministic landscape. Both are needed for a complete account of cardiac influence on the field.

Predictive clinical value. A patient with BPM = 90 and $\dot{H} = +4$ beats/s^2 is approaching threshold; one with BPM = 90 and $\dot{H} = -4$ beats/s^2 is retreating from it. The snapshot is identical; the trajectories are opposite. Cardiac acceleration is therefore an early-warning signal for threshold crossings — detectable at Layer 1 before the emotional field at Layer 2 has crossed its threshold. This has independent support in cardiology: Bauer et al. (2006) demonstrated that *acceleration capacity* and *deceleration capacity* of heart rate — estimates of $\dot{H}$ over a cardiac window — carry prognostic information independent of conventional HRV measures.

The somatic equivalence principle. The cardiac acceleration term $\alpha\,\dot{H}\,\beta$ is structurally identical in the equation to any other forcing term. From the perspective of the field itself — from conscious experience — cardiac-driven activation is indistinguishable from event-driven activation. A sudden heart rate acceleration tilts the landscape by exactly the same mechanism as an external threat or an intrusive memory. The field has no access to the origin of the tilt. This is the formal account of a clinically well-documented phenomenon: anxiety initiated by cardiac irregularity (arrhythmia, postural hypotension, caffeine, exertion) is experienced as emotionally caused, because the somatic signal is identical. Disambiguation requires either external measurement or deliberate interoceptive inquiry that can distinguish the two sources.

Layer 2 — Limbic system / emotional memory. The primary substrate of the Soma-Field Model. The coupling matrix W, memory kernel $K(\tau)$, Hamiltonian $H(\mathbf{e})$, and threshold T all belong here. The limbic layer stores emotional-somatic states and reinstates them in response to partial body cues: a continuous, asymmetric, temporally extended Hopfield network operating on somatic states rather than cognitive patterns. This is the architectural layer that has been absent from every artificial neural network since McCulloch and Pitts (1943) (McCulloch & Pitts, 1943). The cortex has been modelled many times; the limbic system has not.

Structural plasticity under adversity. The Soma-Field framework permits a formal characterisation of the field's resilience under adverse conditions. Define the *plasticity index* Π as a composite of three measurable field properties:

$$\Pi \;=\; \frac{1}{S_{\text{inst}}} + \left.\frac{\partial \|W\|}{\partial t}\right|_{\text{adversity}} + C_{\text{HRV}}^{\text{recovery}}$$

The three terms correspond to: (i) how accessible regulated-state attractors remain under adversity ($1/S_{\text{inst}}$, instanton accessibility — Section 4.4); (ii) how much the coupling matrix can structurally adapt following a threshold crossing ($\partial\|W\|/\partial t$, the plasticity component); and (iii) how quickly the HRV floor recovers after activation ($C_{\text{HRV}}^{\text{recovery}}$, the regulatory resilience component). Complex PTSD is the clinical presentation of chronically low Π across all three terms simultaneously: high barriers to regulated attractors, a rigid W dominated by threat configurations, and impaired C_{HRV} recovery. Structural plasticity is the capacity of the field to update W in the aftermath of adversity without the adversity permanently *becoming* W.

Layer 3 — Neocortex / prefrontal regulatory layer. Top-down modulation of Layer 2, represented as a regulatory term $R_{\text{PFC}}(\mathbf{e}, t)$. The full field dynamics becomes:

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}(t)) + R_{\text{PFC}}(\mathbf{e}, t) + \frac{\sigma_0}{C_{\text{HRV}}}\,\xi(t)$$

R_{PFC} represents voluntary attention, therapeutic technique, and conscious reappraisal acting on the field. It is not a correction of Layer 2 but a modulation of it. Under sustained therapeutic engagement, R_{PFC} participates in the structural modification $W \to W'$ constituting the forward transformation (Section 7).

The **threshold T is the Layer 2 / Layer 3 boundary**: sub-threshold dynamics are processed limbically and remain below conscious awareness; threshold-crossing events enter Layer 3 and become available for narrative, meaning-making, and voluntary response. This is the formal basis for the clinical observation that insight without somatic activation is limited, and somatic activation without Layer 3 engagement cannot produce structural change: the layers are coupled, not independent. R_{PFC} requires a threshold crossing in order to have something to work with.

The two-term Langevin equation introduced in Section 3.3 is the Layer 2 special case ($R_{\text{PFC}} = 0$, $C_{\text{HRV}} = 1$). All subsequent sections develop that special case. The full three-layer equation is the general form.

The Energy Landscape

The Structure of the Emotional Energy Function

The energy function $H(\mathbf{e})$ defines a landscape over the space of possible emotional states. Like a physical landscape of hills and valleys, this landscape has:

- **Valleys (local minima)**: stable emotional states the field naturally moves toward
- **Hills (local maxima)**: unstable configurations the field naturally moves away from
- **Saddle points**: transitional configurations with mixed stability

The key property of an energy function is directionality: the field always moves *downhill*. It always evolves toward lower energy. Therapeutic intervention, in this framework, can be understood as:

1. **Changing the landscape**: modifying W — the coupling matrix — through new relational experience, insight, or somatic work, so that the energy minima are in healthier locations
2. **Adding energy to escape a trap**: helping the field accumulate enough energy to escape a deep but unhealthy local minimum (e.g., the freeze state)
3. **Pointing toward the global minimum**: orienting the field toward regulated calm

Attractor States: Fight, Flight, Freeze, and Regulated Calm

The Soma-Field Model proposes that the major attractor basins of the emotional energy landscape correspond directly to the autonomic states described by Porges' polyvagal theory.

Figure 3a. Topographic (bird's-eye) view of the energy landscape. The field always rolls downhill toward the nearest minimum. Freeze and calm are both low-energy — but freeze is surrounded by high walls. Escape from freeze requires crossing those walls, which means first gaining energy before losing it again. This is the clinical challenge of working with dissociative states.

```
ENERGY
   |
H  |        fight/flight
   |        ┌──┐  ┌──┐
   |        |  |  |  |
   |   _____|  |  |  |_____
   |  |        \/         |
   |  |      saddle        |
   |  |    (transition)    |
```

```
│  │               │   ╔══════════╗
│  │    freeze     │   ║          ║
│  │     ┌─┐       │   ║ regulated║◄── global minimum
│  │_____│ │_______│   ║   calm   ║
│        │             ╚══════════╝
└──────────────────────► EMOTIONAL STATE SPACE
```

Figure 3b. Schematic energy landscape. Fight/flight are high-energy, unstable local minima. Freeze is a low-energy but isolated attractor — easy to enter, hard to escape. Regulated calm is the global energy minimum.

Attractor	Energy State	Polyvagal Correlate	Clinical Presentation
Regulated Calm	Global minimum	Ventral vagal (social engagement)	Present, flexible, connected
Fight	Shallow high-energy minimum	Sympathetic (mobilisation)	Agitation, anger, urgency
Flight	Saddle point / shallow minimum	Sympathetic (mobilisation)	Anxiety, avoidance, rumination
Freeze	Deep isolated minimum	Dorsal vagal (immobilisation)	Dissociation, numbness, collapse

Table 2. Emotional attractors mapped onto Polyvagal states.

The therapeutic significance of this structure is considerable. The freeze state is dangerous not because it is high-energy — it is in fact very low energy — but because it is *isolated*: surrounded by energy barriers that make it difficult to exit. Escape from freeze requires first *increasing* the field's energy (mobilising some arousal) before it can flow toward regulated calm. This corresponds well to the clinical observation that working with dissociated patients requires careful titration of arousal — not too much, not too little — before emotional processing is possible.

The Coupling Matrix as a Personal Signature

The coupling matrix W is not universal. Each person has a unique W, shaped by attachment history, trauma, cultural context, and temperament. A person with a history of developmental trauma may have a W in which anxiety and shame are strongly coupled ($W_{\text{shame, anxiety}} \gg 0$), creating a combined attractor that is particularly deep and sticky. A person with a secure attachment history may have a W in which positive emotions are broadly coupled to one another, creating a wide basin around regulated calm.

This implies that the energy landscape is a therapeutic object in its own right: understanding a patient's W is understanding the structural dynamics of their emotional life.

In the M-theory compactification analogy developed in Appendix A, the coupling topology W corresponds to the shape of the compact G_2 manifold — the seven-dimensional geometry that determines which force-like couplings are allowed and with what strengths. That analogy is here made precise: two people differ not merely in their emotional *parameter settings* but in their coupling *geometry*. Developmental trauma does not set a dial to the wrong value; it deforms the manifold. The therapeutic process of modifying W through relational experience, insight, or somatic work is, in this language, differential geometry: a continuous deformation of the G_2 manifold toward a configuration in which the regulated-calm attractor is globally accessible. The practitioner is, without having been told so, a geometer.

Dissonance and Resolution

The Acoustic Analogy

The Soma-Field Model draws a further structural analogy, this time with acoustics. When two sound waves interact, the quality of their interaction — consonance or dissonance — depends on the phase relationship between them. Consonant intervals (the octave, the fifth) have simple frequency ratios and produce stable, reinforcing interference patterns. Dissonant intervals (the tritone, the minor second) have complex ratios and produce beating, unstable, tension-generating patterns.

The model proposes that the same relationship holds between emotional modes. When two emotional modes are in a compatible relationship — when their interaction is consonant — the field is in a relatively low-energy configuration and moves naturally toward the energy minimum. When they are in an incompatible relationship — when their interaction is dissonant — the field is in a higher-energy configuration, generating a gradient that drives toward resolution.

Dissonance, in this framework, is felt as tension. It is not pathological; it is directional. Dissonance is the field's way of communicating that it is far from equilibrium and that resolution is available.

The Resolution Principle

In music, dissonance resolves to consonance. The tritone — the most dissonant interval in Western tonality — creates a powerful gravitational pull toward resolution. In counterpoint, the rules of voice leading describe the specific paths by which dissonance must resolve. These rules are not arbitrary conventions; they describe the geometry of the acoustic energy landscape.

The same principle applies to emotional dissonance. An unresolved emotional state — grief that has not been fully experienced, anger that has been suppressed, fear that has been dissociated — is a dissonance in the field. It generates a persistent tension gradient. The therapeutic process can be understood as guided voice leading: finding the specific path of resolution that transforms the dissonant configuration into a consonant one.

This provides a formal basis for a widely-held clinical intuition: that emotions need to be *felt through* rather than avoided. Avoidance keeps the field in a dissonant state. The energy minimum — regulated calm — lies on the other side of the dissonance, not around it.

The Soma-Field Instrument

Rationale

The Soma-Field Model is not only a theoretical framework. It motivates a practical therapeutic instrument: a means by which a person can *externalise* their emotional field — make it visible and audible — and interact with it in real time.

The core insight is that the emotional field is normally invisible to its host. It operates below the threshold of conscious awareness, shaping behaviour and physiology without being available for reflection. If its activity could be rendered as a signal — a sound, an image, a pattern — it could become an object of therapeutic attention.

Design

The instrument uses a MIDI controller with 16 rotary knobs as its input interface. Eight emotional dimensions are encoded, each represented by two knobs:

- **Knob 1** of each pair: the somatic (body-level) intensity of that emotional mode
- **Knob 2** of each pair: the cognitive/neural intensity of that emotional mode

This design reflects the two-component structure of the field: body and mind are encoded separately but coupled in the computation. Each knob has a continuous range, allowing fine expression of emotional intensity.

```
┌──────────────────────────────────────┐
│         MIDI CONTROLLER              │
│                                      │
│  [K1][K2]  [K3][K4]  [K5][K6]  [K7][K8]  │
│  emotion1  emotion2  emotion3  emotion4│
│                                      │
│  [K9][K10] [K11][K12][K13][K14][K15][K16] │
│  emotion5  emotion6  emotion7  emotion8│
└──────────────────────────────────────┘
                  │
                  ▼
       ┌───────────────────┐
       │  ENERGY FUNCTION   │
```

```
          │  H(e) computed     │
          │  ∇H(e) computed    │
          └────────────────────┘
                     │
     ┌───────────────┼───────────────┐
     ▼               ▼               ▼
 AUDIO OUTPUT    MIDI OUTPUT    VISUAL OUTPUT
(timbre reflects (pitch/velocity (field map:
 dissonance)     reflects energy) wave topology)
```

Figure 4. The Soma-Field Instrument: input, computation, and multimodal output.

The Feedback Loop

The instrument creates a **closed feedback loop** between the person and their emotional field:

1. The person expresses their current emotional state by adjusting the knobs
2. The system computes the energy function $H(\mathbf{e})$ and its gradient ∇H
3. The energy level, dissonance, and proximity to attractor states are rendered as:
   - **Sound**: harmonic content and timbre reflect the consonance or dissonance of the current state
   - **MIDI output**: pitch rises with tension, resolves as energy decreases
   - **Visual**: a real-time map of the emotional field, showing wave activity, threshold crossings, and the direction of the energy gradient
4. The person hears and sees their emotional field, and adjusts the knobs in response

This loop externalises the emotional field's gradient — the direction in which it is *trying* to move — and makes it available as sensory information. The person becomes not only the source of the emotional signal but also its observer, creating the conditions for reflection and regulation that are at the heart of therapeutic work.

The Pluggable Emotion Model

No single model of the emotions is assumed. The coupling matrix W — the structure that determines how emotional modes interact — is loaded from an external configuration file. Standard models (Plutchik's wheel of emotions, Ekman's basic emotions, the valence-arousal-dominance dimensional model) are provided as defaults. The therapist or client can modify the coupling values to reflect their own understanding of their emotional patterns, or a new model can be substituted entirely. The computational engine is model-agnostic.

Clinical Implications

Assessment

The Soma-Field Model suggests a different orientation for emotional assessment. Rather than asking "What emotion do you feel?" — which presupposes threshold-level conscious awareness — it invites attention to the sub-perceptual field: "What is present in the body right now, even if you cannot name it?" This aligns with Focussing-oriented approaches and with sensorimotor methods that prioritise somatic signal over narrative content.

The energy landscape provides a clinical map. A person chronically in a fight or flight attractor shows a different energy signature from a person in a freeze attractor, even if their presenting narratives are superficially similar. The model suggests that these are structurally different therapeutic challenges: fight/flight require down-regulation, while freeze may first require careful up-regulation before down-regulation becomes possible.

Intervention

The energy function provides a formal basis for several existing clinical interventions:

- **Grounding and titration** (Levine, 2010): adding small, controlled amounts of energy to the field to approach — without flooding — a previously frozen or avoided emotional state
- **Pendulation** (Levine, 2010): oscillating between a dissonant state and a resource state, progressively widening the tolerance window — equivalent to approaching the energy minimum via a series of small excursions
- **Somatic resourcing** (Ogden et al., 2006): establishing a stable low-energy region in the landscape that the field can return to after excursions into high-energy territory
- **Working with the felt sense** (Gendlin, 1978): attending to sub-threshold field activity and allowing it to cross the perception threshold in a supported context

Psychoeducation

The wave model is immediately accessible to clients who have struggled to understand their emotional experience. The statement: *"Your emotions are like waves — they are always there, even when you can't feel them, and they are always moving"* is both technically accurate within the Soma-Field framework and clinically useful as a normalising frame for sub-threshold emotional activity, for the apparently sudden onset of strong feelings, and for the experience of feeling multiple conflicting emotions simultaneously.

The energy landscape metaphor — *"right now the field is in a valley that is hard to leave, but it is not the lowest valley available to you"* — offers a way to discuss the freeze state, dissociation, and emotional stuckness without pathologising, while still acknowledging the structural difficulty of these states and the work required to shift them.

Neurodivergent Conditions as Operator Modifications

A clinically significant extension of the Soma-Field Model concerns neurodivergent conditions — specifically Autism Spectrum Condition (ASC), Attention Deficit Hyperactivity Disorder (ADHD), and Complex Post-Traumatic Stress Disorder (C-PTSD), which frequently co-occur and each present distinct challenges for somatic emotional processing.

The key architectural principle is this: **these conditions are not parameter settings in the model. They are structural modifications to the operators themselves.** This distinction matters both mathematically and clinically. A parameter change ("set the fear threshold lower") is a quantitative adjustment within the existing structure. An operator modification changes the *form* of the dynamics — it alters the governing equations, not merely their coefficients. Each condition wraps the standard pipeline in a different functional modifier, and — critically for the many individuals who carry all three — these modifiers *compose*. The combined condition is not three separate problems; it is the composition of three operators acting on the same underlying field.

The mathematical details of each modifier are given in Appendix B. Clinically, the consequences are as follows.

Complex PTSD introduces a *memory kernel* into the field dynamics: past high-energy states leave decaying echoes that continue to excite the field without new external stimulus. This is why traumatic activation can appear without identifiable trigger — the field is responding to its own history, not its current environment. The standard Hopfield attractor topology is also disrupted: C-PTSD renders the freeze attractor pathologically deep and wide, the window of tolerance (the basin around regulated calm) pathologically narrow, and the coupling matrix asymmetric — a condition under which the field can enter persistent *limit cycles* rather than settling to a stable minimum. Re-experiencing, flashbacks, and hypervigilance are, in this framework, limit-cycle oscillations in the traumatised field.

```
REGULATED FIELD  (symmetric W, no memory kernel)
─────────────────────────────────────────────────────────────────────────────

|              ┌───────┐                    ┌────────────────┐
|    ┌──┐     /         \         ┌──┐     /                  \      ┌─
|   /    \   /           \  ┌┐   /    \   /                   \    /
T ─/──────\─/─────────────\┘─└/──────\─/──────────────────────\──/─ T
|          \               └──┘        \                        \──┘
|           └────────────────────────────────────────────────────────────
└──────────────────────────────────────────────────────────────────────► t
```

```
    ↑ baseline returns to near-zero between episodes
    ↑ each threshold crossing is a discrete, independent event
    ↑ 'regulated calm' is a genuine resting state — the global energy minimum


  C-PTSD MODIFIED FIELD  (asymmetric W, memory kernel K(t-s) present)
  ──────────────────────────────────────────────────────────────────────────────

   │┌──────────┐          ┌────────┐          ┌─────────────────────────┐
  T│           \  ┌──┐  /           \  ┌──┐  /                          ─ T
   ││           \/    \/             \/    \/
   ││  ← even the troughs stay near T or above: baseline is elevated
   └────────────────────────────────────────────────────────────────────► t
    ↑ memory kernel: each activation feeds energy back into the next
    ↑ field rarely returns to true rest — past states re-enter present dynamics
    ↑ almost entirely above T: activation is the default, not the exception
    ↑ 'regulated calm' requires a non-perturbative transition (the instanton):
      small steps do not reach it; a qualitatively different move is needed
```

Figure 5. The same emotional field mode under two dynamic regimes. Top: regulated dynamics — the field oscillates and returns to a low baseline between episodes; conscious emotion (above T) is episodic and resolves. Bottom: C-PTSD-modified dynamics — the memory kernel elevates the baseline so that the field rarely returns to rest; episodes bleed into one another; the system cycles rather than settles. The mathematical basis for this comparison is given in Appendix B.2.

Developmental timing and what can be recovered. The character of the C-PTSD modification depends critically on *when* it occurred — the developmental age τ_d at which the primary traumatic modification took place.

For **late trauma** (τ_d large — adult or post-verbal): a coupling matrix W_0 formed before the event. The modification is additive: $W = W_0 + \delta W_{\text{trauma}}$. A counterfactual pre-trauma self exists, encoded in explicit narrative memory. Therapeutic processing can target δW specifically, and the goal of recovering proximity to W_0 is formally coherent.

For **early trauma** (τ_d small — pre-verbal, perinatal): the coupling matrix W was *formed under the modification*. There is no W_0. The asymmetric coupling and the memory kernel coefficients are the baseline architecture, not additions to one. A counterfactual pre-trauma self was never encoded — it does not exist as a recoverable state.

This is a formal statement of a clinical fact that somatic therapists recognise but rarely have a mechanistic basis for: early trauma cannot be *processed away* in the sense of recovering a prior self, because no prior self was formed. The therapeutic goal is not subtraction ($W \to W_0$, which is undefined) but **forward transformation**: constructing a W' that supports a wider window of tolerance, different attractor topology, and lower memory-kernel amplitudes. This is a different mathematical operation — and requires a different therapeutic model.

The Soma-Field Instrument can reflect this distinction directly: a user whose primary modification is pre-verbal initialises with a *structural* coupling matrix (the modification *is* the baseline), not a neurotypical matrix with an added modifier. The formal basis for this parameterisation is given in Appendix B.2.1.

ADHD raises the effective *thermal noise* of the field — the amplitude of the sub-perceptual fluctuations — and simultaneously reduces the damping coefficient that slows the field's response to the energy gradient. The result is a field that explores its energy landscape rapidly and unpredictably, is easily displaced from shallow attractor basins by small perturbations (distractibility), but also achieves states of intense concentration (hyperfocus) when the coupling to a high-salience stimulus temporarily deepens a specific attractor basin far beyond its resting depth. ADHD is not a deficit of attention; it is a high-temperature, low-damping emotional field with a stimulus-dependent attractor structure.

Autism Spectrum Condition modifies the *projection kernels* — the functions that determine how the continuous somatic field is sampled to produce the discrete state vector — and the *sparsity* of the coupling matrix. Interoceptive research in autism (Garfinkel et al., 2016) documents significant differences in the processing of internal body signals; in model terms, certain somatic regions are over-represented (heightened sensory sensitivity) and others under-represented (reduced interoceptive clarity, contributing to alexithymia). The coupling matrix in ASC tends toward greater sparsity — fewer strong cross-modal emotional couplings — a pattern consistent with monotropism (Murray, 2018): the field settles deeply into individual attractors but transitions between them require proportionally more energy. Intense interests, emotional consistency within a context, and difficulty with unexpected transitions all follow from this attractor topology.

For the Soma-Field Instrument, the practical implication is significant. Rather than asking a neurodivergent user to configure their experience through knob adjustments, the system can instantiate the appropriate operator modifications as a named profile — *"load C-PTSD modifier"*, *"load ADHD modifier"* — each of which transforms the pipeline at the correct mathematical level. The user then interacts with a field that already reflects their structural reality, rather than one calibrated for a neurotypical baseline.

A further clinical implication deserves explicit statement. The Soma-Field Model locates interoceptive accuracy in the field itself: whether a somatic signal has exceeded its perceptual threshold T_i is a property of the field state, not a property of the clinician's assessment of the patient's credibility. A patient reporting an acute somatic state is reporting a threshold-crossing event. The model provides no mechanism by which external disbelief suppresses that crossing. Modified projection operators — as occur in ASC — produce *different* somatic self-reports; the model gives no reason to assume they produce *less accurate* ones. The clinical literature documents a systematic tendency to interpret unusual interoceptive self-reports from neurodivergent patients as indicative of psychogenic origin rather than genuine somatic signal (Nicolaidis et al., 2015). The Soma-Field Model predicts that this interpretive pattern constitutes a category error: it confuses operator modification with signal absence. The practical consequences — missed diagnoses, deferred treatment, and the iatrogenic reinforcement of existing trauma — are well-documented and, within this framework, mathematically predictable.

Limitations and Future Directions

The Soma-Field Model is a theoretical framework and must be evaluated as such. Its current form makes several idealisations that require scrutiny.

The coupling matrix W is treated as a fixed parameter, but emotional coupling is dynamic: it changes with context, relationship, and developmental history. A more complete model would treat W as a slowly-evolving quantity, shaped by the field's own history — a form of synaptic plasticity applied to the emotional domain.

The threshold T_i is treated as a fixed property of each emotional mode, but experimental evidence suggests that thresholds are modulated by attentional focus, arousal level, and interpersonal context. A person in a safe therapeutic relationship will typically have lower thresholds — more material reaches conscious awareness — than the same person in an unsafe context.

The acoustic analogy, while structurally productive, requires empirical grounding. The claim that emotional dissonance and acoustic dissonance share formal properties is a hypothesis, not an established finding. Empirical work comparing physiological measures of emotional tension with acoustic analysis of synchronised vocal or musical output would be a productive direction for testing this claim.

The instrument described in Section 6 is a prototype concept. User studies with clinical populations, and collaboration with practising therapists, will be required to assess its therapeutic utility and to identify appropriate clinical contexts.

Future theoretical work should address the relational field: the observation, familiar in systemic and relational approaches to psychotherapy, that emotional fields are not bounded by individual bodies but are co-generated in the space between people. The coupling matrix W of a relationship may be as clinically significant as the W of an individual.

Axiomatic QFT status (update, 2026). A subsequent paper in this series (P14, *The Universal Somatic Field as a Euclidean Quantum Field Theory*) proves that the free-field USF satisfies all five Osterwalder–Schrader axioms, placing it within the rigorous framework of constructive quantum field theory. The proof is machine-verified in Lean 4 with zero sorries. Reflection positivity (OS3) guarantees the legitimacy of the Minkowski continuation proved in the temporal-dynamics companion paper. The interacting (Hopfield-coupled) theory is addressed in P15.

Conclusion

The Soma-Field Model proposes a formally grounded account of emotional dynamics that is consistent with the clinical observations of somatic psychotherapy, polyvagal theory, and Focussing-oriented practice. Its central claims — that emotions are a persistent distributed field, that conscious experience is a threshold crossing, and that emotional dynamics are governed by an energy function that drives the field toward stable attractor states — are not novel as clinical intuitions. What is novel is the formal structure that unifies them, and the instrument that the structure motivates.

The model does not resolve the philosophical question of what emotions fundamentally *are*. It offers instead a working representation: one that is precise enough to be computationally implemented, close enough to existing clinical frameworks to be therapeutically applicable, and open enough to be modified as understanding deepens. It invites the therapist to think of the consulting room as a space in which two emotional fields interact — each shaping the other's energy landscape — and of therapeutic work as the art of attending to that interaction with enough precision and care to guide both fields toward lower energy, toward greater coherence, toward regulated calm.

The wave is always there. Therapy is learning to listen to it.

A note on provenance. The Soma-Field Model was not developed from a position of theoretical neutrality. The author carries, as primary data, a lifetime of direct experience of the dynamics described above. The neurodivergent operator modifications of Appendix B are not theoretical abstractions: the C-PTSD memory kernel of B.2 was installed pre-verbally, at approximately eighteen months of age, during a developmental trauma that predates language acquisition entirely. No narrative trace of the origin event exists — there was no verbal capacity with which to encode one. Only the field echo remains, and a measurable physical asymmetry in the body that received it. The ASD and ADHD operator modifications of Appendix B.4 and B.3, respectively, were the instruments by which the model was subsequently constructed: the monotropic attractor structure of B.4 provided the capacity for sustained engagement with an entirely unfamiliar theoretical domain; the high-temperature field dynamics of B.3 drove rapid traversal across it.

The proximate cause is described in full in the companion patient-facing publication. Briefly: an acute somatic emergency in 2025 — a genuine threshold-crossing event, later confirmed as cerebral hypoxia secondary to Long Covid — was attributed, at clinical presentation, to psychiatric origin. The present paper is, among its other functions, a formal response to that attribution.

The causal chain is as follows. A pre-verbal trauma in approximately 1968 installed the C-PTSD operator modifications described in Appendix B.2. The ASD and ADHD modifications of Appendix B.3 and B.4 shaped the system across the intervening decades. Fifty-seven years later, that system's accurate interoceptive signal was dismissed as psychiatric noise. The paper which formally demonstrates that this dismissal constitutes a category error was produced, as a direct causal consequence, by the same operator stack that it describes. The paper is the fixed point of its own subject matter. The author considers this observation methodologically significant.

Publication Claim Registry

To support claim-level review rather than all-or-nothing acceptance, this manuscript registers its highest-impact claims with scope labels and disconfirmation tests.

Claim ID	Claim	Scope	Evidence in this work	Disconfirmation criterion
SF-1	Conscious percept is a propagator pole of the soma-field	S1 Structural	Formal derivation in Sections 2-3	Inability to express percept dynamics as Green's-function response under stated operator
SF-2	Emotional attractors are Hopfield-energy minima	S2 Predictive	Energy model and trajectory framework	Constructed update rule under model assumptions with systematic energy ascent

Claim ID	Claim	Scope	Evidence in this work	Disconfirmation criterion
SF-3	Threshold governs felt vs sub-felt emotional activity	S2 Predictive	Threshold operator and clinical mapping	Reliable high-amplitude mode activity with no threshold-dependent behavioural or physiological signature
SF-4	Topological barriers explain classical therapeutic plateaus	S2 Predictive	Formal treatment plus linked companion experiments	Controlled demonstration that matched low-noise classical dynamics crosses registered barriers at equivalent rate
SF-5	Quantum extension yields topological reachability advantage	S2 Predictive	QUANT-EXP-1 companion evidence and linked artifacts	Controlled replication showing no reachability advantage over matched classical baseline

Scope labels: S1 = structural; S2 = predictive; S3 = independently replicated. Current publication target for core claims is S2.

Claim-Evidence-Result Matrix

To make review traceable, each core claim is paired with concrete evidence outputs and current result status.

Claim ID	Evidence artifact(s)	Current result status
SF-1	Sections 2-3 derivation of field/propagator structure	structural derivation complete
SF-2	Energy formulation + instrument runtime equations	predictive structure complete
SF-3	Threshold operator definition + clinical interpretation sections	predictive mapping complete
SF-4	Barrier analysis; companion paper *Quantum Soma and the Penrose Gap* (doi:10.5281/zenodo.20351230)	**confirmed (QUANT-EXP-1 PASS)**
SF-5	QUANT-EXP-1 experiment outputs (see supplementary archive, doi:10.5281/zenodo.20351230)	**confirmed: cold 0/200, CI [0.000, 0.019]; quantum peak 0.408–0.410; all hardening checks PASS**

This matrix is intended for reviewer navigation and is updated as companion results are expanded or independently replicated.

Replication Package Requirements

To make SF-2 through SF-5 externally testable, each release tagged for review must ship a minimal replication package that can be executed without private context.

Required contents:

1. simulation code and support modules (see supplementary archive, doi:10.5281/zenodo.20351230),
2. full parameter snapshot (W, $\mathbf{b}$, γ, D, θ, temperature policy),
3. raw trajectory logs with timestamped attractor labels,
4. analysis scripts that produce the reported summary tables,
5. frozen output artifacts (CSV/plots) referenced in this manuscript.

A claim remains S2 until an independent operator reproduces directionally consistent outcomes from this package under the same declared protocol.

Reviewer-Risk Objections and Responses

To reduce ambiguity in peer review, the highest-probability objections are mapped to bounded responses and concrete upgrade paths.

Reviewer objection	Current response in this manuscript	Remaining action to reach stronger status
"This is an analogy, not a formal model."	Sections 2-4 define operators, dynamics, and testable predictions; Section 9.1 registers disconfirmation criteria claim-wise.	Promote more claims from S2 to S3 via independent replication.
"Evidence is pilot-stage and may not generalize."	Section 9.2 explicitly labels pilot support and companion-only scope.	Add multi-operator replication and blinded protocol variants.
"Quantum advantage may be implementation-specific."	SF-5 includes a controlled disconfirmation criterion against matched classical baselines.	Publish full benchmark harness with pre-registered acceptance thresholds.
"Clinical interpretation may exceed data scope."	Scope labels (S1/S2/S3) and claim registry separate structural from predictive claims.	Add prospective cohort evidence before any clinical-effectiveness claim.

Independent Replication Ledger Linkage

S2 to S3 promotion for this manuscript is governed by an independent replication ledger maintained in the supplementary archive (doi:10.5281/zenodo.20350515).

Tracked claim IDs in ledger scope: SF-2, SF-3, SF-4, SF-5.

Promotion gate: a claim is upgraded only when at least one ledger row records an independent operator PASS with a reproducible package hash and linked raw/derived evidence artifacts.

Acknowledgements

This work exists because ten years of psychotherapy moved the barriers far enough that two events in early 2026 could cross them. The theory is, among other things, a record of that.

Conclusion: The Field Is the Territory

We began with a question that respectable science rarely asks aloud: why does anything feel like anything? Twelve chapters later, we have not merely answered the question — we have shown that it was never as mysterious as it seemed. The mystery arose from a mismatch: we had a picture of the universe as made of matter, governed by forces, described by equations — and we had experience, sitting outside that picture, irreducible, private, undeniable. The Universal Somatic Field dissolves the mismatch by putting experience *inside* the picture. Not as a late addition, not as an emergent byproduct, but as a primary physical field, present from the beginning, governing dynamics from the quantum scale to the civilisational.

What We Have Established

The framework that runs through this book has three load-bearing claims, each of which we have examined in turn.

The first is **ontological**: the somatic field is a physical field in the same sense that the electromagnetic field is a physical field. It has energy, it obeys equations, it propagates through space, and it can be coupled to measuring instruments. The distinction between the physical and the experiential is not a distinction between different kinds of thing; it is a distinction between the third-person description of the field and the first-person inhabiting of it.

The second is **formal**: the somatic field equation is derivable — not postulated — from M-theory compactification, and its attractor structure is the emotional landscape. This is not a metaphor improved by physics vocabulary. The derivation is in the papers; the Lean 4 proofs are in the repository; the formal identities are machine-checked.

The third is **empirical**: the framework makes predictions that can be tested. The quantum annealing experiment has been run. The BRECVEMA mechanisms map onto field dynamics. The Arnold tongue width is measurable from EEG. The WKB prediction for emotional transitions gives a testable rate formula. Science lives or dies by its predictions, and the USF framework is alive in this sense: it predicts things that could, in principle, go wrong.

What Remains

No framework is complete at first statement. Three classes of open questions follow from the material in this book.

The first is **measurement**. The somatic field is real, but the instruments for measuring it directly do not yet exist. Current instruments (EEG, MEG, fMRI, skin conductance) measure correlates. The field-theoretic framework points toward what a direct measurement would require: a sensor coupled to the tensor components of the somatic field, not merely to its projections onto the scalar electromagnetic field. This is a technology programme, not a physics programme. The physics is done; the engineering has not yet started.

The second is **biological detail**. The framework derives the broad strokes — the Hopfield potential, the Arnold tongue, the WKB barrier formula — from first principles, but fills in the biological coupling constants from order-of-magnitude estimates and consistency conditions. A full specification of the somatic field for a human nervous system requires knowing the coupling constants precisely. That requires experiment.

The third is **cosmological**. The derivation of the cosmological constant as the vacuum expectation value of the somatic tensor trace is tantalising and formally correct, but the numerical value requires a level of control over the compactification geometry that current string phenomenology cannot provide. This is not a refutation; it is an invitation to future work.

The Invitation

This book was designed to be read by someone who had no prior knowledge of the framework and who came to it with some scepticism — the right starting attitude for encountering a large claim. If the reading has gone well, the scepticism will have softened into curiosity. Not agreement, necessarily; but the recognition that the claim is serious, that the derivations are real, that the predictions are testable, and that the framework does something that no existing framework does: it gives a mathematical account of why the world feels like anything.

That account is now in your hands. The field equations are written down. The proofs are checked. The experiments have begun.

What happens next depends on who reads this, and what they do with it.

Juslin, P. N., & Sloboda, J. A. (Eds). (2011). *Handbook of music and emotion: Theory, research, applications*. Oxford University Press.

McCulloch, W. S., & Pitts, W. (1943). A logical calculus of the ideas immanent in nervous activity. *The Bulletin of Mathematical Biophysics*, *5*(4), 115–133. https://doi.org/10.1007/BF02478259

Ramachandran, V. S., & Hirstein, W. (1998). The perception of phantom limbs: The D.O. Hebb lecture. *Brain*, *121*(9), 1603–1630. https://doi.org/10.1093/brain/121.9.1603

Seth, A. K. (2021). *Being you: A new science of consciousness*. Dutton.

Veneziano, G. (1968). Construction of a crossing-symmetric, Regge-behaved amplitude for linearly rising trajectories. *Il Nuovo Cimento A*, *57*(1), 190–197. https://doi.org/10.1007/BF02824451